

Digital Trade and Multidimensional Economic Complexity

Viktor Stojkoski

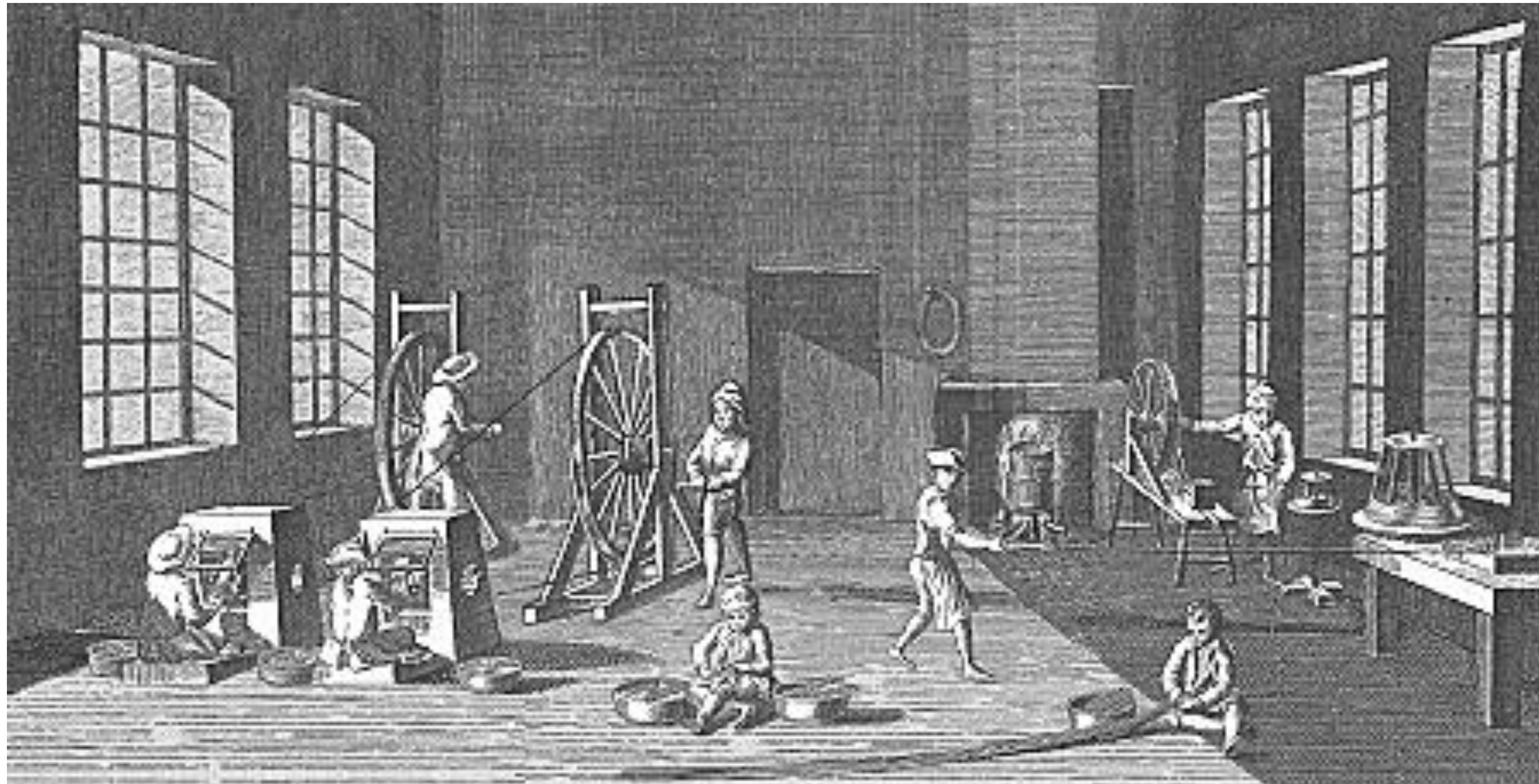


Faculty of Economics - Skopje
Center for Collective Learning

December 2025

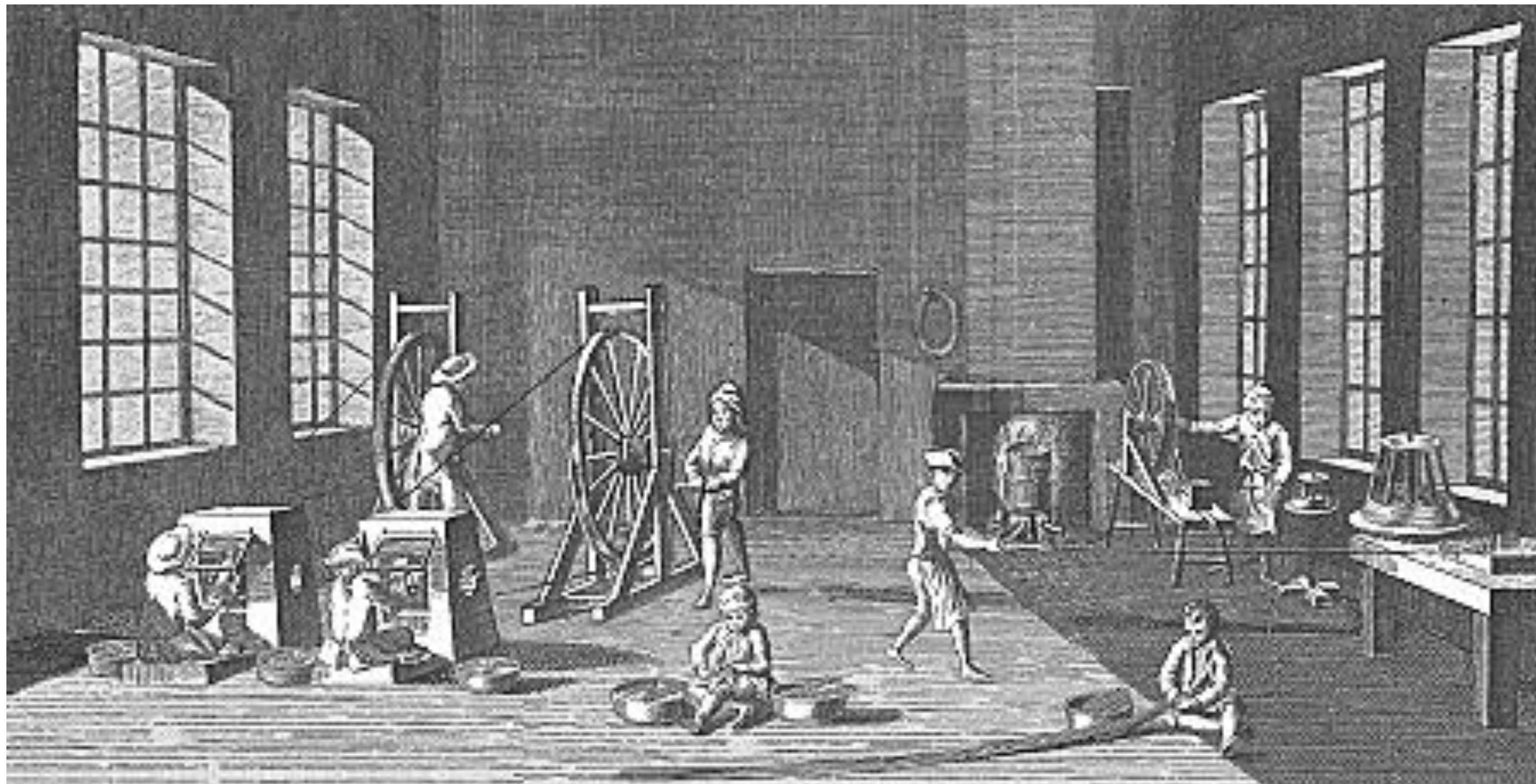
The Division of labor is at the root of Economic Development

“Individuals, organizations, and nations are endowed with or acquire specialized capabilities, and either form combinations or trade to take advantage of the capabilities of others in addition to their own.” – *Adam Smith, The Wealth of Nations*



The Division of labor is at the root of Economic Development

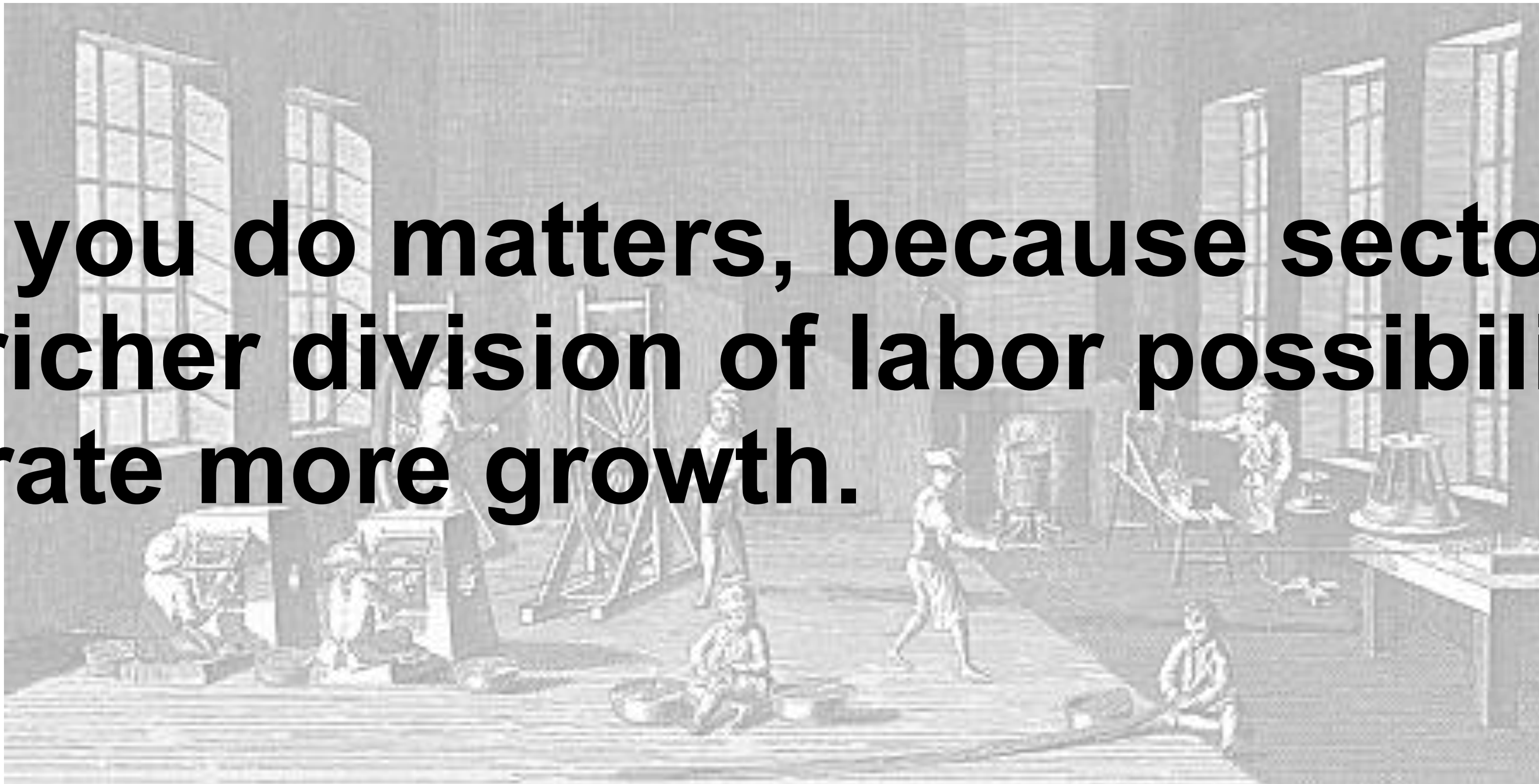
Some activities allow much more division of labor and innovation than others (e.g., manufacturing vs. agriculture).— Adam Smith, *The Wealth of Nations*



The Division of labor is at the root of Economic Development

Some activities allow much more division of labor and innovation than others (e.g., manufacturing vs. agriculture).— Adam Smith, *The Wealth of Nations*

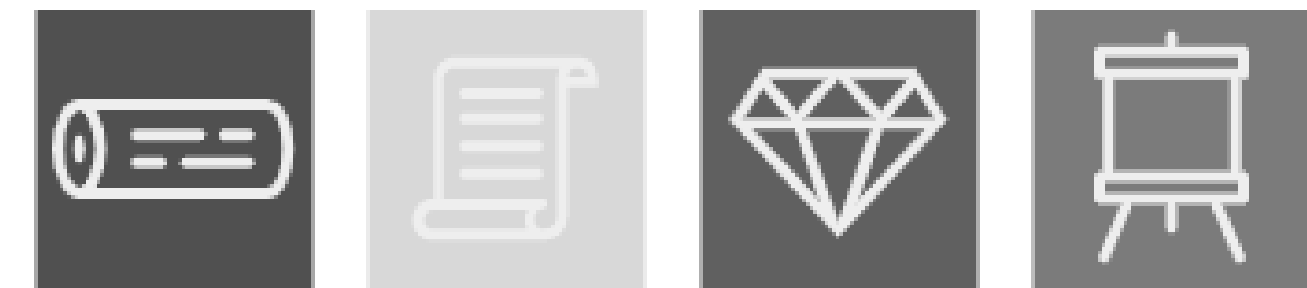
What you do matters, because sectors with richer division of labor possibilities generate more growth.



Economic Complexity

Helps us quantify which sectors support more economic development

- Capabilities is **non-fungible**: It behaves like **ingredients**, i.e. it is **combinatorial** instead of aggregative.
- Economic complexity methods involve the use of machine learning and network techniques to predict and explain **the structure of economic output**.



Measuring Economic Complexity

Reduced-dimensionality representations of specialization matrices that explain the geography of hundreds of economic activities

Measuring Economic Complexity

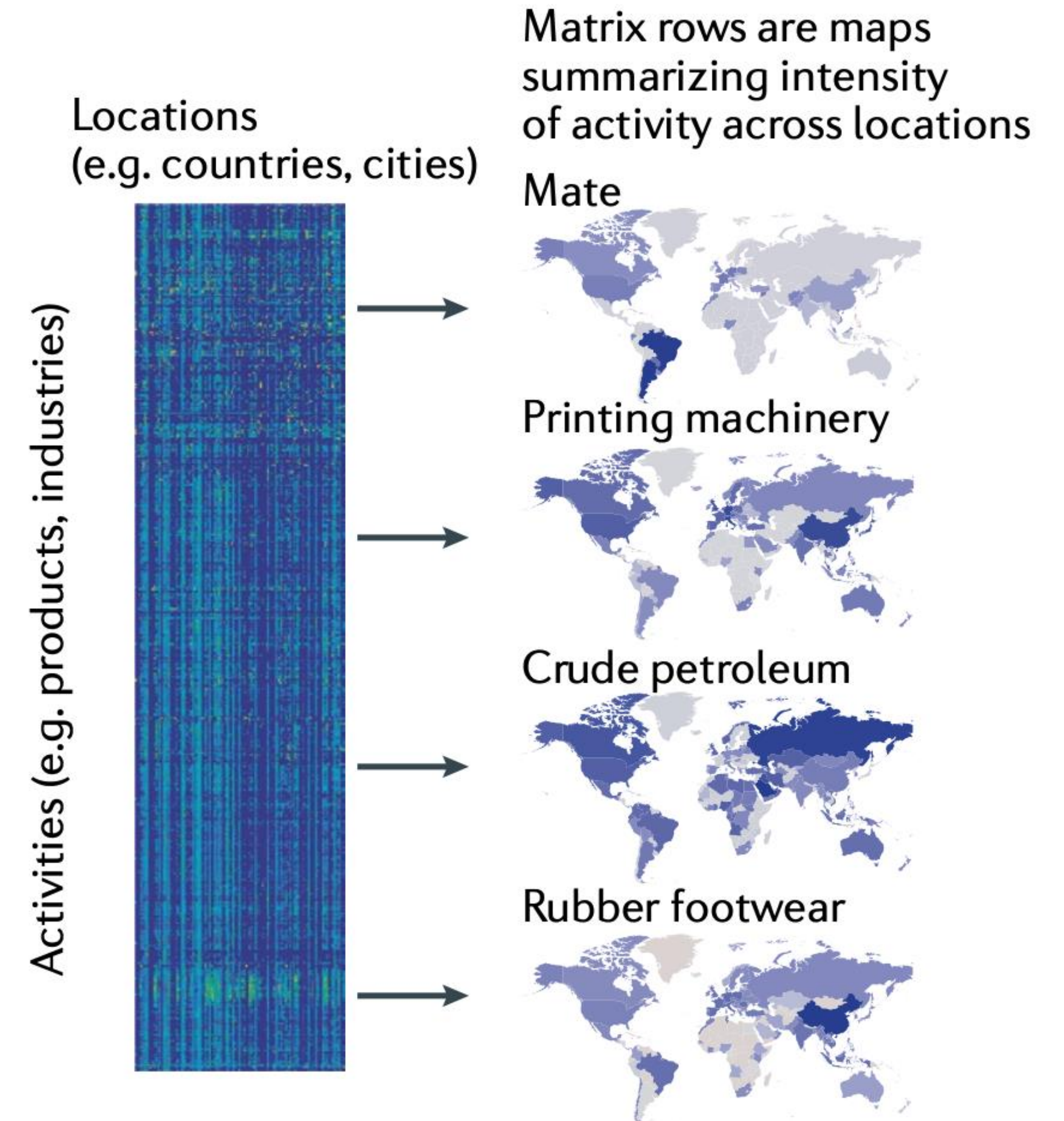
Reduced-dimensionality representations of specialization matrices that explain the geography of hundreds of economic activities

- Location by activity Specialization matrix:

$$M_{cp} = \begin{cases} 1 & \text{if Specialized} \\ 0 & \text{otherwise} \end{cases}$$

- e.g. South Korea is specialized in exporting Cars if it exports more than what is expected in the world

a Specialization matrix (R)



Measuring Economic Complexity

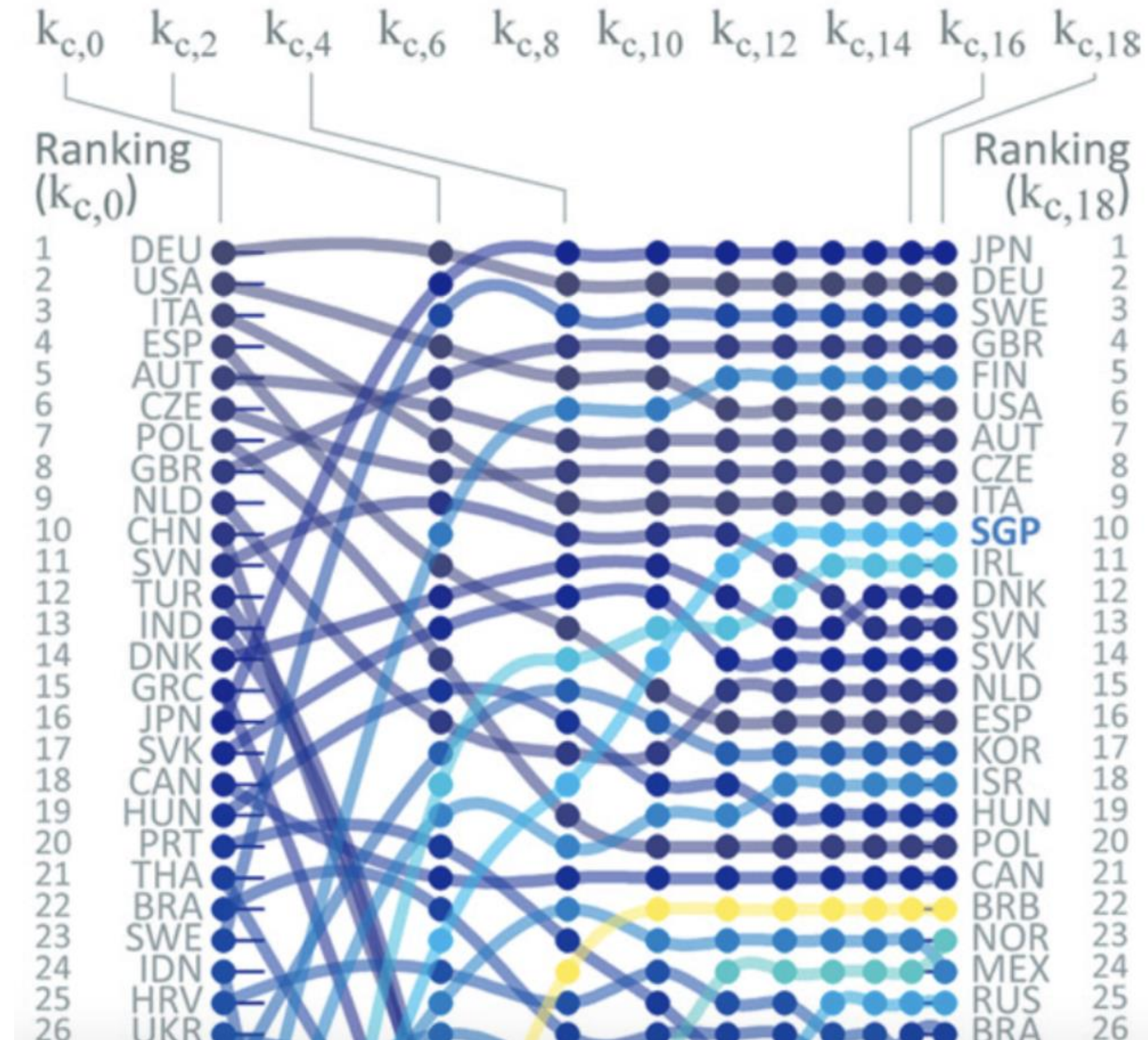
Reduced-dimensionality representations of specialization matrices that explain the geography of hundreds of economic activities

- The **complexity of a location** is the **average** of the complexities of the activities in which it is specialized

$$K_c = \frac{1}{M_c} \sum_p M_{cp} K_p$$

- The **complexity of an activity** is the **average** of the complexities of the locations specialized in it

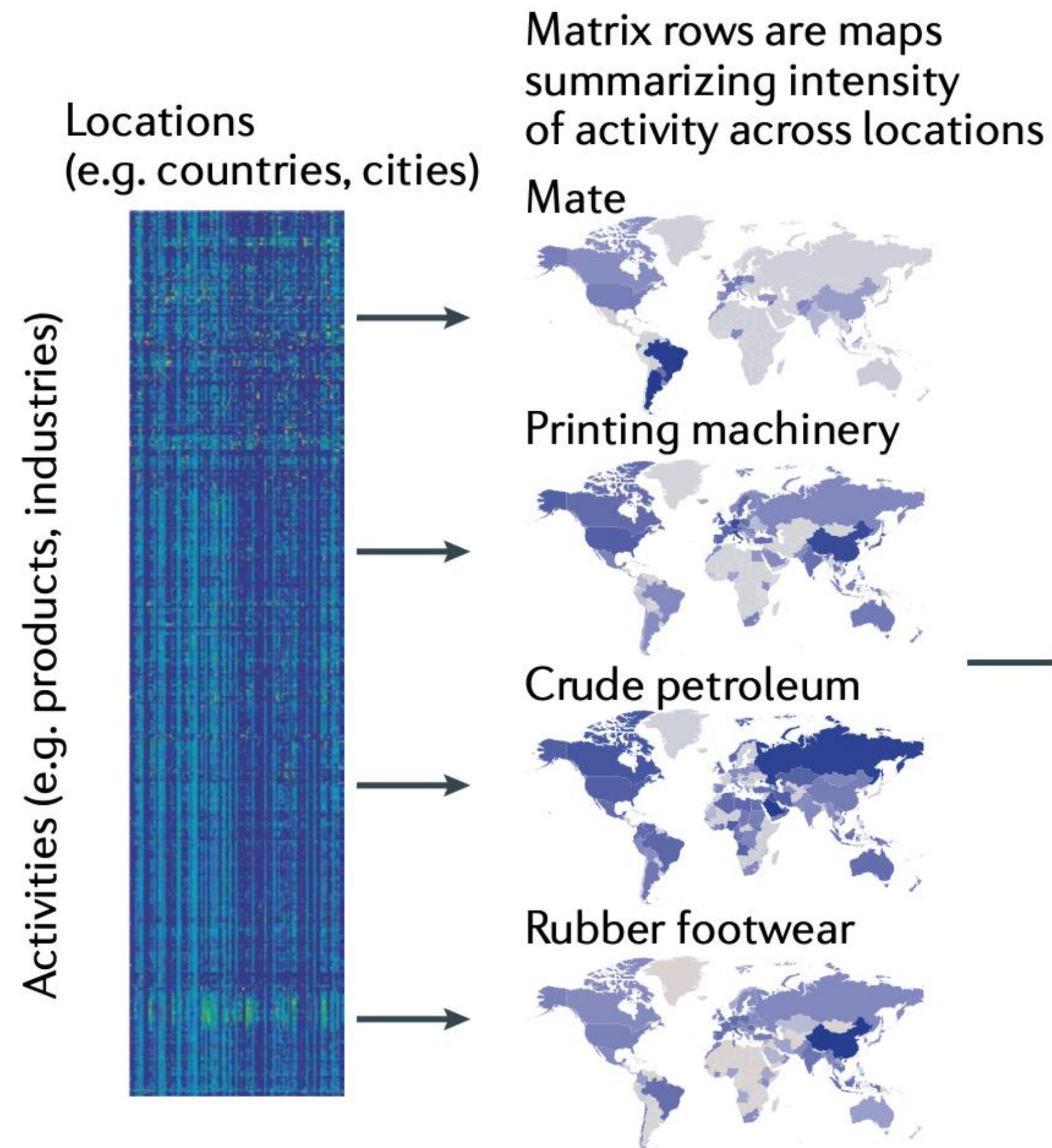
$$K_p = \frac{1}{M_p} \sum_c M_{cp} K_c$$



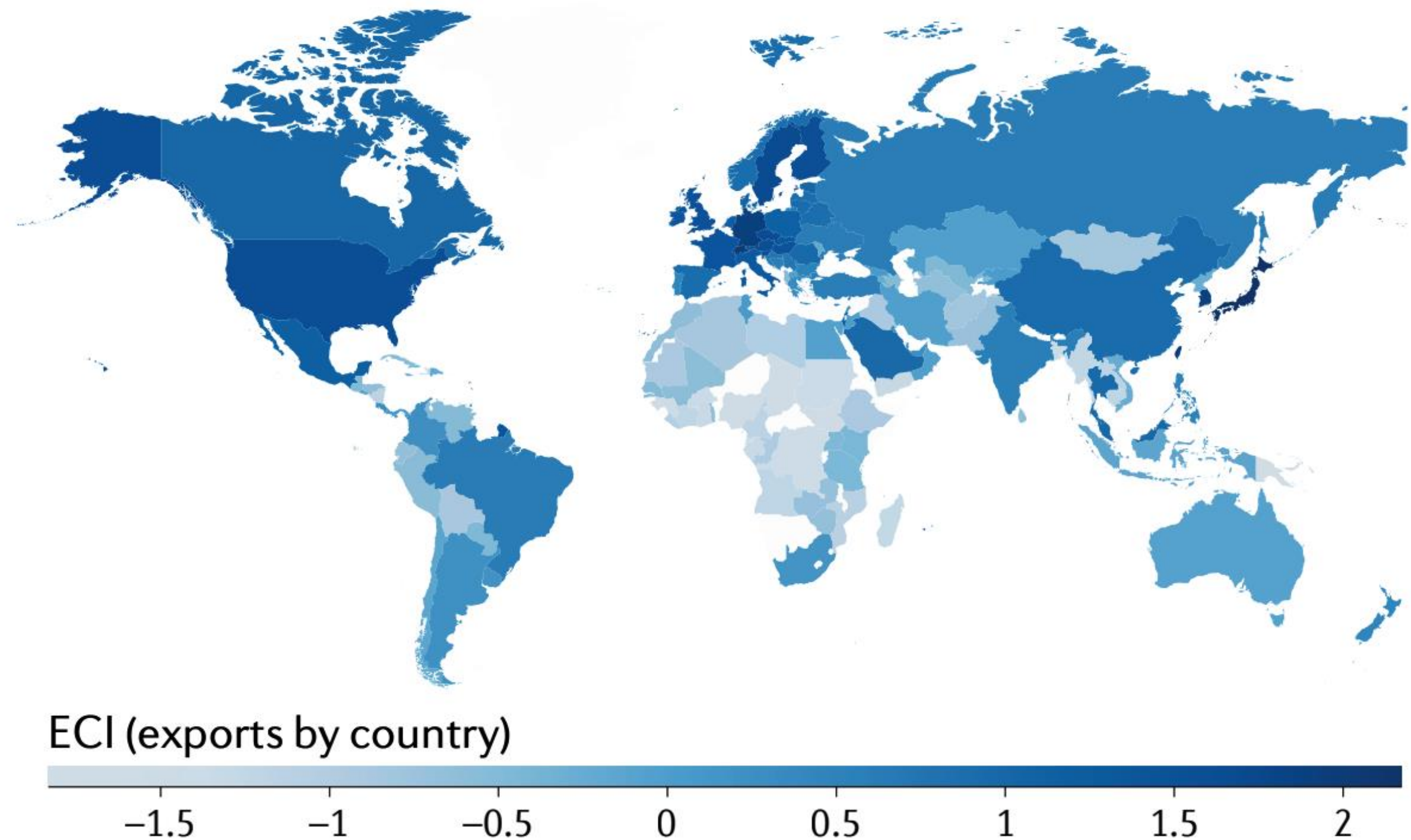
Measuring Economic Complexity

Reduced-dimensionality representations of specialization matrices that explain the geography of hundreds of economic activities

a Specialization matrix (R)



Economic complexity is an approximation to the map that best explains all other specialization maps



Economic complexity rankings (trade)

2000

3	 GERMANY	1.75
4	 SWEDEN	1.70
5	 UNITED STATES	1.66
6	 UNITED KINGDOM	1.64
7	 FINLAND	1.59
8	 AUSTRIA	1.46
9	 IRELAND	1.44
10	 FRANCE	1.38
11	 NETHERLANDS	1.24
12	 BELGIUM	1.21
13	 ITALY	1.19
14	 DENMARK	1.19
15	 ISRAEL	1.18
16	 CZECHIA	1.07
17	 SLOVENIA	1.05
18	 CANADA	1.03
19	 SPAIN	1.00
20	 NORWAY	0.96
21	 SINGAPORE	0.88
22	 CHINESE TAIPEI	0.86
23	 SLOVAKIA	0.73
24	 HUNGARY	0.71
25	 NEW ZEALAND	0.64
26	 RUSSIA	0.63
27	 MEXICO	0.60
28	 SOUTH KOREA	0.60
29	 BRAZIL	0.59

2010

3	 SWITZERLAND	1.85
4	 SWEDEN	1.67
5	 CHINESE TAIPEI	1.62
6	 FINLAND	1.60
7	 UNITED KINGDOM	1.59
8	 UNITED STATES	1.59
9	 CZECHIA	1.52
10	 AUSTRIA	1.52
11	 SINGAPORE	1.52
12	 SOUTH KOREA	1.49
13	 FRANCE	1.42
14	 SLOVENIA	1.36
15	 IRELAND	1.36
16	 BELGIUM	1.30
17	 HUNGARY	1.29
18	 ITALY	1.27
19	 ISRAEL	1.24
20	 SLOVAKIA	1.20
21	 NETHERLANDS	1.18
22	 DENMARK	1.04
23	 MEXICO	1.00
24	 POLAND	0.98
25	 NORWAY	0.95
26	 CANADA	0.94
27	 BELARUS	0.86
28	 SPAIN	0.85
29	 MALAYSIA	0.84

2021

3	 CHINESE TAIPEI	1.93
4	 SOUTH KOREA	1.82
5	 GERMANY	1.81
6	 SINGAPORE	1.74
7	 CZECHIA	1.56
8	 SWEDEN	1.54
9	 AUSTRIA	1.50
10	 UNITED STATES	1.46
11	 UNITED KINGDOM	1.43
12	 SLOVENIA	1.42
13	 FINLAND	1.42
14	 HUNGARY	1.36
15	 FRANCE	1.35
16	 SLOVAKIA	1.31
17	 IRELAND	1.29
18	 BELGIUM	1.28
19	 ITALY	1.27
20	 ISRAEL	1.21
21	 HONG KONG	1.16
22	 NETHERLANDS	1.09
23	 MEXICO	1.09
24	 MALAYSIA	1.09
25	 CHINA	1.07
26	 ROMANIA	1.07
27	 DENMARK	1.02
28	 POLAND	1.01
29	 THAILAND	0.97

Why are economic complexity measures useful?

The geography of economic activities determines the inclusive green growth potential

- Economic complexity is related to:

- 1. Economic growth**
- 2. Income inequality**
- 3. Emission intensity**

Why are economic complexity measures useful?

The geography of economic activities determines the inclusive green growth potential

- Economic complexity is related to:

- Economic growth
- Income inequality
- Emission intensity

IDEA:
Higher complexity economies can participate in sophisticated sectors that support higher wages.

	Annualized growth in GDP pc (by decade)				
	(1978-1988, 1988-1998, 1998-2008)				
VARIABLES	(1)	(2)	(3)	(4)	(5)
Initial Economic Complexity Index (ECI)	0.04430*** (0.009)	0.03005*** (0.007)	0.04240*** (0.008)	0.04143*** (0.010)	0.04389*** (0.009)
[ECI] X [Income per capita, log]	-0.00371*** (0.001)	-0.00244*** (0.001)	-0.00345*** (0.001)	-0.00354*** (0.001)	-0.00381*** (0.001)
Increase exports (goods and services)		0.04549*** (0.007)			
- in constant dollars (as a share of initial GDP)					
Exports to GDP			0.00009 (0.000)		
Export Concentration				-0.00890 (0.008)	
Population, log					0.00168 (0.001)
Initial Income per capita, log	-0.00638*** (0.001)	-0.00562*** (0.001)	-0.00729*** (0.001)	-0.00611*** (0.001)	-0.00558*** (0.001)
Increase in natural resource exports	0.03682*** (0.010)	0.00169 (0.005)	0.03441*** (0.008)	0.03699*** (0.010)	0.03758*** (0.010)
- in constant dollars (as a share of initial GDP)					
Constant	0.08251*** (0.011)	0.06741*** (0.011)	0.08616*** (0.011)	0.08145*** (0.011)	0.04878** (0.022)
Observations	291	260	284	291	291
R ²	0.434	0.584	0.449	0.436	0.440
Year FE	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

(Hidalgo and Hausmann, 2009; Hausmann et al., 2014; Chavez et al., 2017; Dogan et al., 2022; Domini, 2019; Koch, 2021; Lo Turco and Maggioni, 2020; Ourens, 2012; Stojkoski et al., 2016; Stojkoski and Kocarev, 2017)

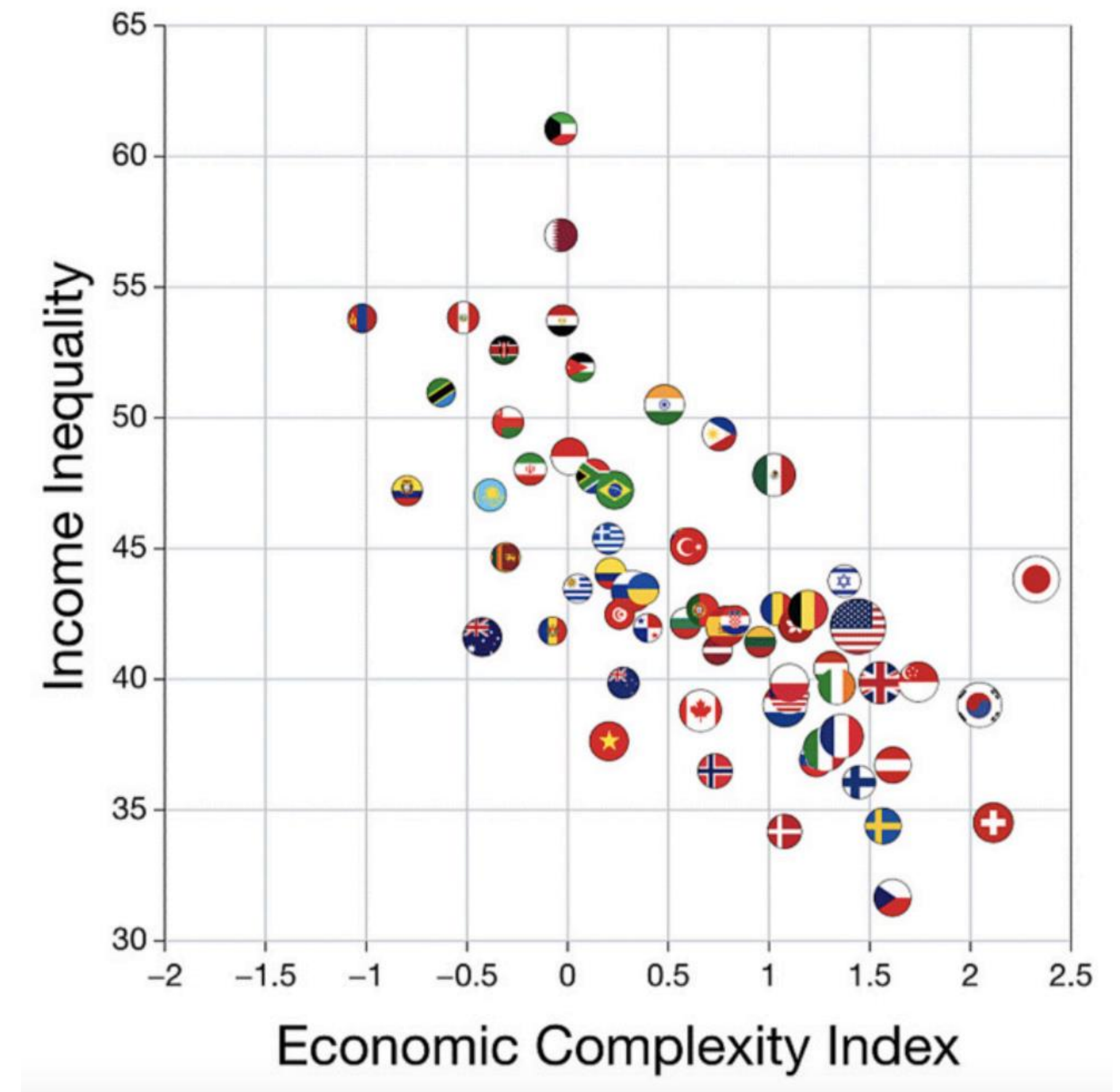
Why are economic complexity measures useful?

The geography of economic activities determines the inclusive green growth potential

- Economic complexity is related to:
 1. Economic growth
 - 2. Income inequality**
 3. Emission intensity

IDEA:

Firms operating in knowledge intense activities promote inclusive institutions because of their need to attract and retain talent.



(Barza et al., 2020; Basile and Cicerone, 2022; Ben Saad and Assoumou-Ella, 2019; Chu and Hoang, 2020; Fawaz and Rahnema-Moghadamm, 2019; Hartmann et al., 2017; Sbardella et al., 2017)

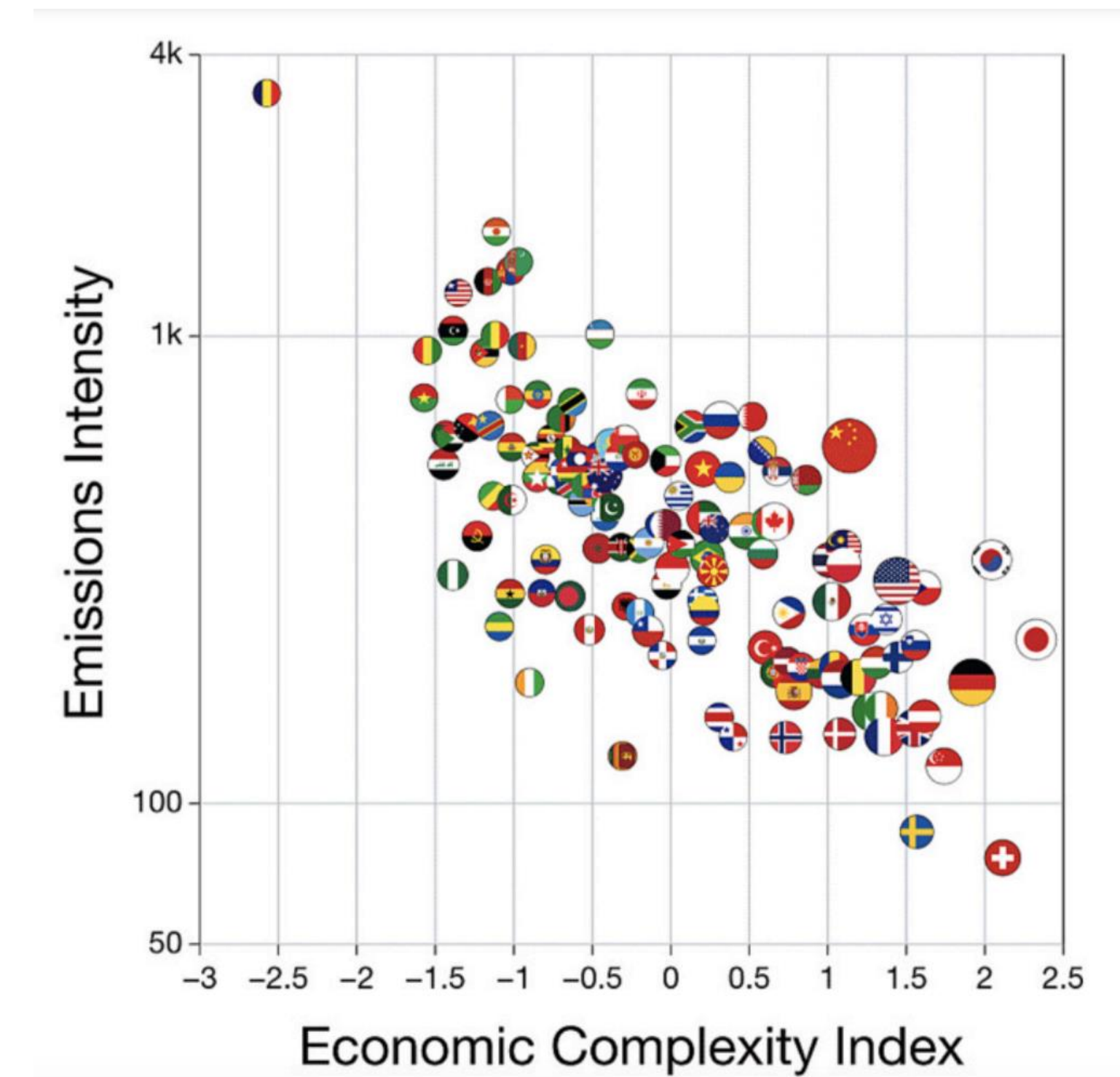
Why are economic complexity measures useful?

The geography of economic activities determines the inclusive green growth potential

- Economic complexity is related to:
 1. Economic growth
 2. Income inequality
 3. **Emission intensity**

IDEA:

The emissions required to produce a unit of GDP by extracting tin ore are larger than the emissions required to produce a unit of GDP by manufacturing metal-cutting machines.



Limitations of Economic Complexity measures

“They don’t measure **diversification**, complexity, etc.”



Limitations of Economic Complexity measures

“They don’t measure **diversification**, complexity, etc.

We need new measures...

And over the past decade there have been a lot of papers that try to improve the measurement of economic complexity.

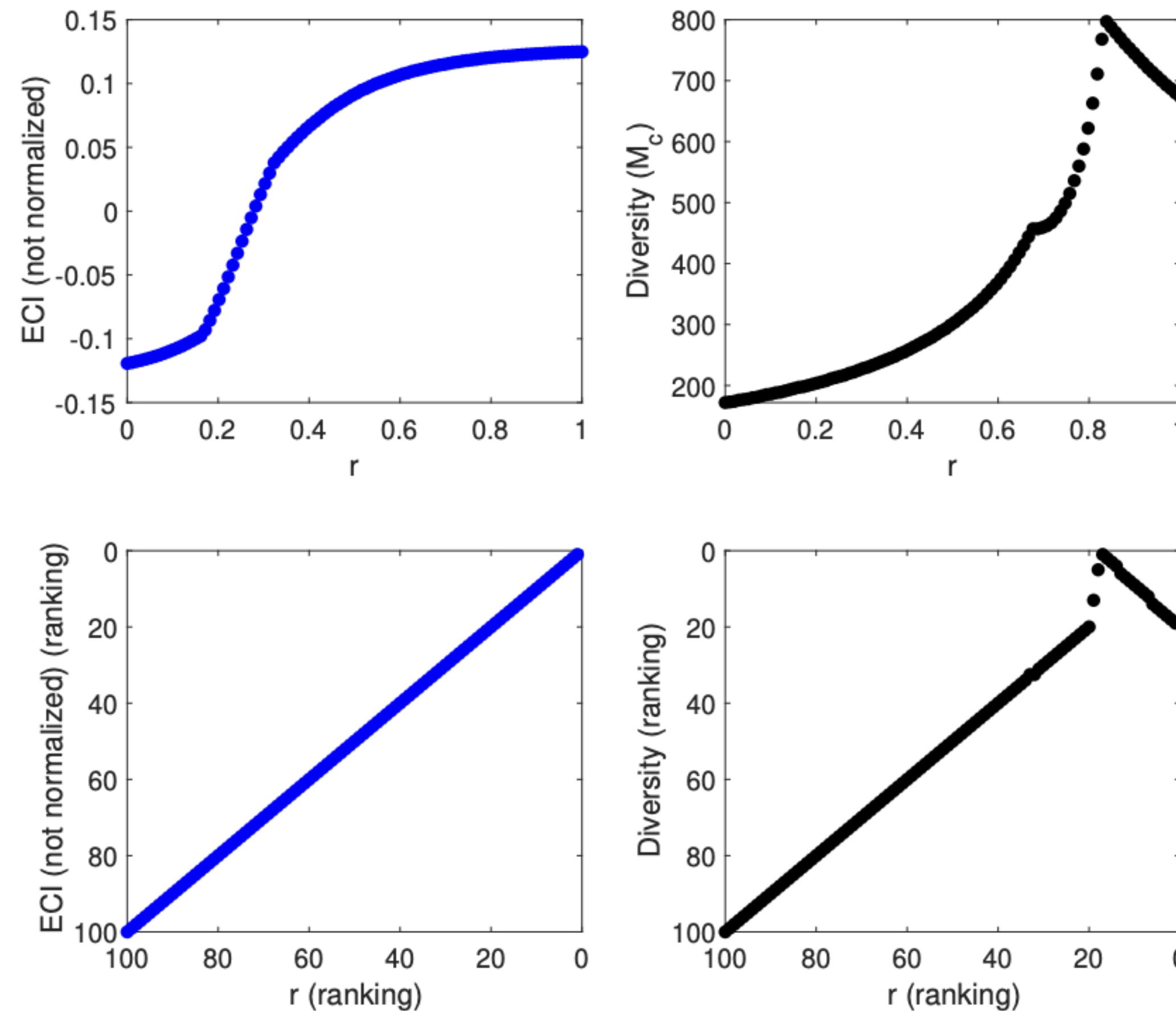
Limitations of Economic Complexity measures

“They don’t measure **diversification, complexity, etc.”**

More recently, in a theoretical model it was shown that Economic Complexity is proportional to the capabilities in an economy, whereas measures correlated with diversification have lower correlation.

Limitations of Economic Complexity measures

“They don’t measure **diversification**, complexity, etc.”



r = probability to have a capability

Limitations of Economic Complexity measures



Limitations of Economic Complexity measures

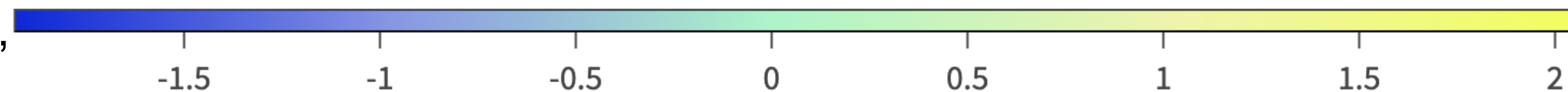
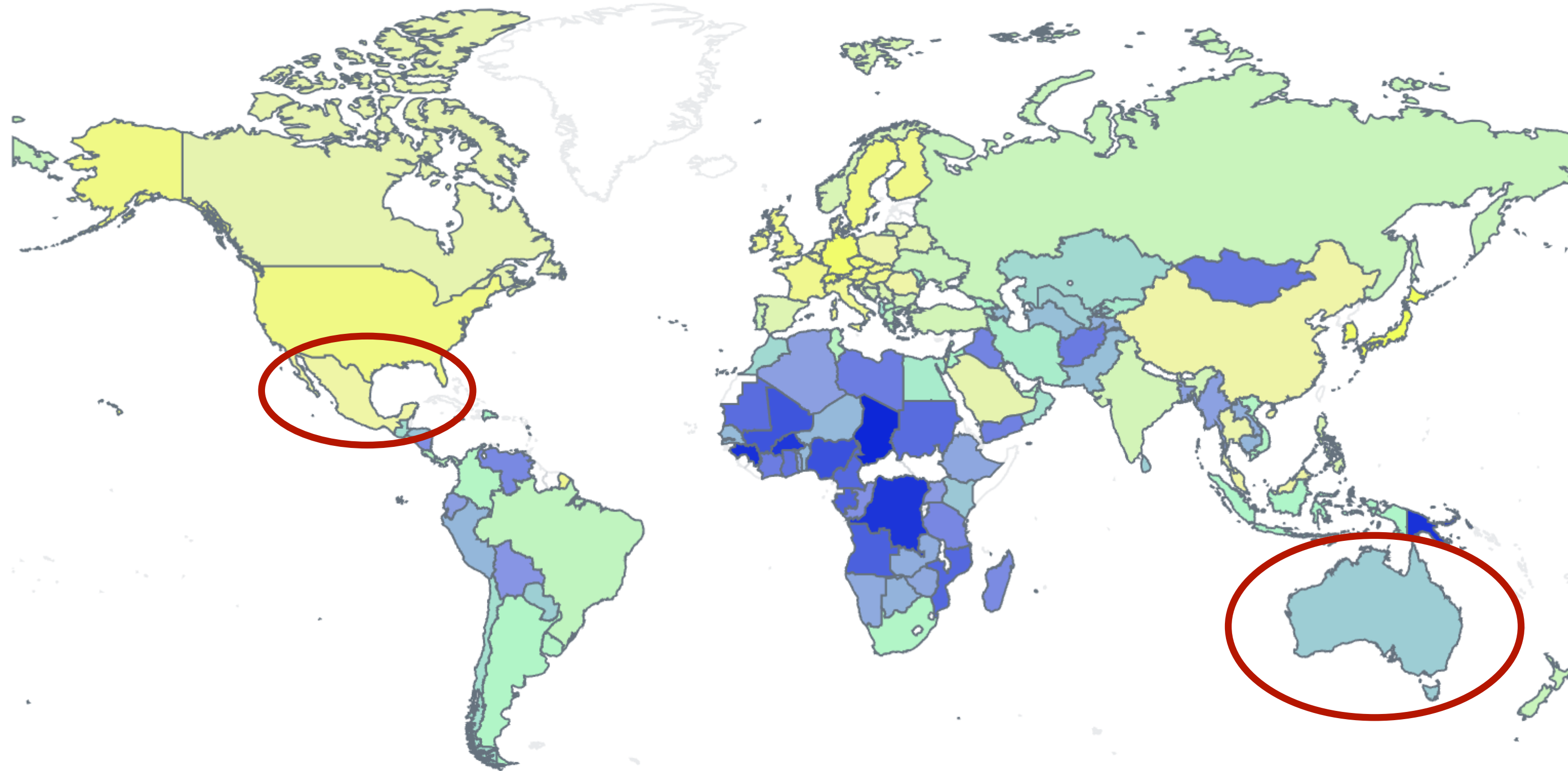
Second, because the data is collected by customs offices, it includes only goods and not services. This is an important drawback, as services are becoming a rising share of international trade. Unfortunately, the statistical efforts of most countries of the world have not kept up with this reality. Finally, the data does not include information on non-tradable activities. These are an important part of the economic eco-system that allows products and services to be made. Our current research is focused on finding implementable solutions to these limitations, and we hope we will be able to present them in future versions of this Atlas. ●

Biases in measures of exports-based measures of economic complexity

Mexico: Ranks high in trade complexity due to integration in the USA value chains.

Australia: Ranks low in trade complexity because of distance to advanced markets.

Economic Complexity Index Trade (ECI Trade)



Economic Complexity

Trade

Technology

Research

Multidimensional Economic Complexity and Inclusive Green Growth

1 JPN
2 SWE
3 DEU
4 SWE
5 SGP
6 CZE
7 KOR
8 AUT
9 GBR
10 FIN
11 SVN
12 USA
13 FRA
14 IRL
15 HUN
16 SVK
17 FRA
18 ISR
19 NLD
20 BEL
21 DNK

1 SWE
2 ITA
3 AUT
4 DEU
5 CHE
6 FIN
7 CHN
8 AUS
9 ESP
10 NOR
11 GBR
12 NZL
13 TUR
14 CAN
15 NLD
16 ZAF
17 FRA
18 RUS
19 BEL
20 BRA
21 DNK

1 USA
2 GBR
3 AUS
4 CAN
5 ISR
6 IRL
7 NLD
8 NOR
9 SWE
10 NZL
11 ITA
12 DEU
13 HKG
14 BEL
15 FIN
16 FRA
17 ESP
18 CHE
19 AUT
20 DNK
21 GRC

Viktor Stojkoski, Philipp Koch, Cesar Hidalgo

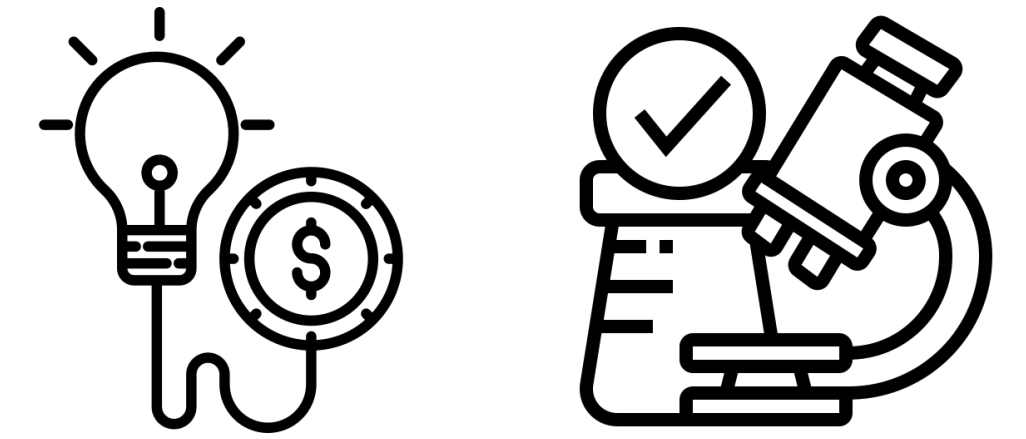
Our Resolution

“Looking at complexity with 4d glasses!”



*Adding the dimensions of **Research, Technology, and Digital Products** to the measurement of economic complexity*

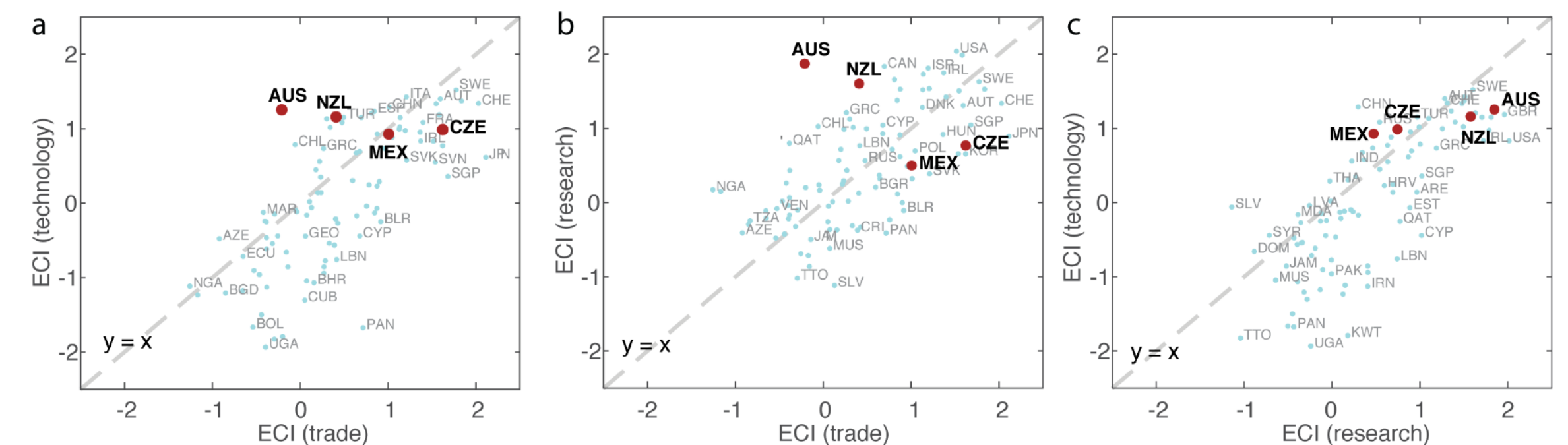
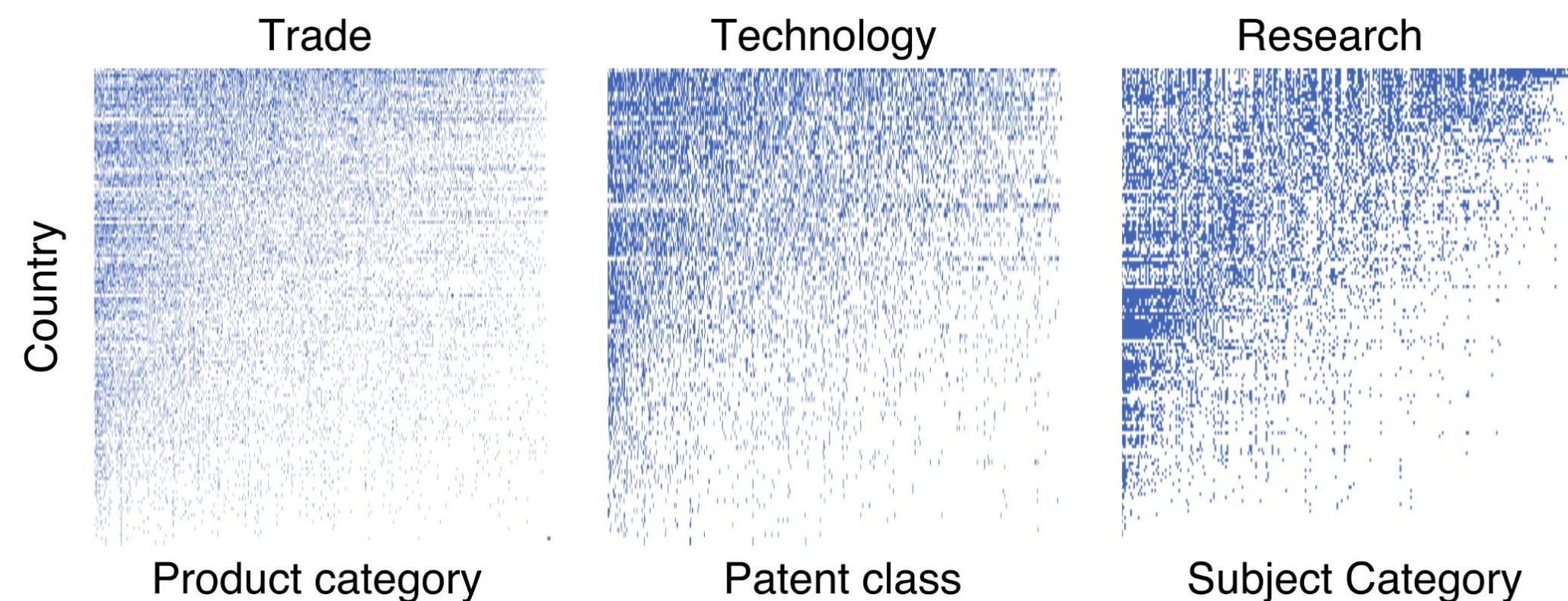
Multidimensional Economic Complexity and Inclusive Green Growth



Objective: We enhance economic complexity metrics by introducing a multidimensional approach.

Methods: We estimated three independent economic complexity indexes (ECI) based on data regarding exports, patents, and research papers for over 100 countries.

These metrics were then used to explain international variations in three key indicators: economic growth (changes in GDP per capita), income inequality (Gini index), and the intensity of greenhouse gas emissions (emissions per unit of GDP).



Why is this important?

A need to understand how different forms of specializations combine to explain international variations in inclusive green growth:

(i) improving the welfare of the population through economic growth



(ii) supporting low levels of income inequality



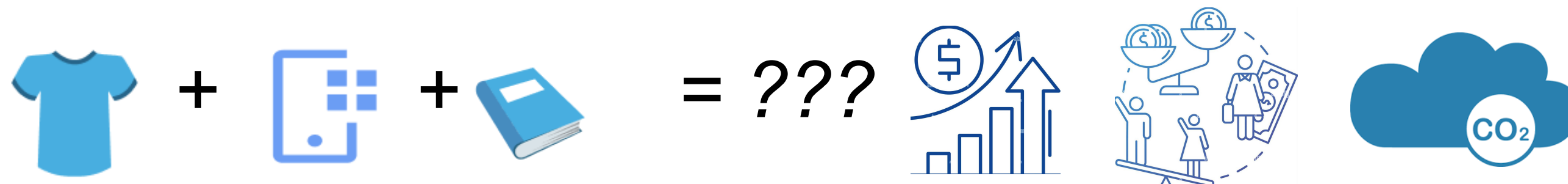
(iii) lowering the intensity of greenhouse gas emissions



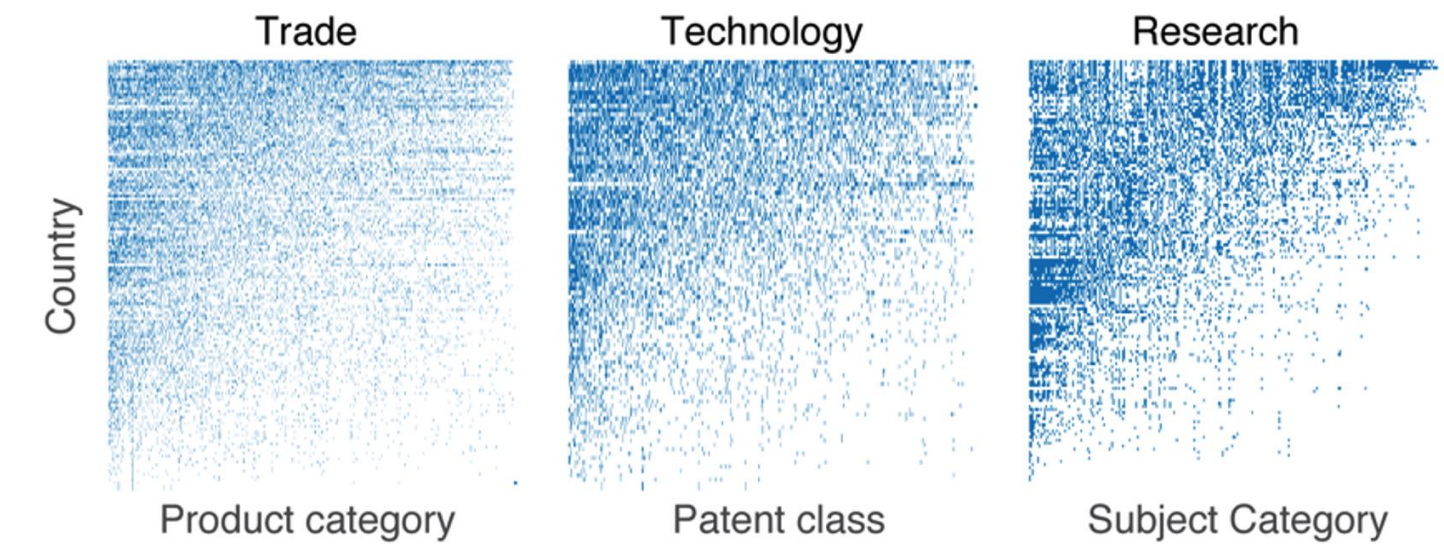
Why is this important?

A need to understand how different forms of specializations combine to explain international variations in inclusive green growth:

- *Does research complexity help reduce GHG emissions?*
- *Does technology complexity improve the explanatory power of trade complexity to explain income inequality?*
- *Are there any interactions between the dimensions in explaining inclusive green growth?*



Data



Data for **150 countries** and **24 years (1996 - 2019)**.

• Product exports:



Exports

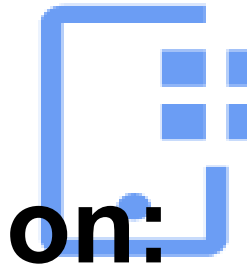
-> **Source:**

Observatory of Economic Complexity

-> **Classification:** HS4

-> **# Activities:** 1241 (product categories)

• Technological innovation:



Patent applications

-> **Source:**

WIPO International Patent System

-> **Classification:** CPC 4 digit

-> **# Activities:** 668 (patent classes)

• Research output:



Published documents

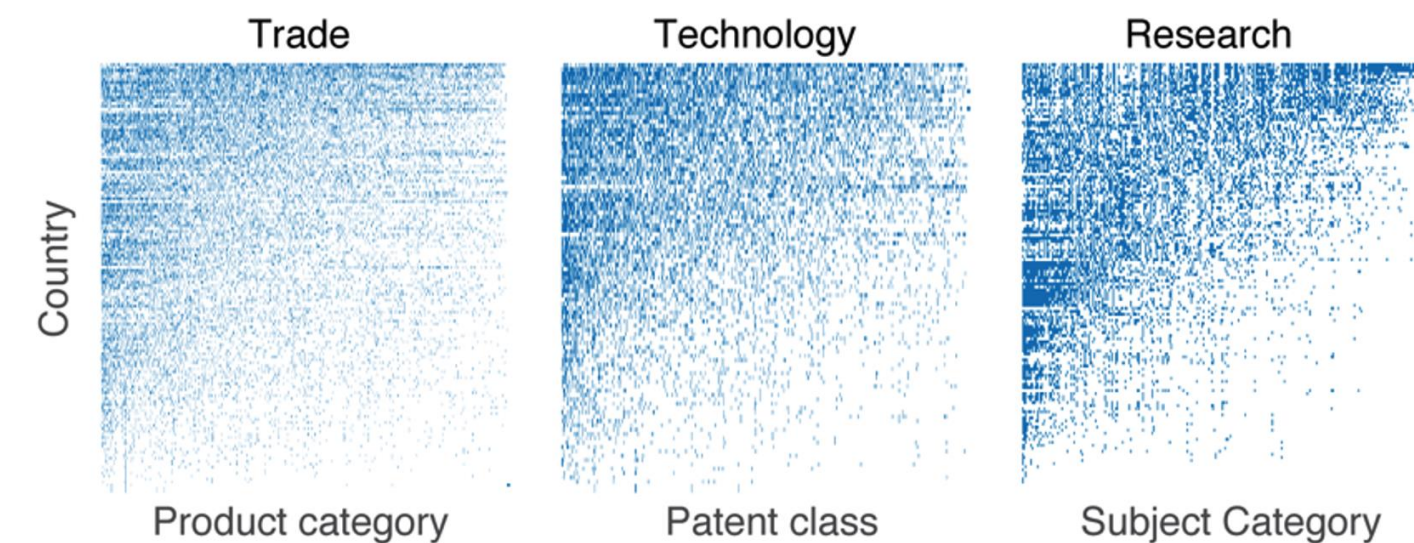
-> **Source:**

SCImago Journal & Country Rank

-> **Classification:** Scopus

-> **# Activities:** 313 (research categories)

Method



Specialization matrix M for each dimension d :

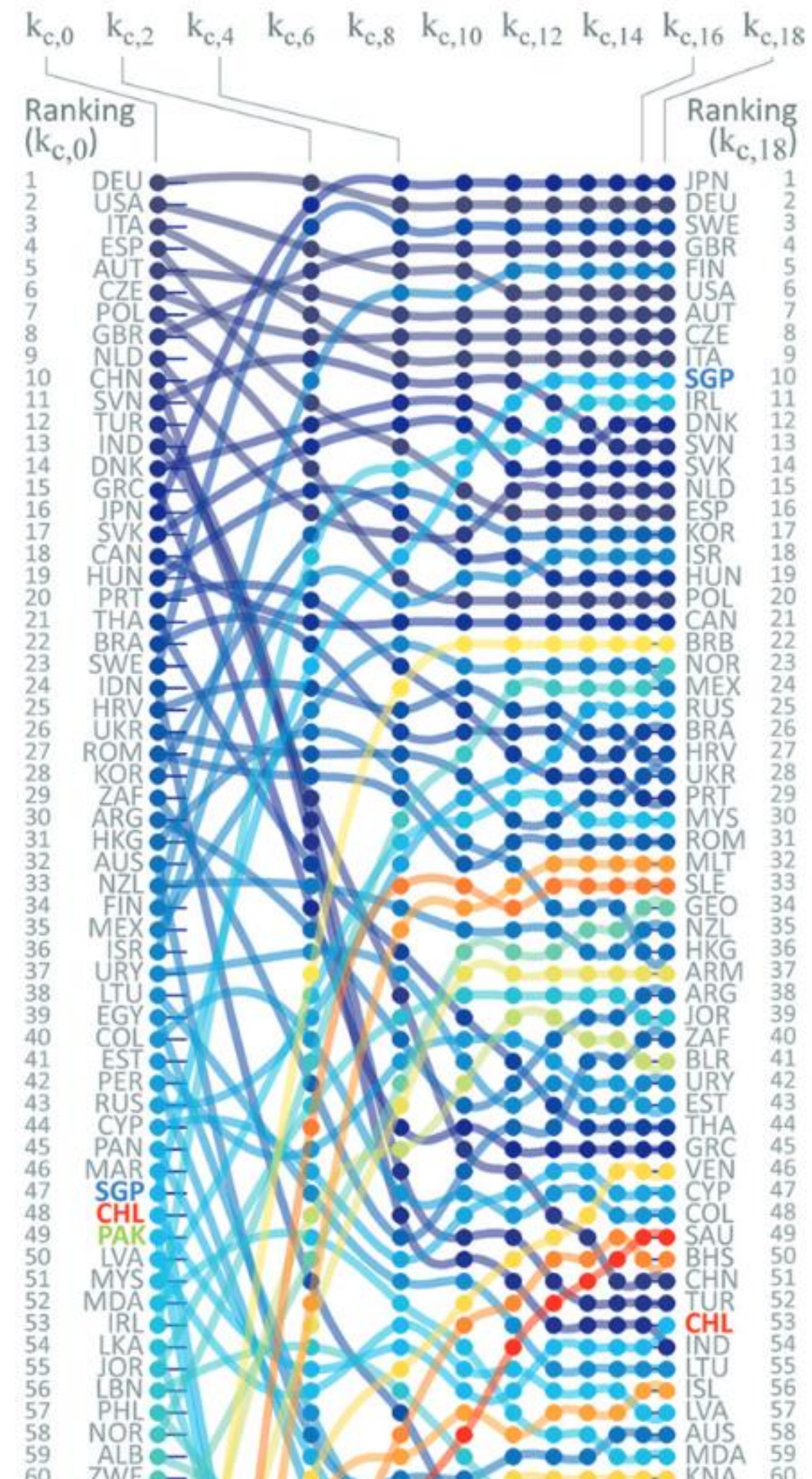
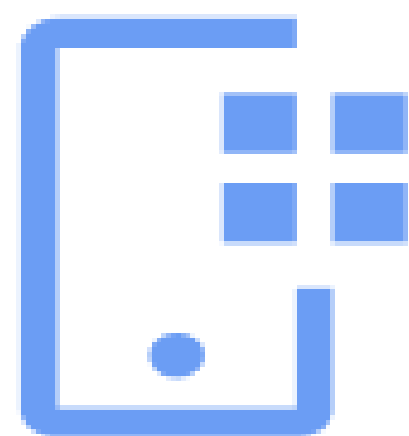
$$M_{cp}^d = \begin{cases} 1, & \text{if } c \text{ is specialized in } p \\ 0, & \text{otherwise} \end{cases}$$

Specialization definition: Revealed Comparative Advantage Index.

For each dimension we estimate **Economic Complexity Index (ECI)**.

For each country we have **3 complexity indexes**:

- 1) ECI (trade)
- 2) ECI (technology)
- 3) ECI (research)

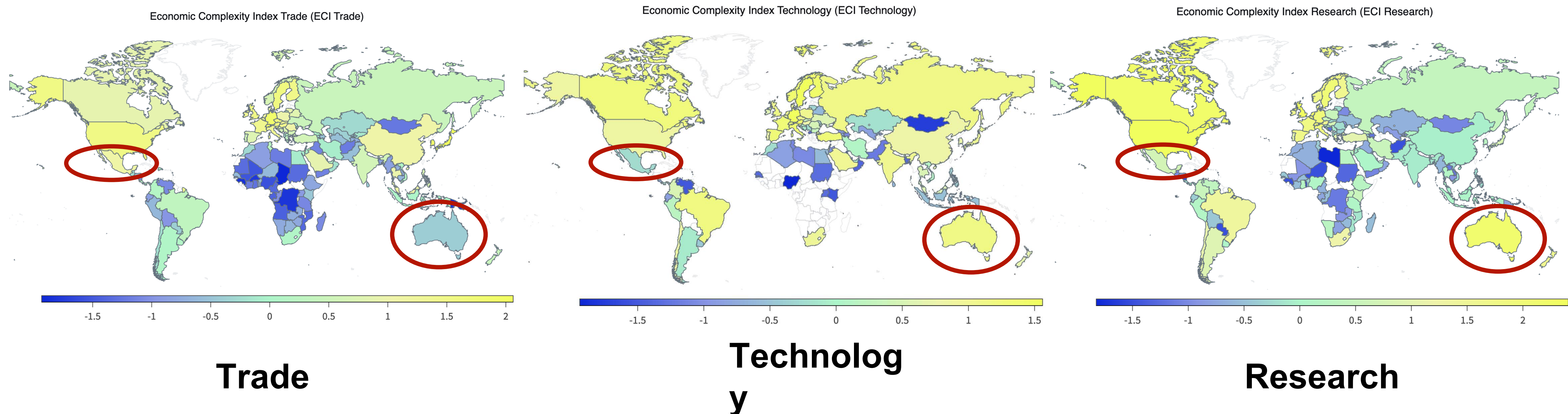


Key Finding 1: Complementarity of complexity metrics

Trade data misses aspects of economies that can be partly recovered using patents and research data.

Mexico: Ranks high in trade complexity, *but lower in technology and research* due to integration in the USA value chains.

Australia: Ranks low in trade complexity because of distance to advanced markets, *but excels in technology and research*.



Multidimensional economic complexity and inclusive green growth

Criteria for selecting final multidimensional model

Linear regression analyses of the form:

outcome ~ ECI + interactions + controls

Can one of these combinations explain better a macroeconomic outcome than only trade?

Final multidimensional model:

1. Largest increase in the explanatory power over the baseline model

(Adjusted Coefficient of determination - R²).

2. Every ECI coefficient (and interaction) needs to be significant.

3. The model needs to be robust and perform better than other metrics for concentration, intensity and diversity **+ IV**.

Multidimensional economic complexity and inclusive green growth

Instrumental variable

- For each country, we use the average complexity of the three economies that have the most similar specialization pattern and with which the country does not share a land or maritime border.

$$\phi_{cc'}^d = \frac{\sum_p M_{cp}^d M_{c'p}^d}{\max(M_c^d, M_{c'}^d)},$$

- Countries with similar specialization patterns should also have similar complexity as the target country, but the complexity of the similar economies should not affect the macroeconomic outcome of the target country and it should not be related to other country specific features.
- We exclude neighboring countries as potential similar countries because it is known that local conditions usually diffuse geographically, and thus may bias our instrument.

Multidimensional economic complexity and economic growth

Specification

Linear regression analyses of the form:

$$\text{outcome} \sim \text{ECI} + \text{interactions} + \text{controls}$$

outcome - 10 year annualized GDP per capita (PPP) growth

controls - log of initial GDP per capita (PPP) (**Sollow term**)

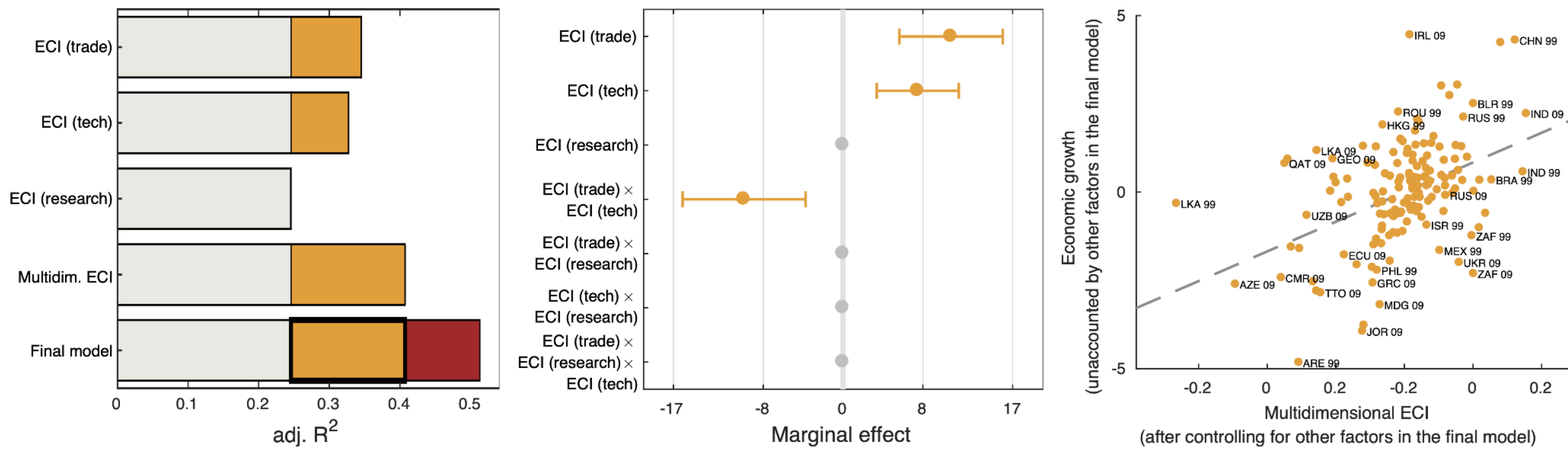
two panels: 1999-2009; 2009-2019.



Key Finding 2: Trade and technology complexity explain economic growth

Combining trade and technology complexity provides a more accurate explanation of 10-year annualized economic growth compared to trade complexity alone

- 1.Trade and technology complexity are positively related with economic growth, exhibiting a substitute relationship.
- 2.Countries with higher trade complexity than technology complexity experience greater growth, whereas countries scoring low in both dimensions tend to exhibit lower growth.
- 3.Research complexity is not correlated with economic growth.



Multidimensional economic complexity and income inequality

Specification

Linear regression analyses of the form:

$$\text{outcome} \sim \text{ECI} + \text{interactions} + \text{controls}$$

outcome - Gini coefficient

controls - log of GDP per capita (PPP) & its square (**Kuznets term**)

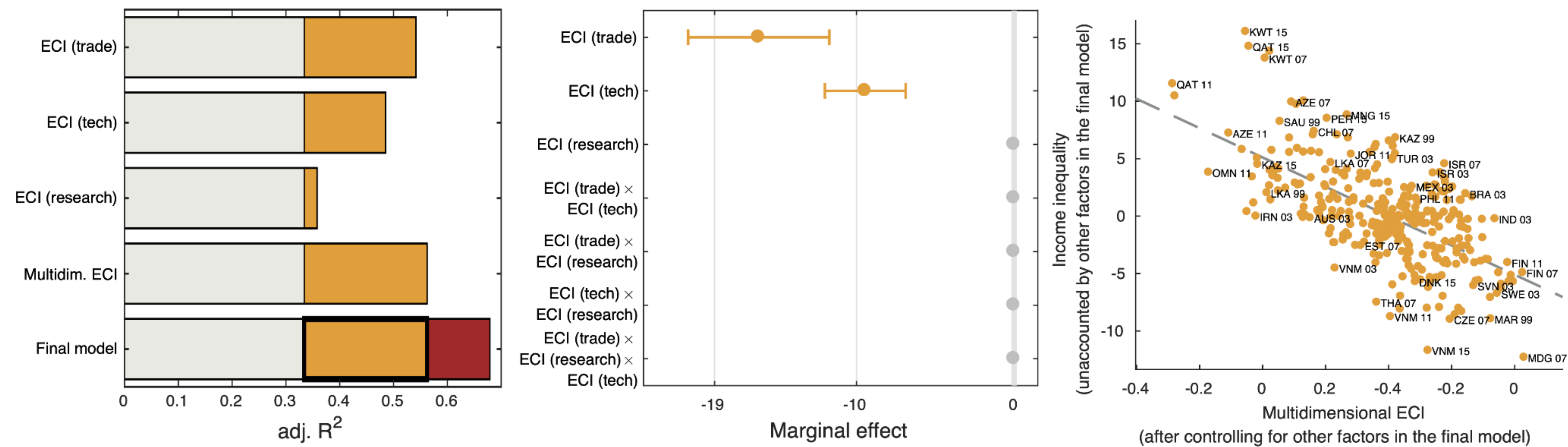
two panels: five panels: 1996-99, 2000-03, 2004-07, 2008-11, & 2012-15.



Key Finding 3: Trade and technology complexity explain income inequality

Trade and technology complexity also explain international differences in the Gini index, a measure of income inequality.

- 1. Countries scoring high in both trade and technology complexity typically exhibit lower income inequality.
- 2. As with economic growth, research complexity is limited in explaining income inequality.



Multidimensional economic complexity and emission intensity

Specification

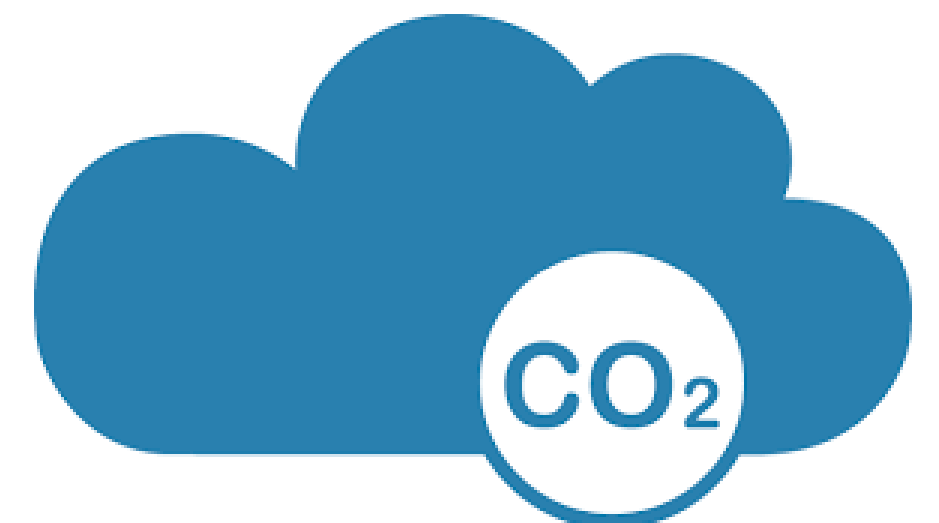
Linear regression analyses of the form:

$$\text{outcome} \sim \text{ECI} + \text{interactions} + \text{controls}$$

outcome: log of greenhouse gas emissions (in kt CO₂e) divided by GDP

controls: log of GDP per capita (PPP) (**Romero & Grankow**)

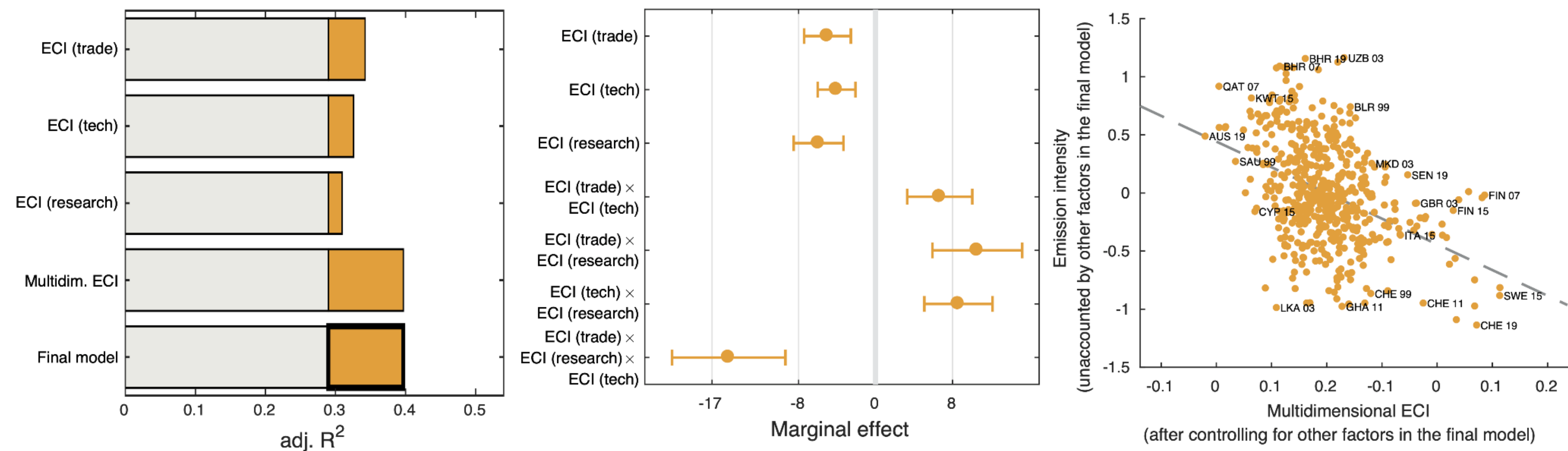
six panels: 1996-99, 2000-03, 2004-07, 2008-11, 2012-15, & 2016-19.



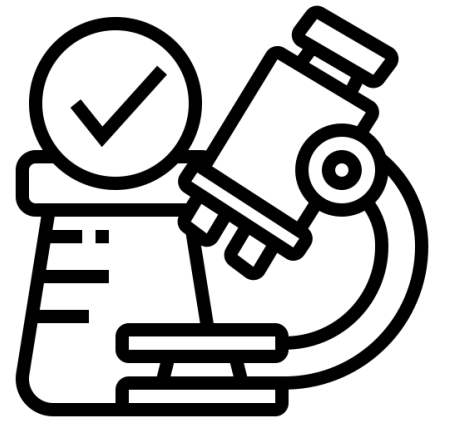
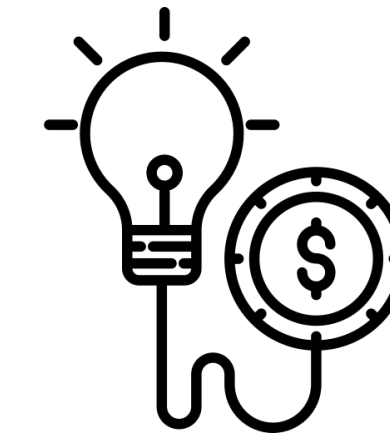
Key Finding 4: Countries scoring high in all three metrics have lower emission intensities

We find that countries scoring high in all three dimensions tend to have lower emission intensities (emissions per unit of GDP).

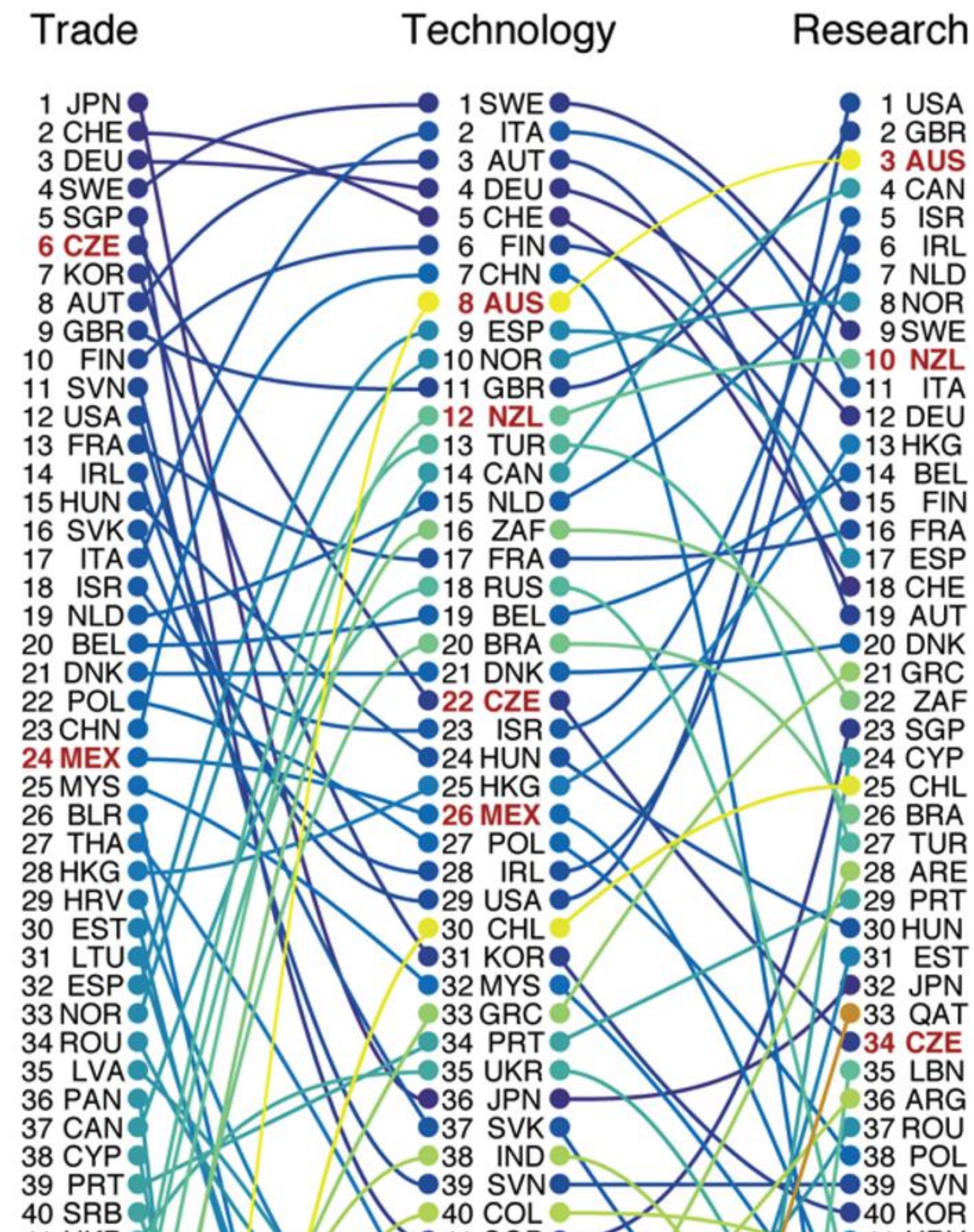
- 1. The relationship holds only for production emissions, not consumption emissions (production corrected for trade).
- 2. This arises because economic complexity measures are based on outputs and lack information on consumption.



Multidimensional Economic Complexity and Inclusive Green Growth Discussion

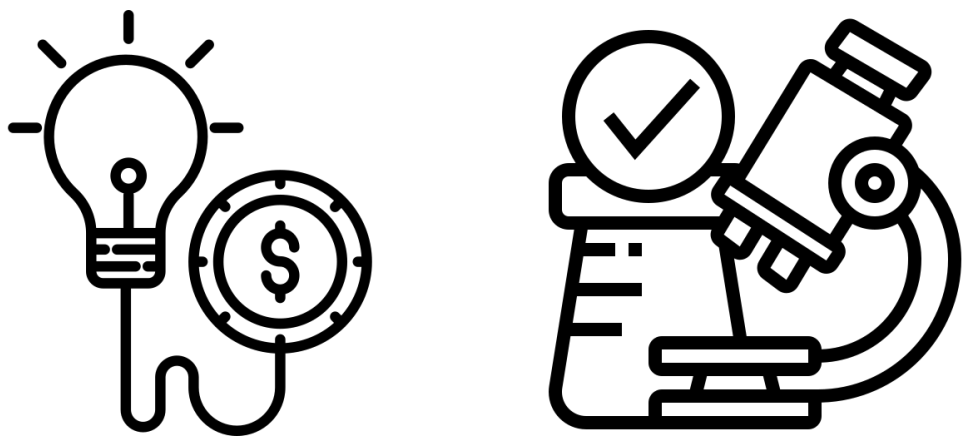


- The different dimensions of economic complexity interact and can be used to improve the explanatory power of the product ECI in inclusive green growth.
- The findings underscore the importance of a multidimensional approach to economic complexity for strategic diversification and development.
- By balancing multiple targets and opportunities, it provides a more holistic landscape for inclusive green development, emphasizing that economic complexity transcends trade sophistication.



Using Multidimensional Economic Complexity for

Forecasting Growth



Forecasting Annual Growth by 2032 for 87 countries

Rank ↑↓	Country ↑↓	Predicted Annual GDP per capita growth ↑↓	ECI Trade ↑↓	ECI Research ↑↓	ECI Tech ↑↓	GDP per capita (in PPP, constant 2021 USD) ↑↓
1	India	4.2%	0.59	0.38	0.85	8088
2	Philippines	3.8%	0.64	0.47	0.48	8797
3	Uganda	3.2%	0.28	0.61	0.22	2688
4	South Africa	3.1%	0.45	0.75	0.83	14173
5	Vietnam	3.0%	0.52	0.34	0.61	12230
6	Brazil	3.0%	0.49	0.77	0.89	17673
7	China	2.9%	0.75	0.36	0.8	20407

29	Bulgaria	2.0%	0.61	0.37	0.67	29405
30	Slovakia	2.0%	0.76	0.46	0.68	37791
31	Japan	2.0%	1	0.67	0.85	44549
32	Hungary	2.0%	0.77	0.68	0.79	38648
33	Romania	1.9%	0.73	0.52	0.71	37969
34	Sudan	1.9%	0.13	0.36	0.22	3420
35	Chile	1.9%	0.4	0.77	0.81	29050
36	South Korea	1.9%	0.94	0.52	0.77	48571

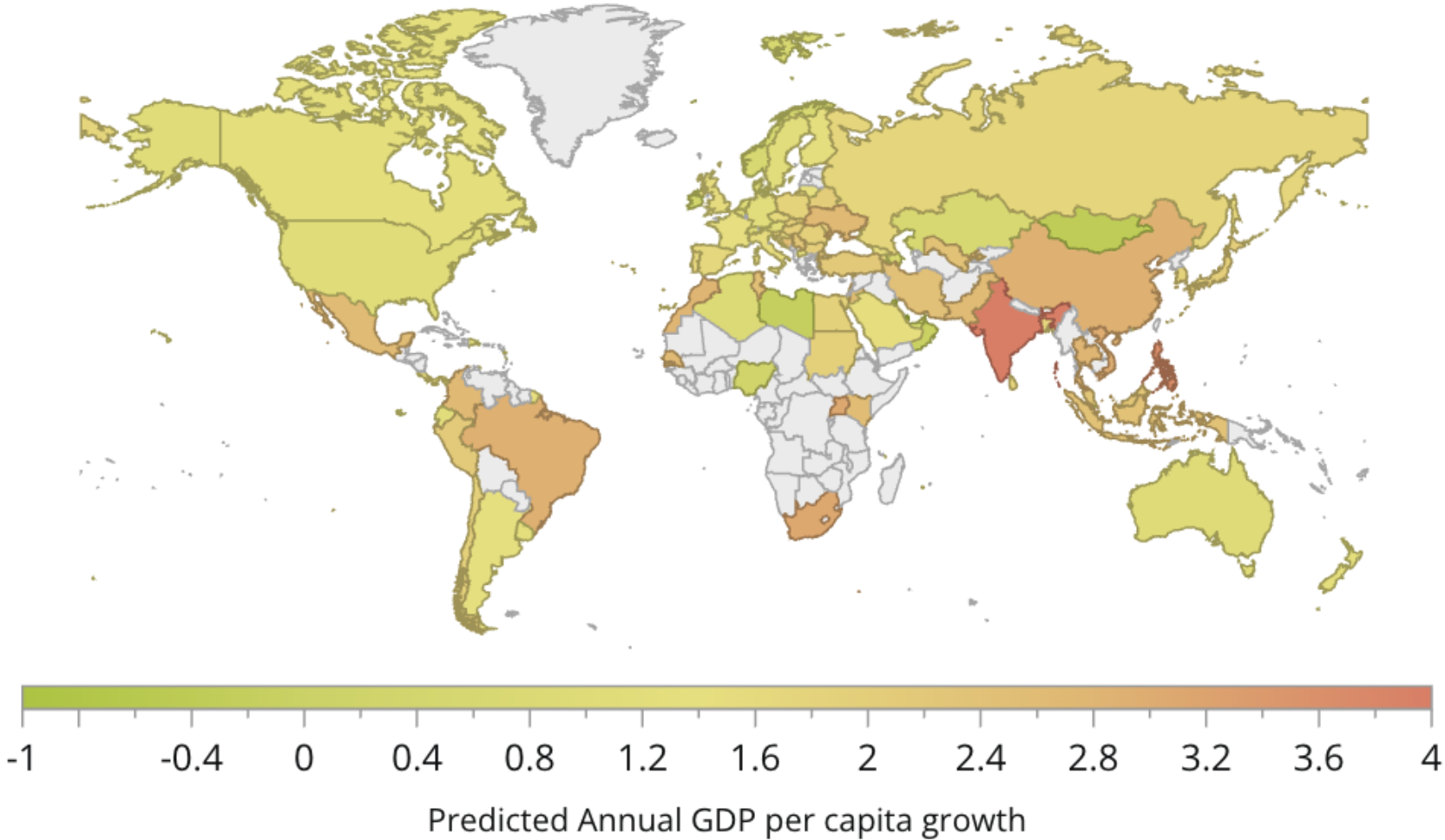
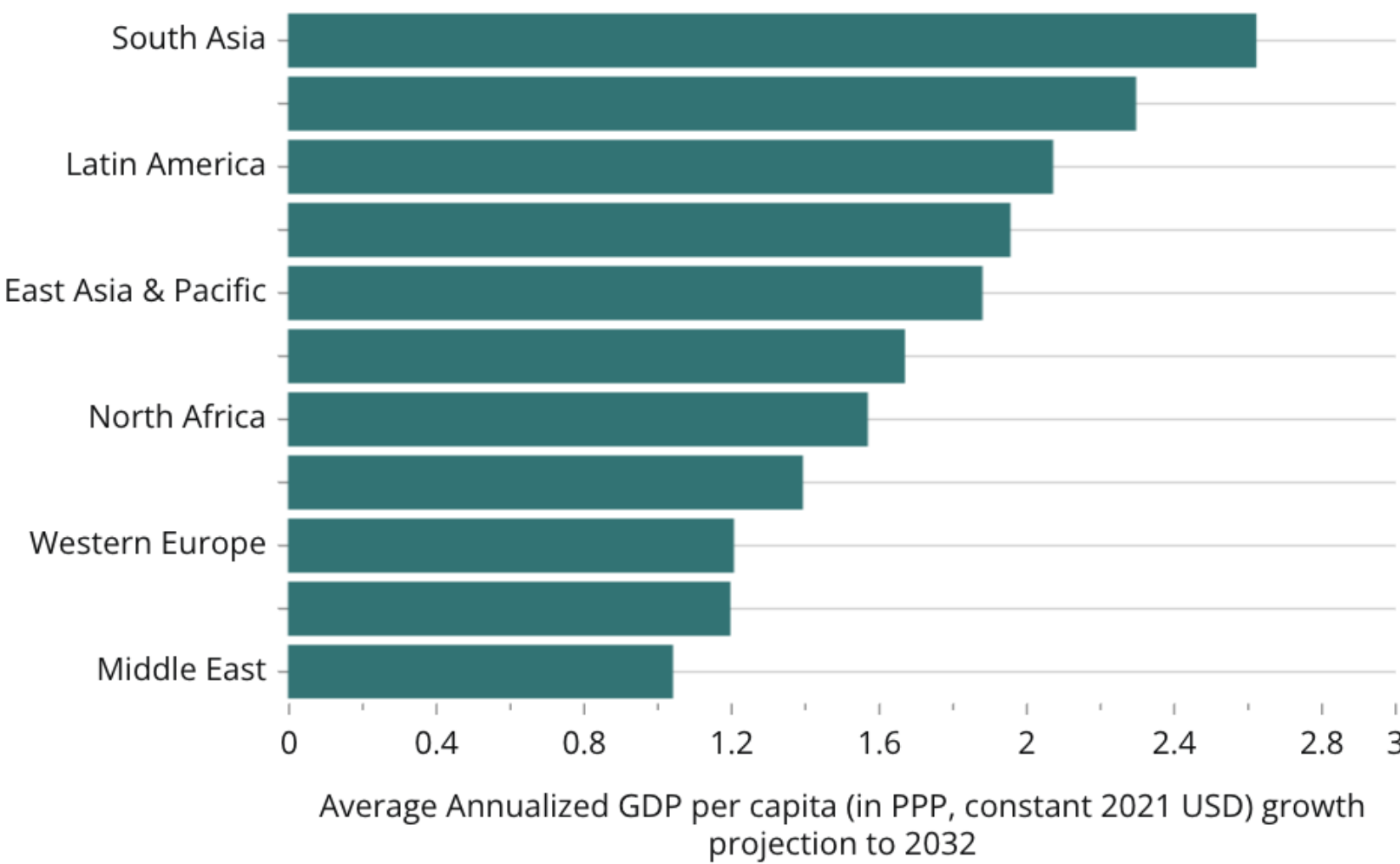


Figure 2. Multidimensional ECI growth projections by country for 2032.

Estimating Digital Product Trade Through Corporate Revenue Data

Viktor Stojkoski, Philipp Koch, Eva Coll, Cesar A. Hidalgo

Award-Winning TV Shows



Trending Now



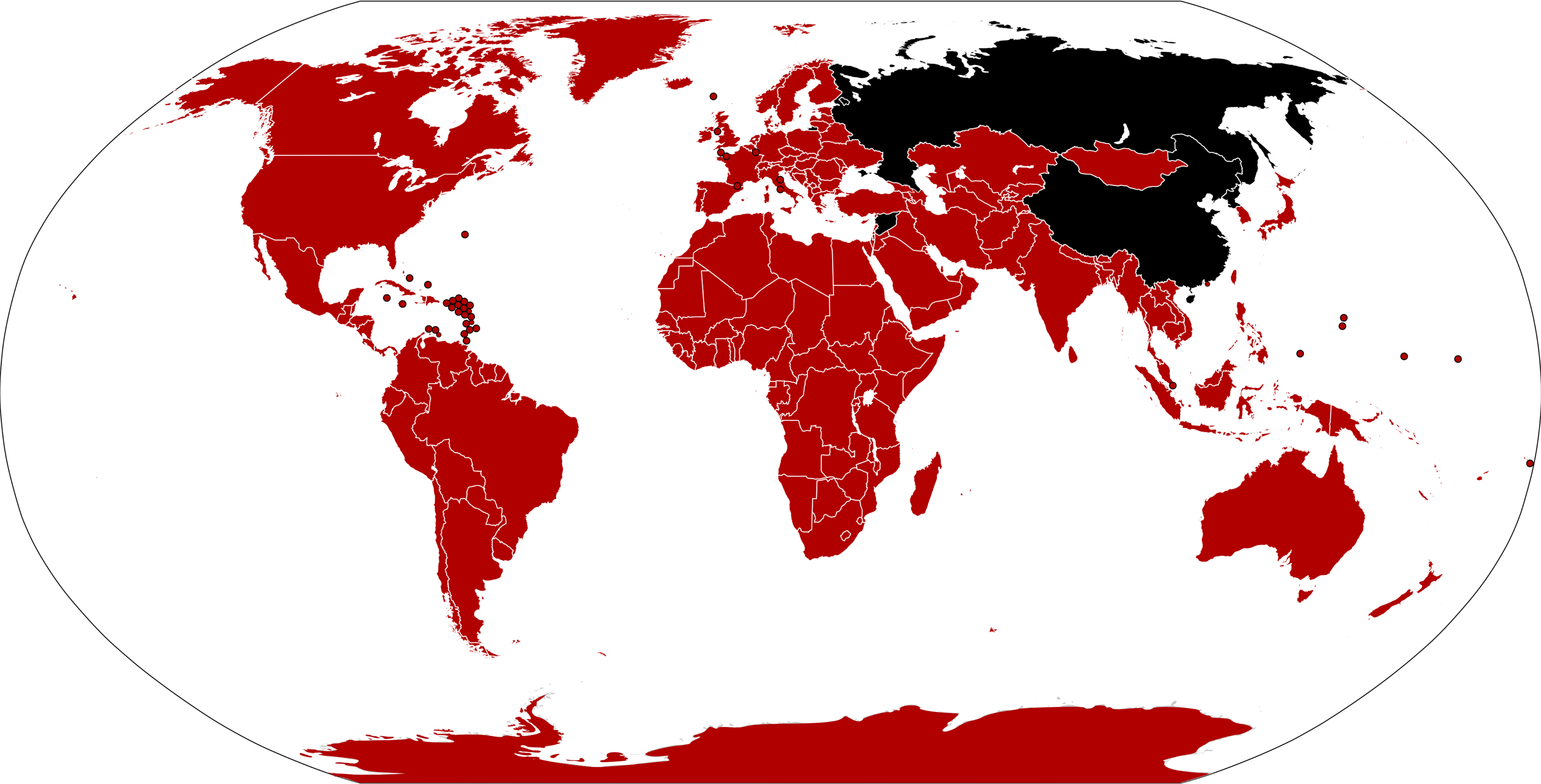
Critically Acclaimed TV Dramas



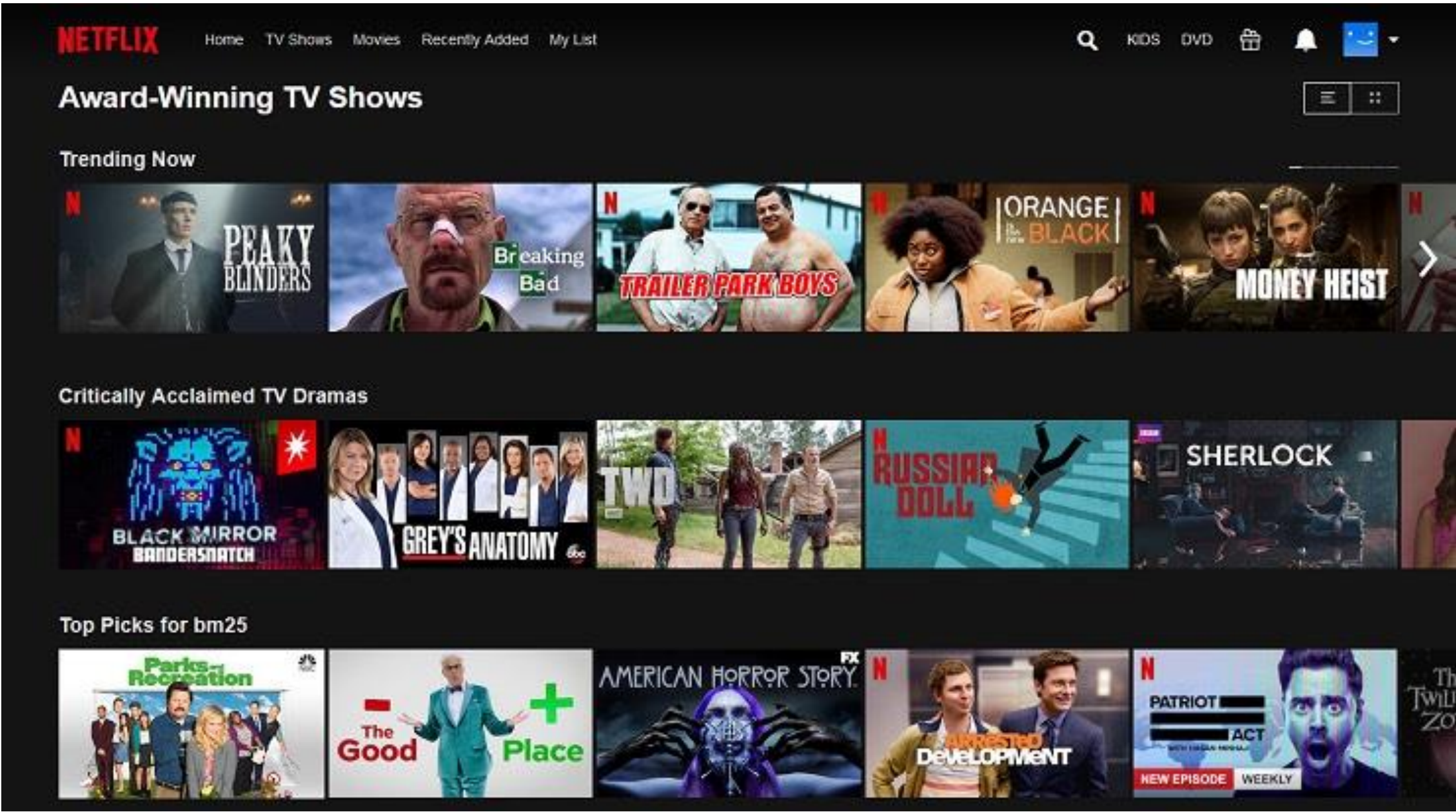
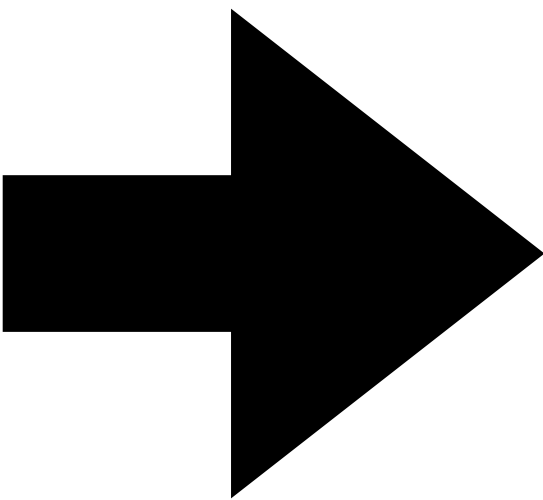
Top Picks for bm25



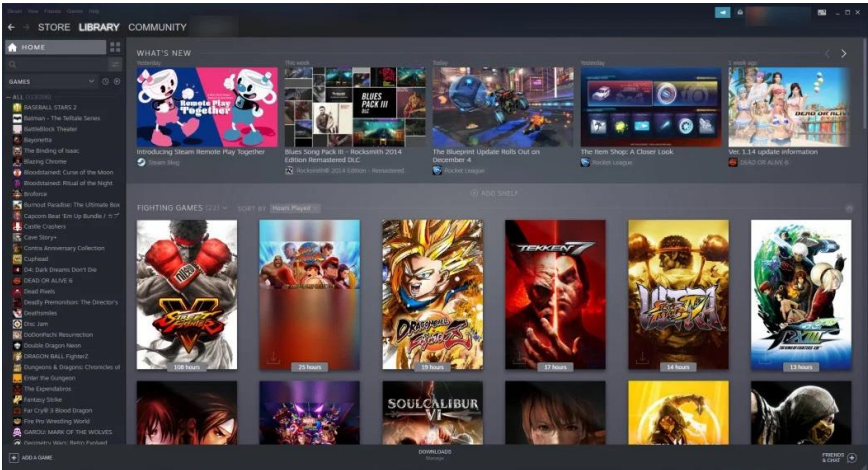
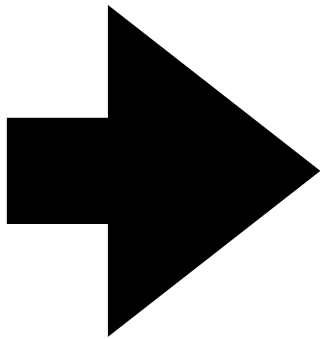
Availability of Netflix, as of March 2022



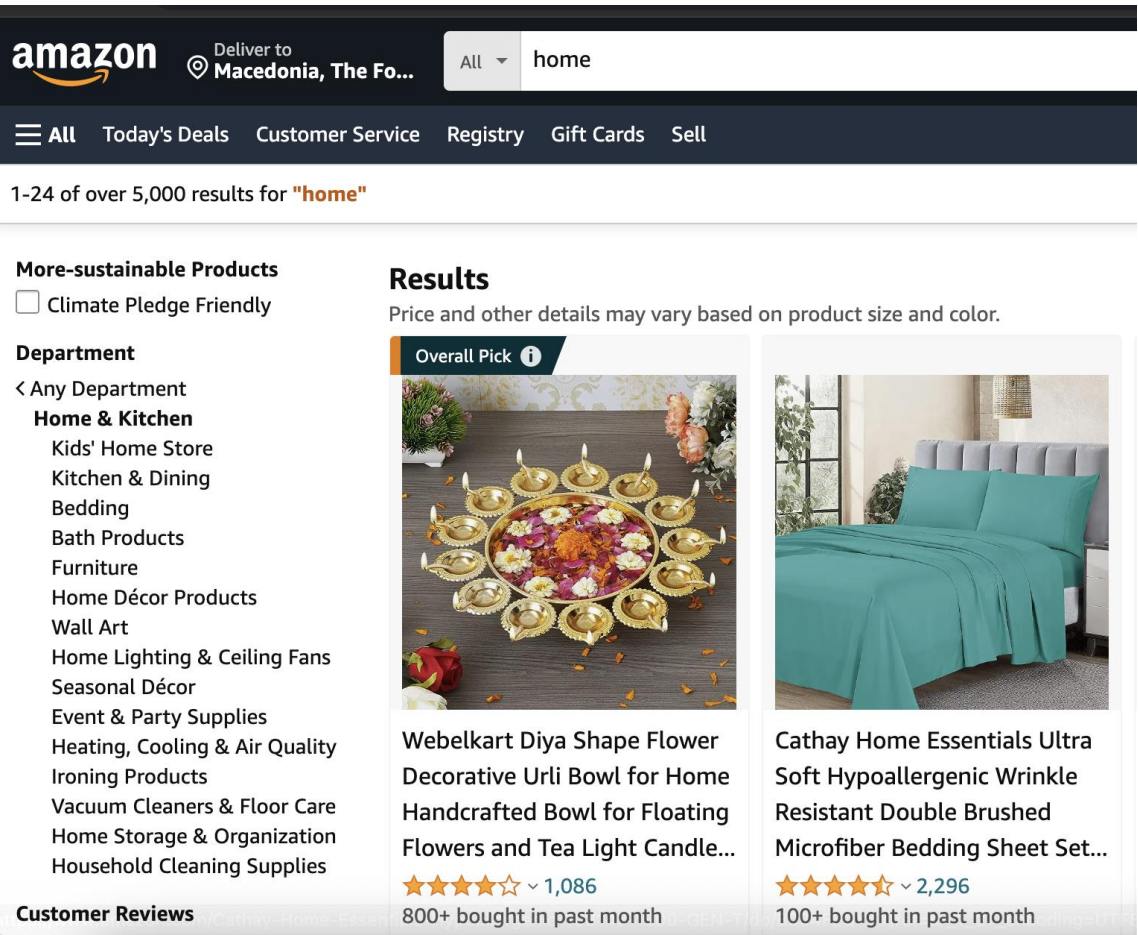
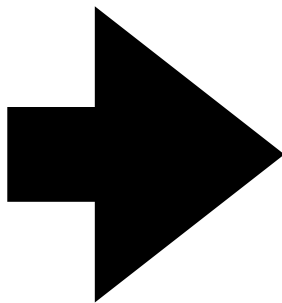
Movies

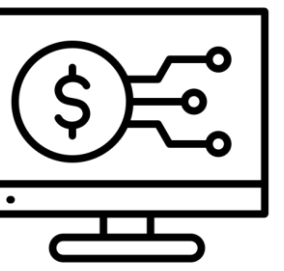


Video games / Software



Markets

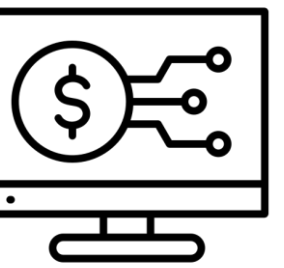




How do we define digital products?

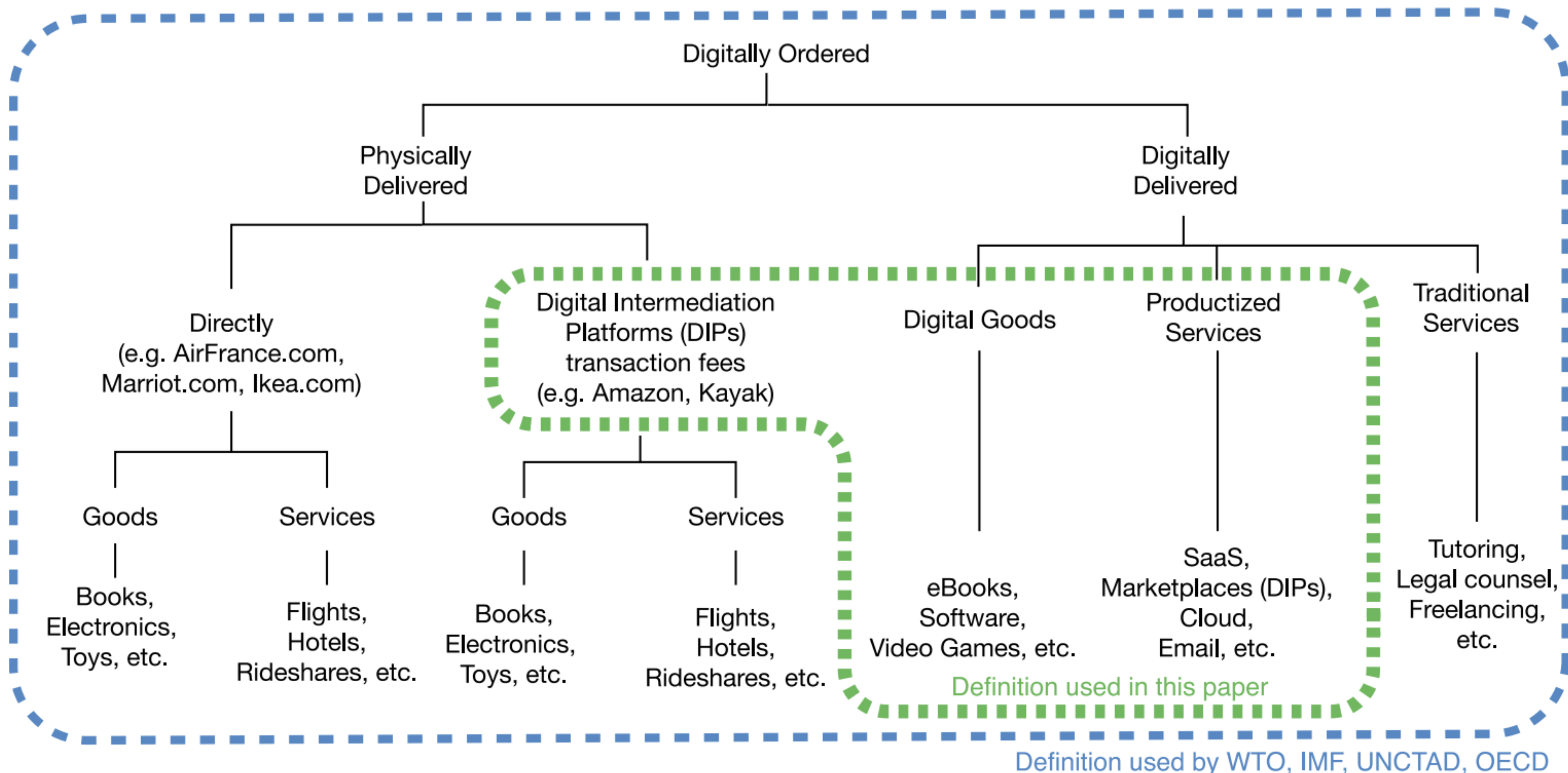
- WTO, IMF, UNCTAD, OECD define digital trade as all transactions that are either:
1) **digitally ordered** or 2) **digitally delivered**.





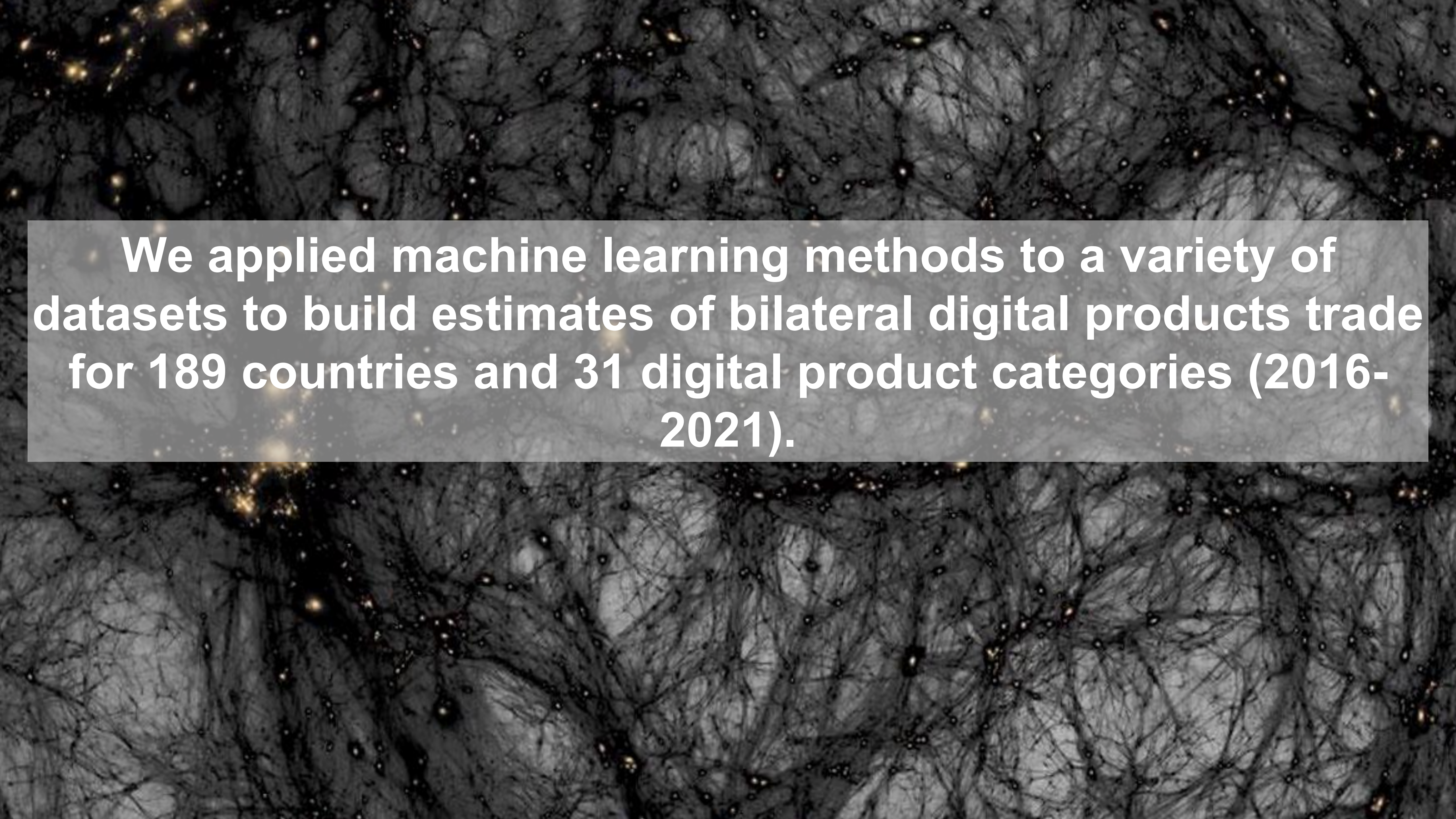
How do we define digital products?

- WTO, IMF, UNCTAD, OECD define digital trade as all transactions that are either 1) digitally ordered or digitally delivered



The background of the slide is a detailed visualization of the cosmic web, showing a complex network of dark matter filaments and galaxy clusters. Numerous bright, yellowish-white points of light, representing galaxies, are scattered throughout the dark, textured fabric of the universe.

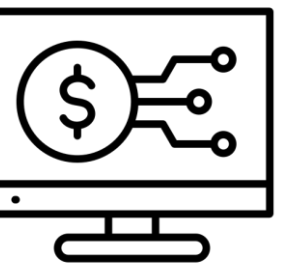
**Digital Products are International Trade
“Dark Matter”**

The background of the slide is a deep space image showing the cosmic web, with dark, filamentary structures and numerous bright, distant galaxies and star clusters.

We applied machine learning methods to a variety of datasets to build estimates of bilateral digital products trade for 189 countries and 31 digital product categories (2016-2021).

The background of the image is a detailed visualization of the cosmic web, showing a complex network of dark matter filaments and galaxy clusters. Numerous bright, yellowish-white points of light, representing galaxies, are scattered throughout the dark, textured fabric of the universe. A semi-transparent grey rectangular box is centered horizontally across the middle of the image, containing the text.

**BUT WHY DO WE NEED TO UNDERSTAND
DIGITAL TRADE?**



Understanding Trade Balances

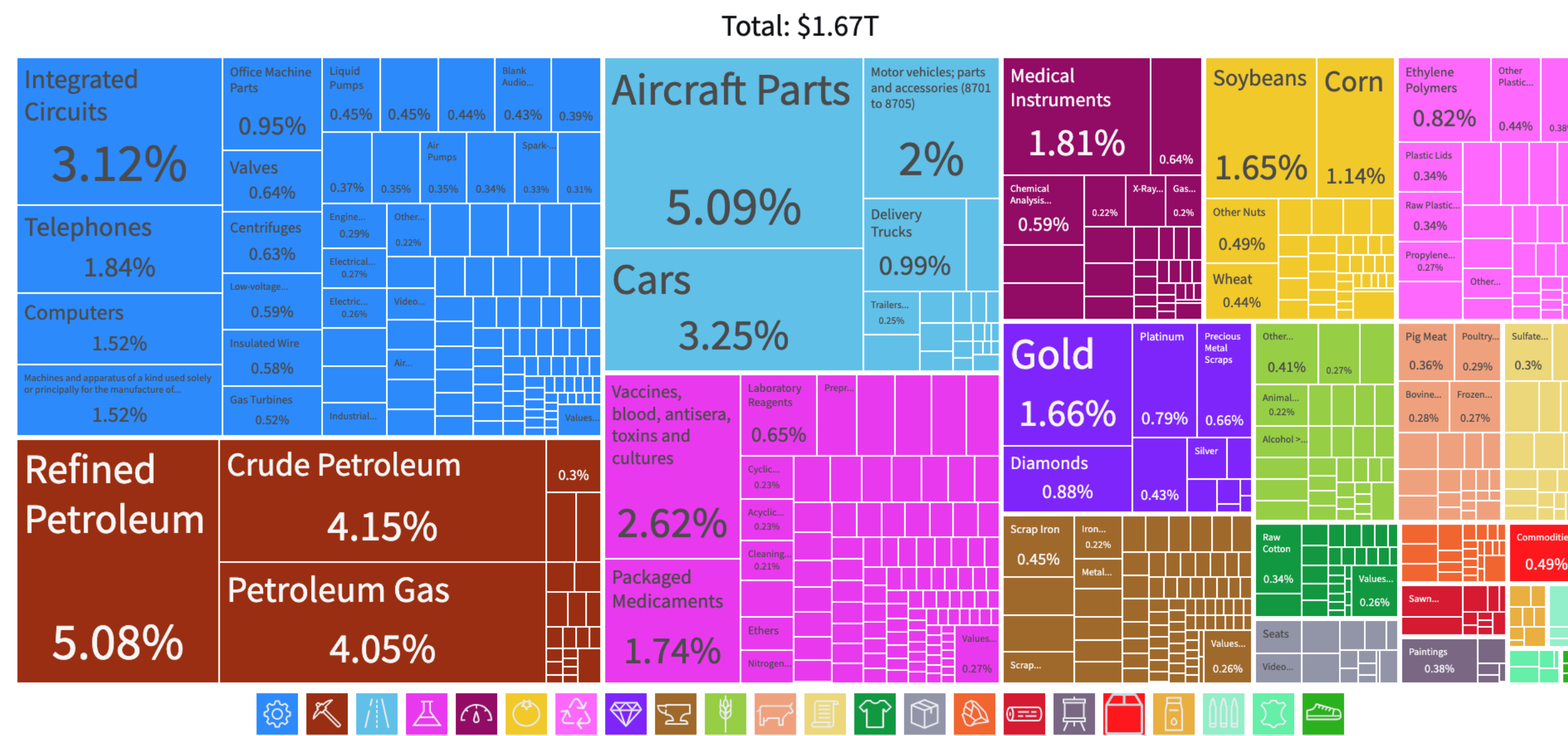
In 2021 USA exported \$ 1.63T products but imported \$ 2.73T.

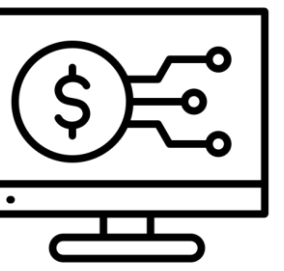
- What about Facebook Advertising, Google Cloud, Stripe Online Payments?
- Can these exports **change the geography of trade balances?**
- USA is just one example of a country that exports “bits” and imports “atoms”.

< 2020

What does United States export? (2021)

2022 >





Understanding the Twin Transition

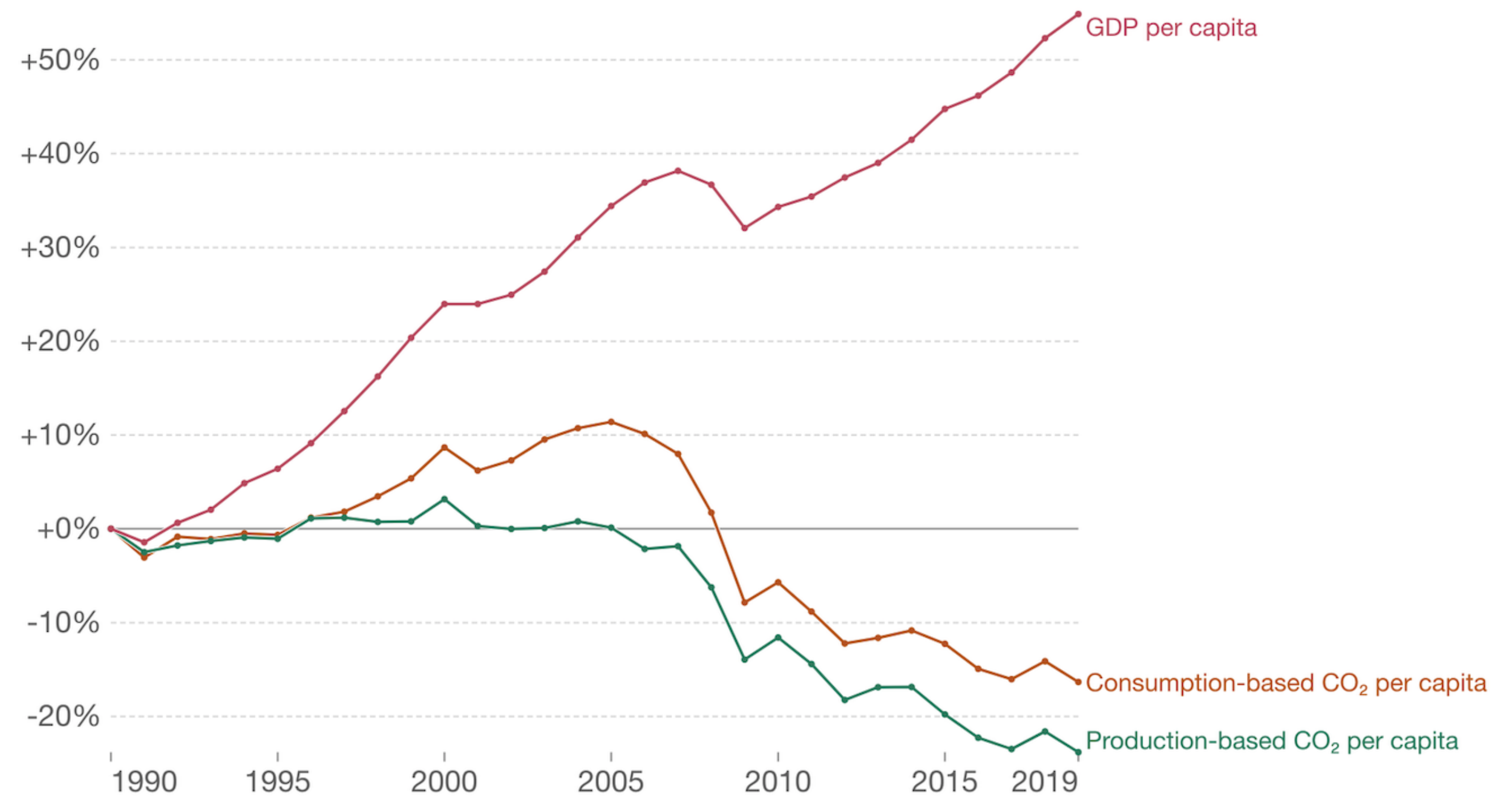
Many developed economies have decoupled growth and emissions.

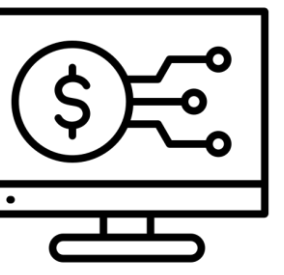
- **Digitization has a key role in supporting sustainable growth** during this decisive decade for action on climate change.
- Can a digital trade dataset help us understand **how economies can improve digital function and drive sustainability?**

Change in per capita CO₂ emissions and GDP, United States

Consumption-based emissions¹ are national emissions that have been adjusted for trade. This measures fossil fuel and industry emissions². Land use change is not included.

Our World
in Data

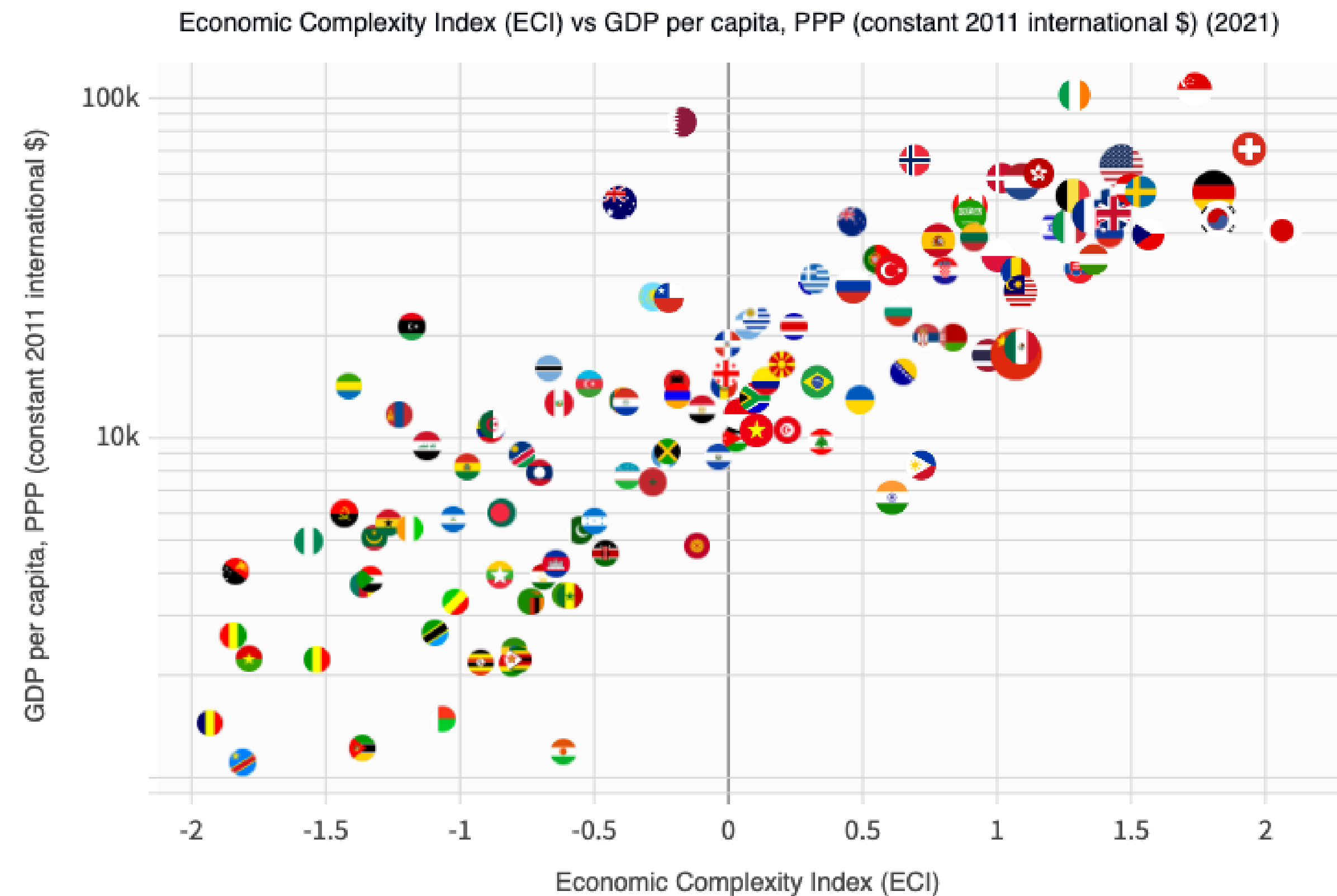




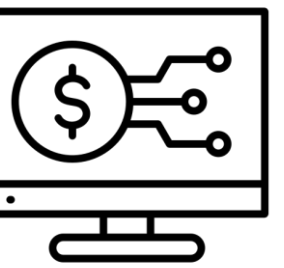
Understanding Economic Complexity

Adding digital trade to the landscape of economic complexity

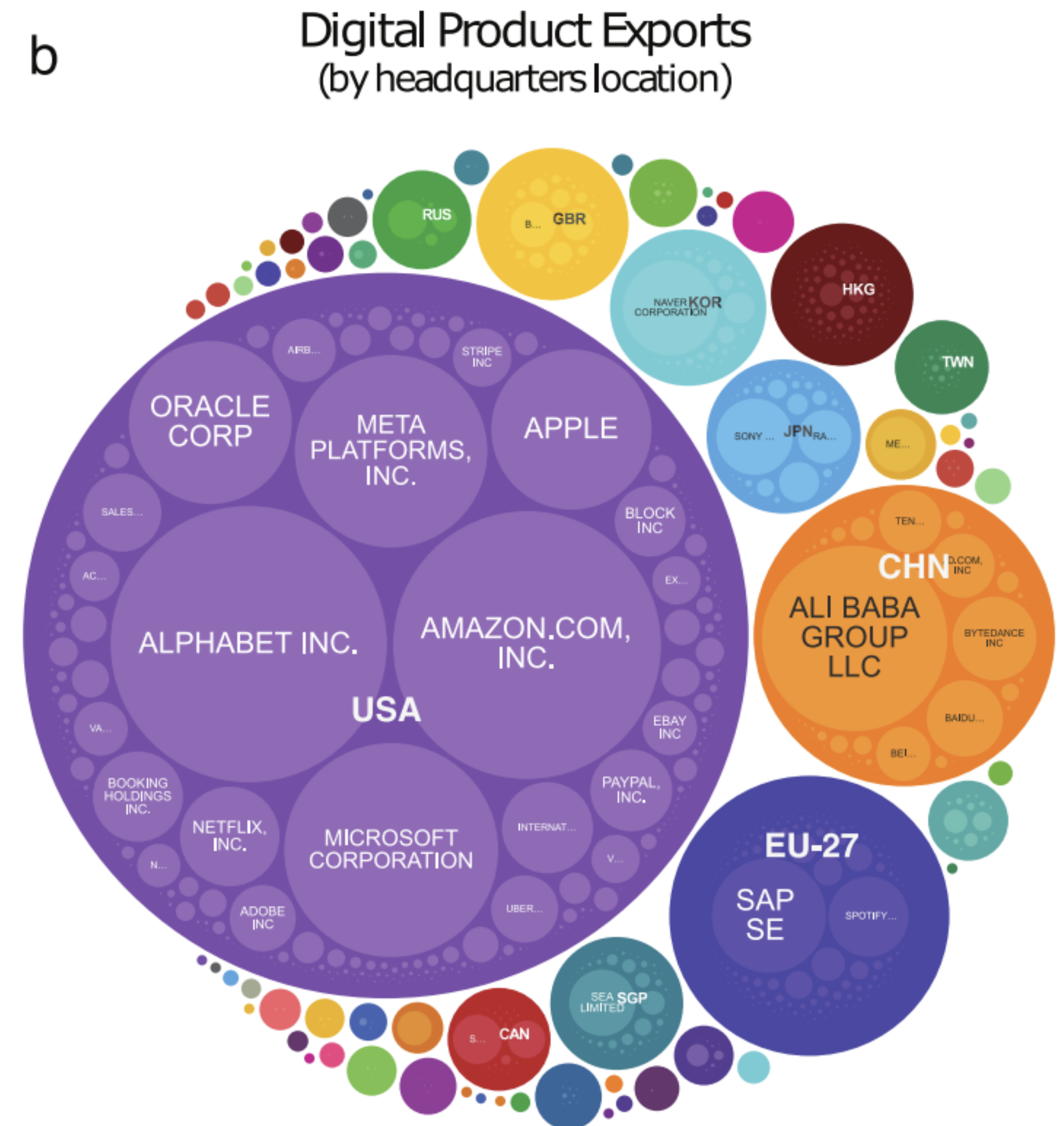
- Economic complexity measures are used frequently to explain variations in growth, inequality, and emissions, but may suffer from incomplete accounts of world trade.
- Can digital trade improve measurements of economic complexity?



Contribution



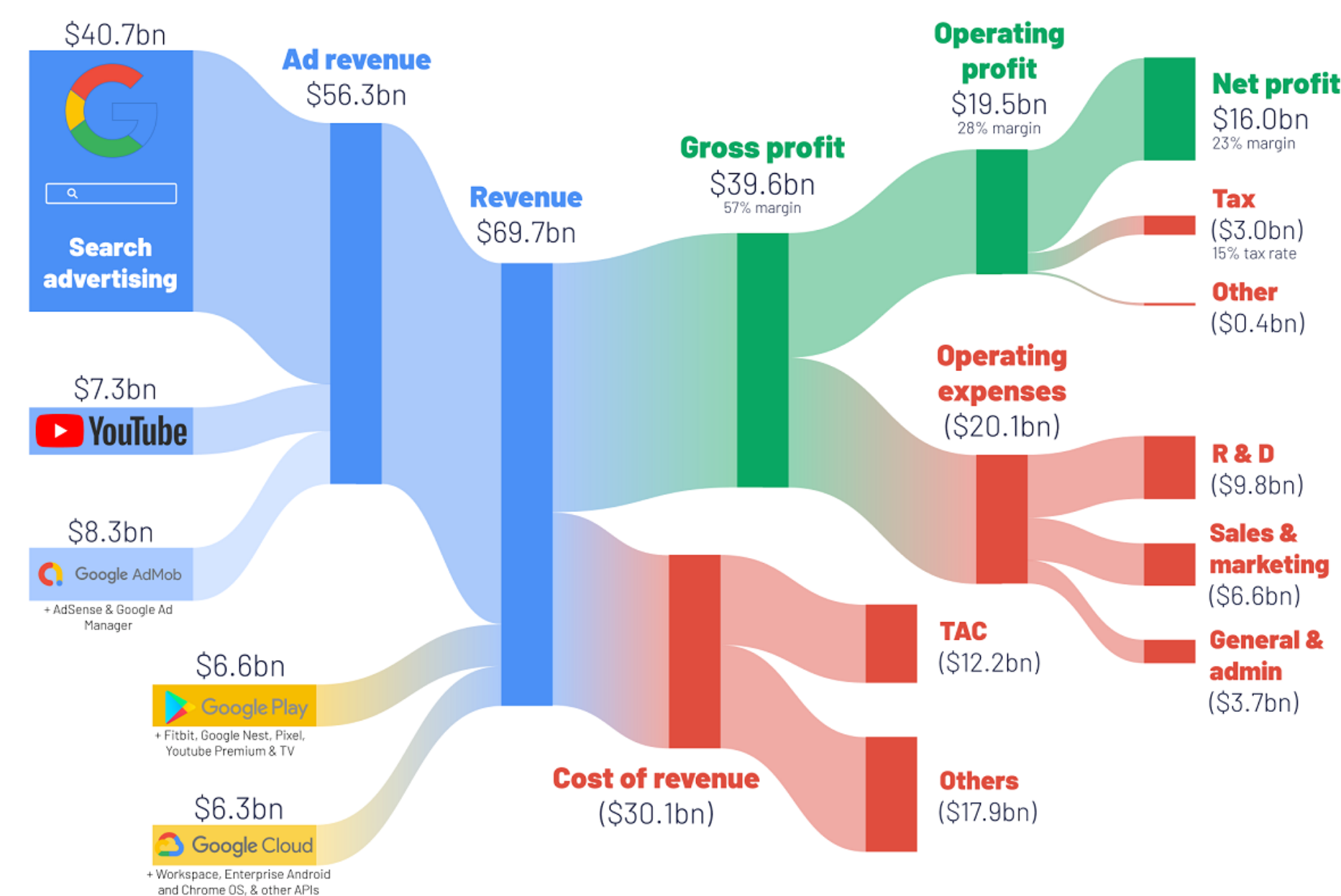
- We apply machine learning and optimal transport to a variety of datasets to build estimates (in USD) of bilateral digital trade for **189 countries** and **31 digital product/service categories**.
- We use these estimates to:
 1. Compare the geography of digital and physical trade.
 2. Investigate the environmental and structural implications of digital trade.



Construction of the Dataset

Step 1: Collecting **ground truth data** about digital corporate revenues and consumption patterns across countries.

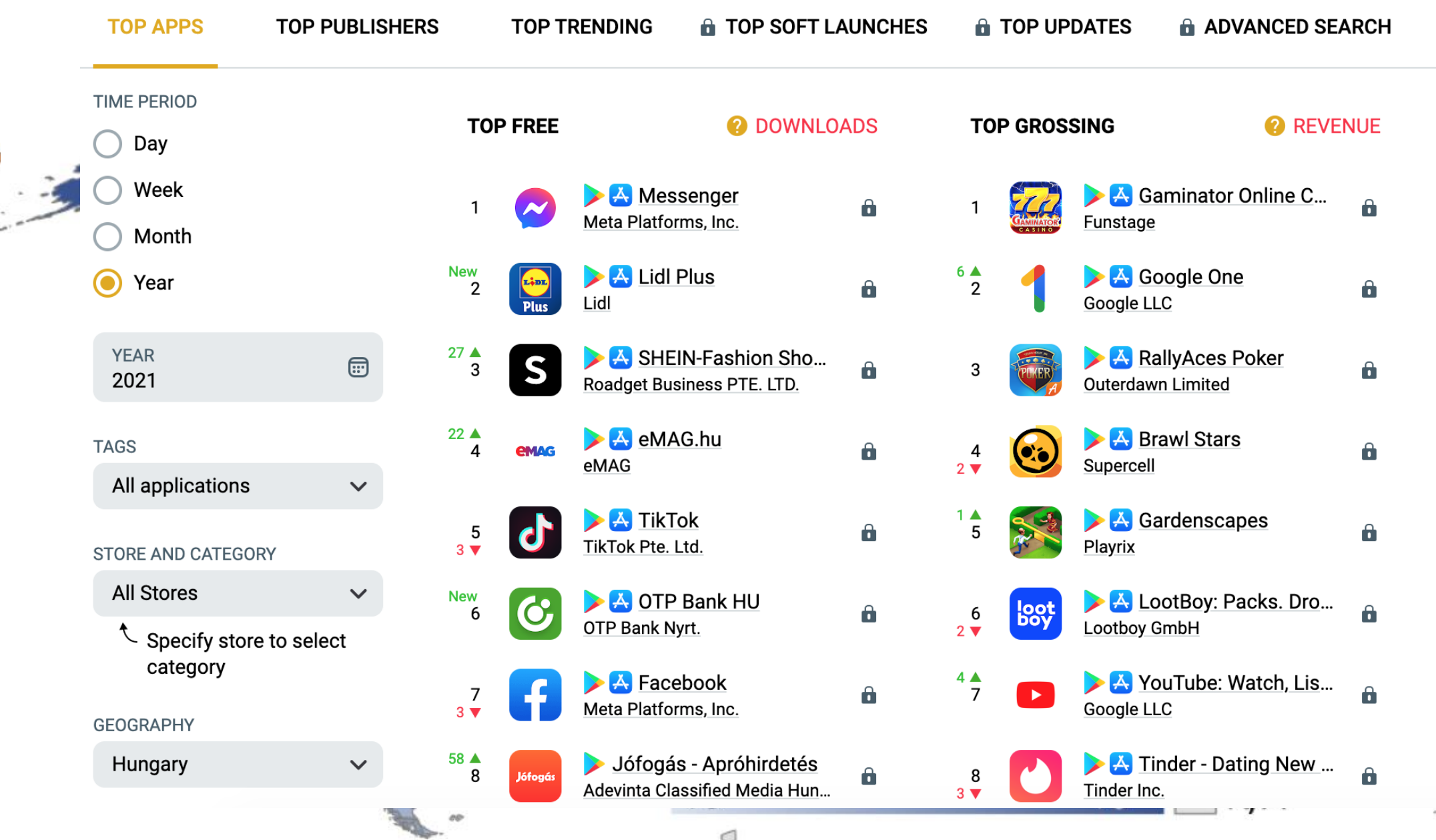
Digital Revenue



Source: Alphabet June 2022 quarterly results

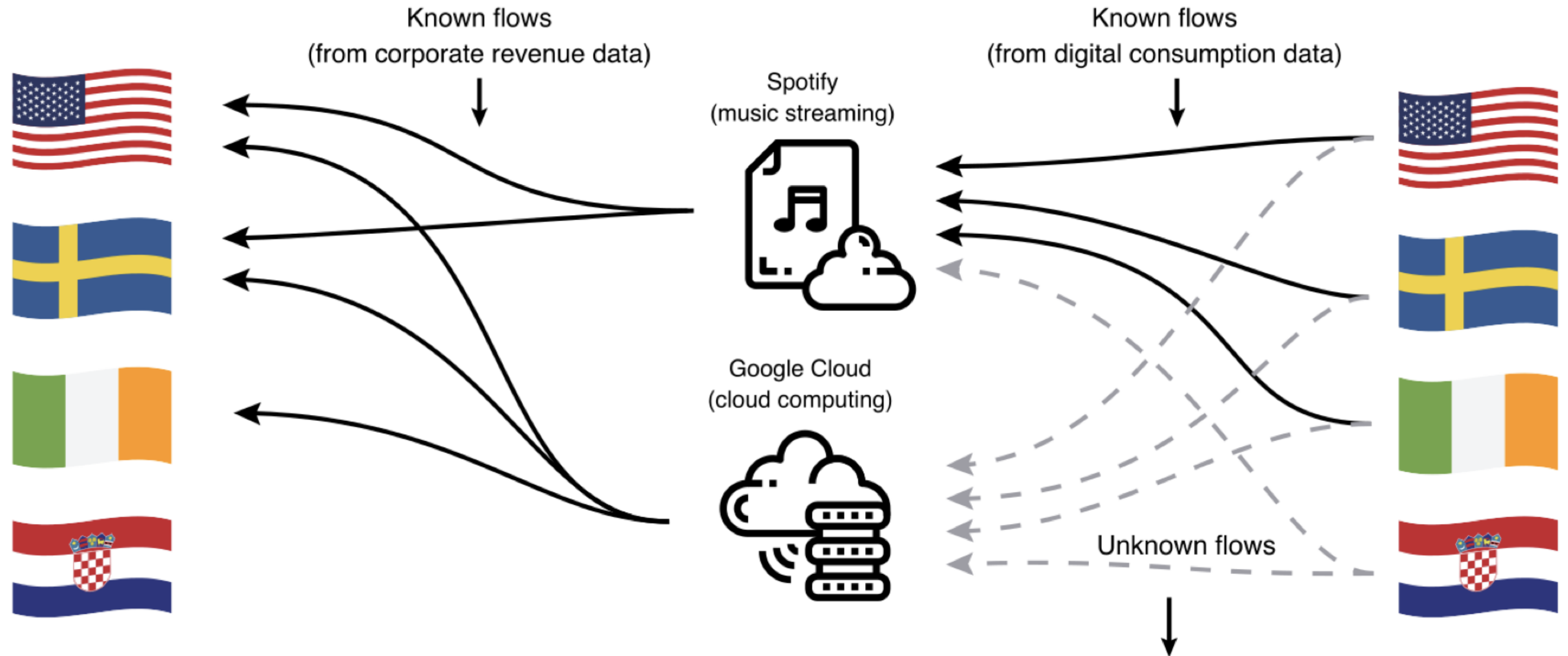
genuine impact

Digital Consumption



Construction of the Dataset

Step 2: Using **machine learning** to fill in missing consumption patterns.



Decision tree predicting missing digital consumption flows using data on GDP, Language, History, Distance, Regions, Corporate Revenues, & ICT.

Construction of the Dataset

a)

Alphabet 2023 Annual report

Disaggregated Revenues

The following table presents revenues disaggregated by type (in millions):

	Year Ended December 31,		
	2021	2022	2023
Google Search & other	\$ 148,951	\$ 162,450	\$ 175,033
YouTube ads	28,845	29,243	31,510
Google Network	31,701	32,790	31,312
Google advertising	209,497	224,473	237,855
Google subscriptions, platforms, and devices	28,032	29,055	34,688
Google Services total	237,529	253,528	272,543
Google Cloud	19,206	26,280	33,088
Other Bets	753	1,068	1,527
Hedging gains (losses)	149	1,960	236
Total revenues	\$ 257,637	\$ 282,836	\$ 307,394

The following table presents revenues disaggregated by geography, based on the addresses of our customers (in millions):

	Year Ended December 31,					
	2021		2022		2023	
United States	\$ 117,854	46 %	\$ 134,814	48 %	\$ 146,286	47 %
EMEA ⁽¹⁾	79,107	31	82,062	29	91,038	30
APAC ⁽¹⁾	46,123	18	47,024	16	51,514	17
Other Americas ⁽¹⁾	14,404	5	16,976	6	18,320	6
Hedging gains (losses)	149	0	1,960	1	236	0
Total revenues	\$ 257,637	100 %	\$ 282,836	100 %	\$ 307,394	100 %

⁽¹⁾ Regions represent Europe, the Middle East, and Africa (EMEA); Asia-Pacific (APAC); and Canada and Latin America ("Other Americas").



b)

Structured Revenue Data

year	firm	origin	destination	category	revenue
2023	Alphabet	USA	USA	Cloud Computing	XB USD
2023	Alphabet	USA	USA	Digital Advertising	XB USD
...	...	...	...	...	...
2023	Alphabet	USA	EU	Digital Advertising	XB USD
2021	Alphabet	USA	EU	Video Streaming	XB USD
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
2023	Alphabet	USA	Other Americas	Cloud Computing	XB USD
2023	Alphabet	USA	Other Americas	Digital Advertising	XB USD
...	...	...	...	...	...



Machine Learning

c)

Augmented Revenue Data

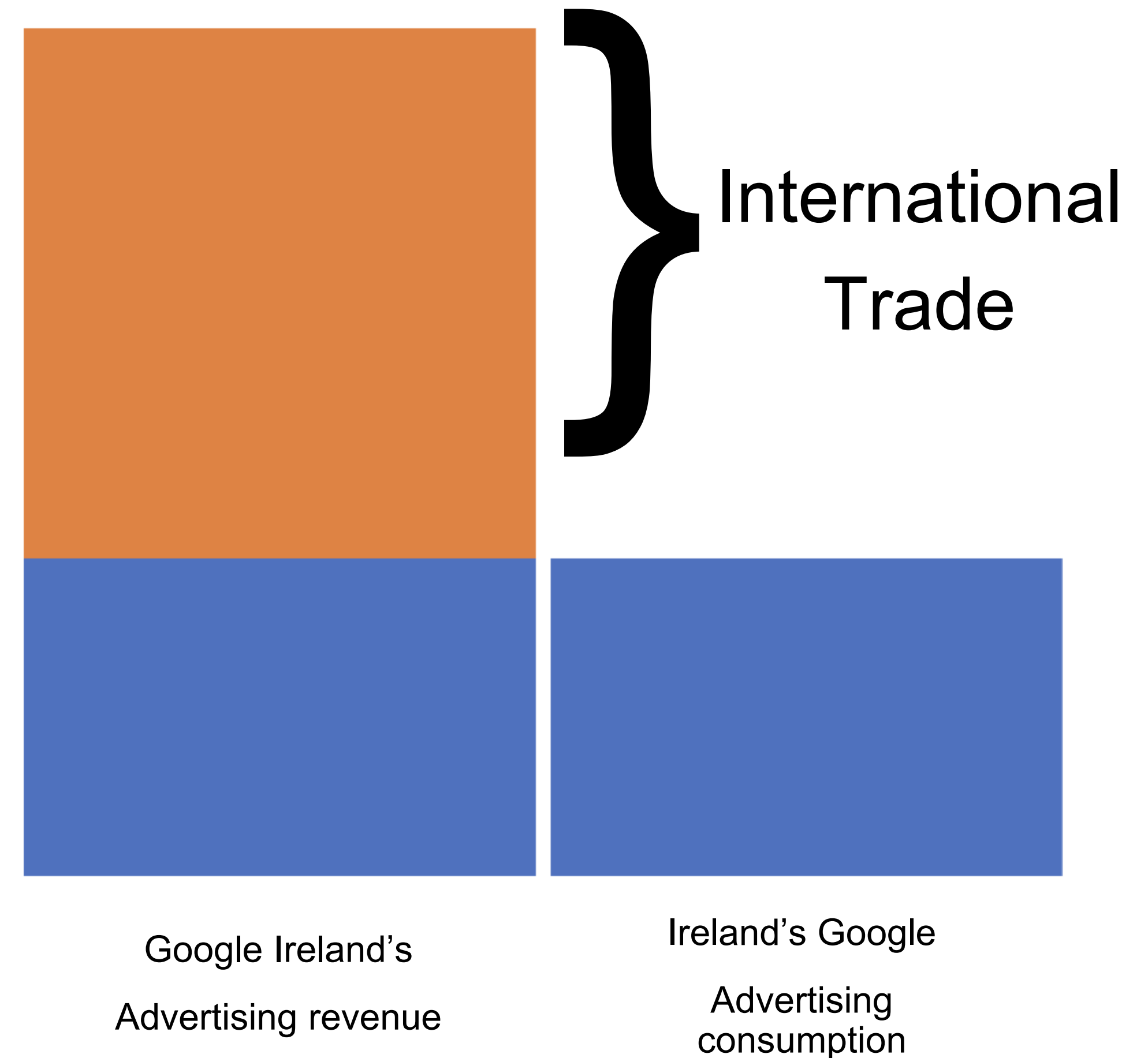
year	firm	origin	destination	category	revenue
2023	Alphabet	USA	USA	Cloud Computing	XB USD
2023	Alphabet	USA	USA	Digital Advertising	XB USD
...	...	...	...	...	...
2023	Alphabet	USA	Germany	Digital Advertising	XB USD
2021	Alphabet	USA	North Macedonia	Video Streaming	XB USD
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
2023	Alphabet	USA	Chile	Cloud Computing	XB USD
2023	Alphabet	USA	Brazil	Digital Advertising	XB USD
...	...	...	...	...	...

Construction of the Dataset

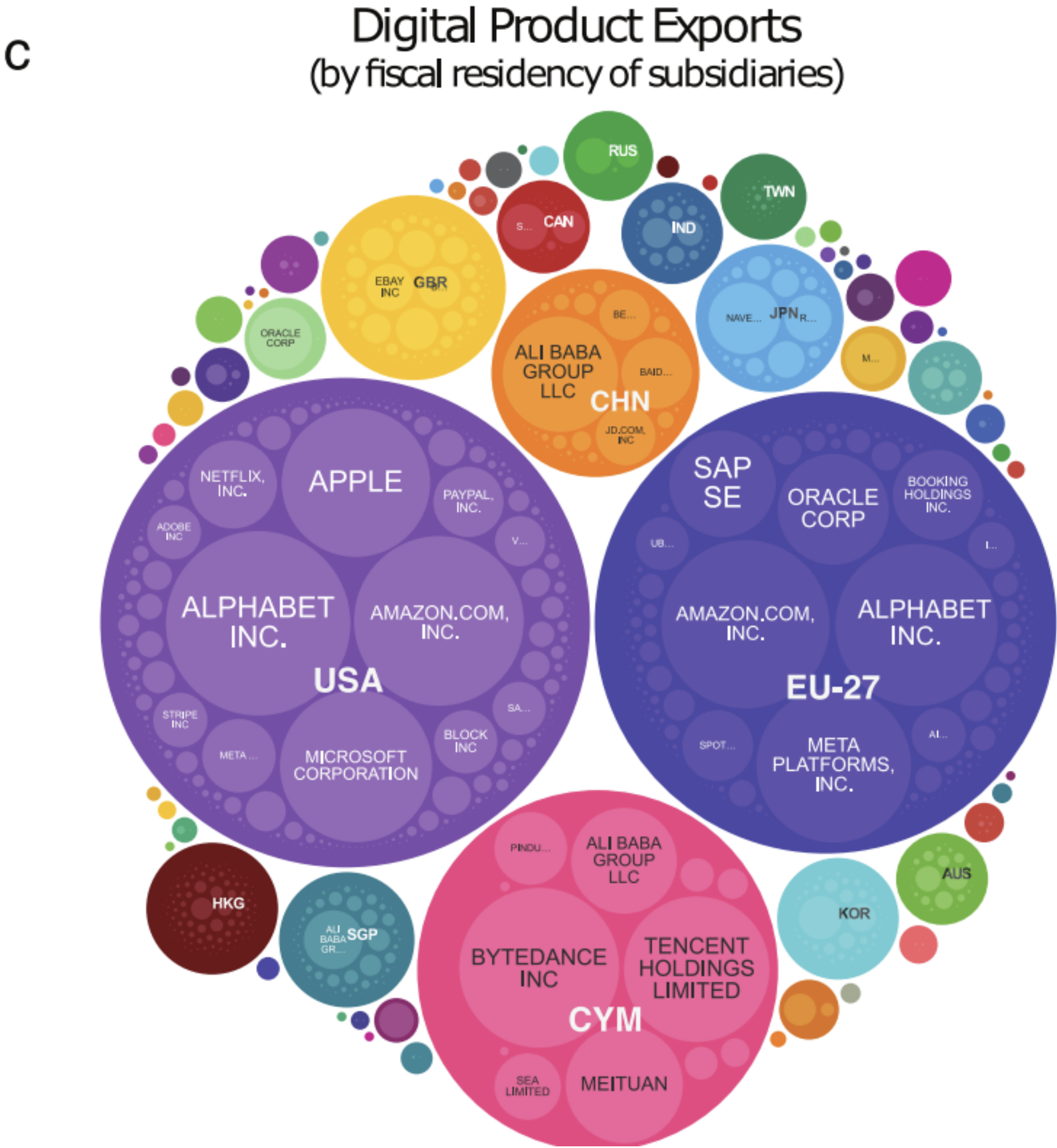
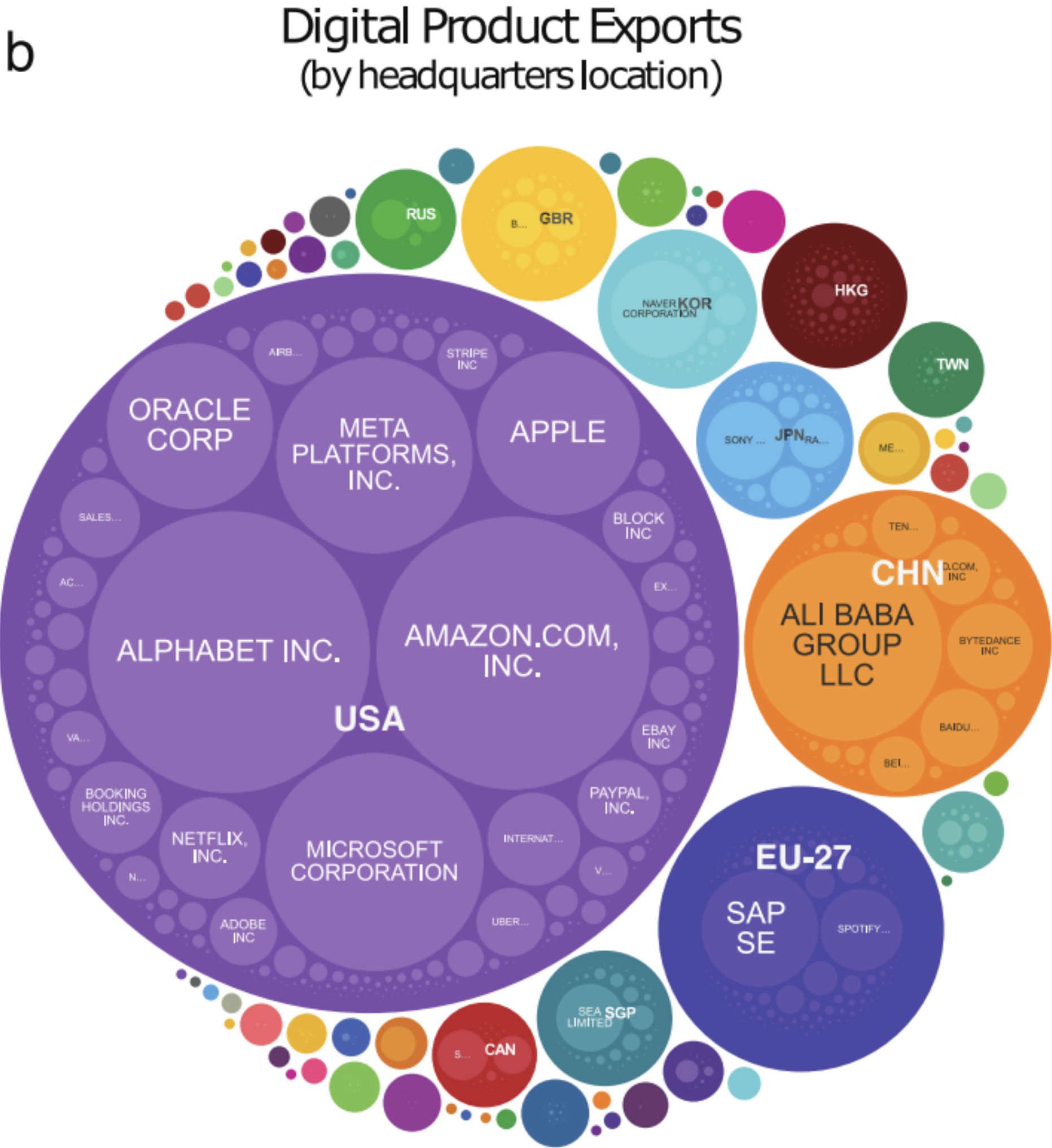
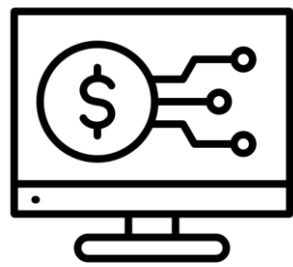
Step 3: Assigning domestic revenues/consumption and **international trade**.

Optimal transport: Trade as little as possible***

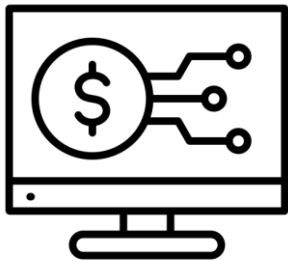
- **Relevant for customs purposes.**
- Assigning all revenue to headquarters: **relevant for the geography of knowledge.**



The dataset



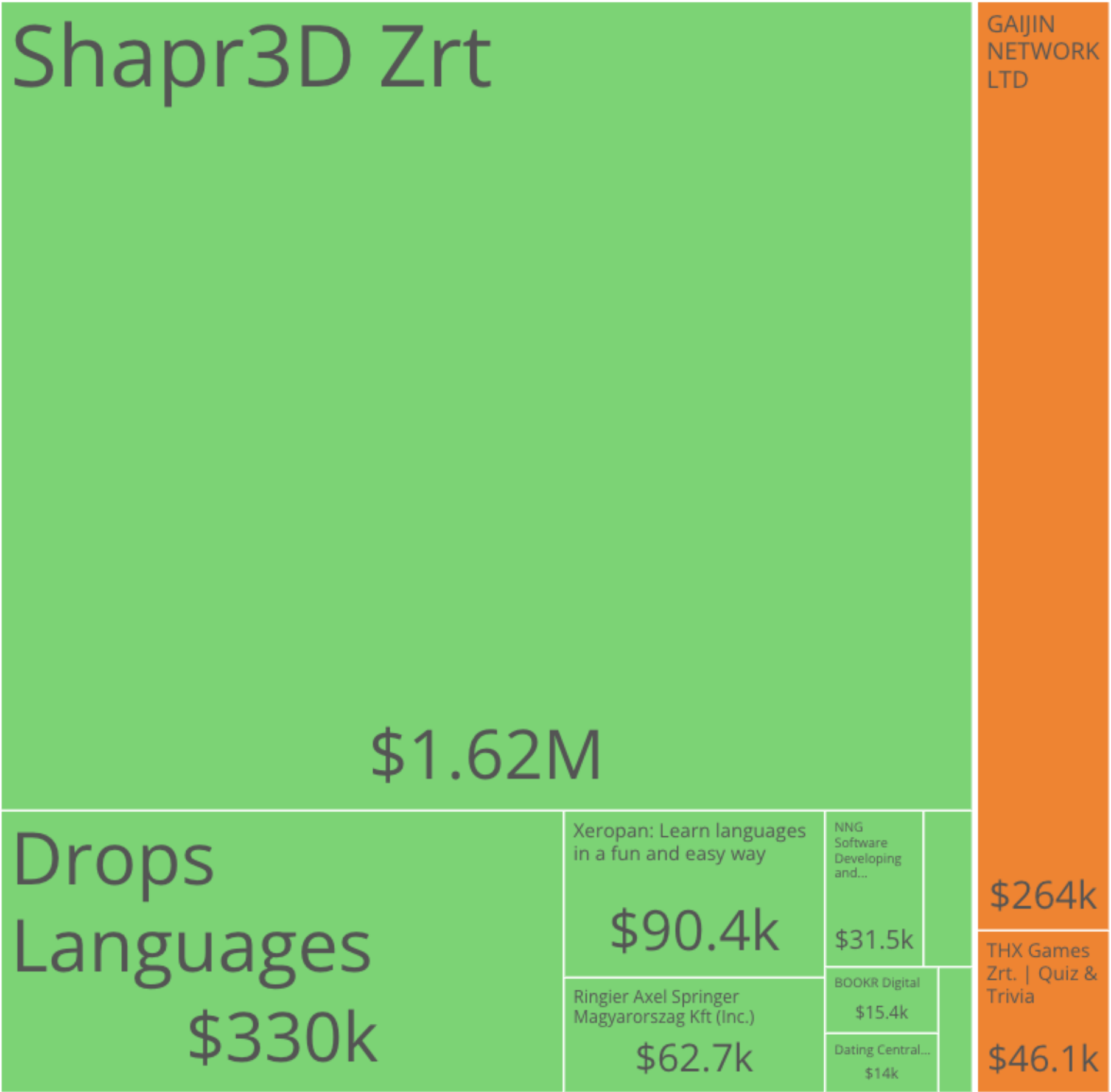
The dataset - Hungary



Hungary digital exporters

[Click to Select a Firm]

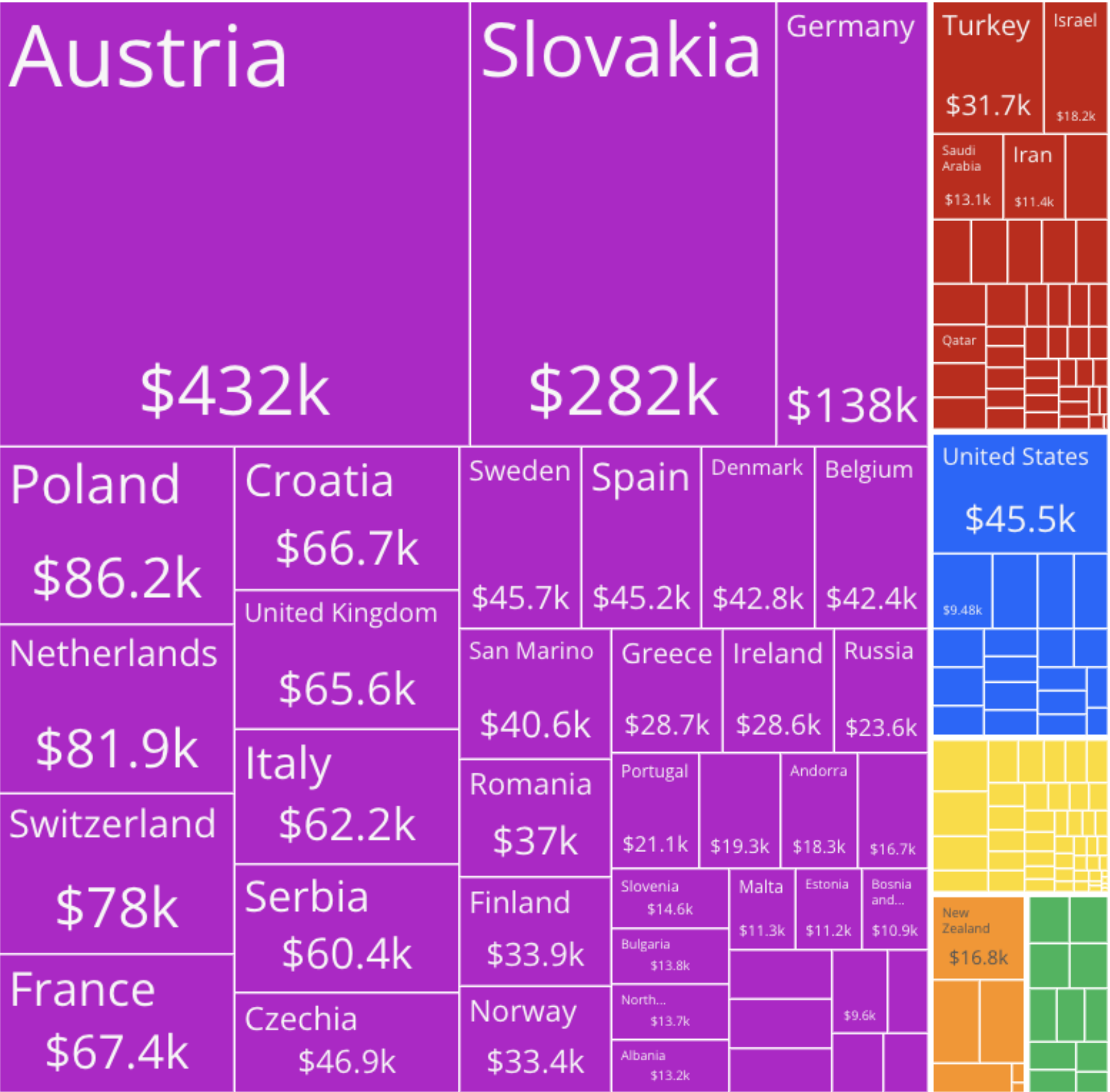
Total: \$2.5M



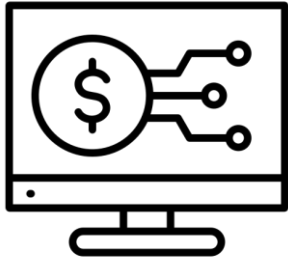
Hungary digital exports destinations

[Click to Select a Country]

Total: \$2.5M

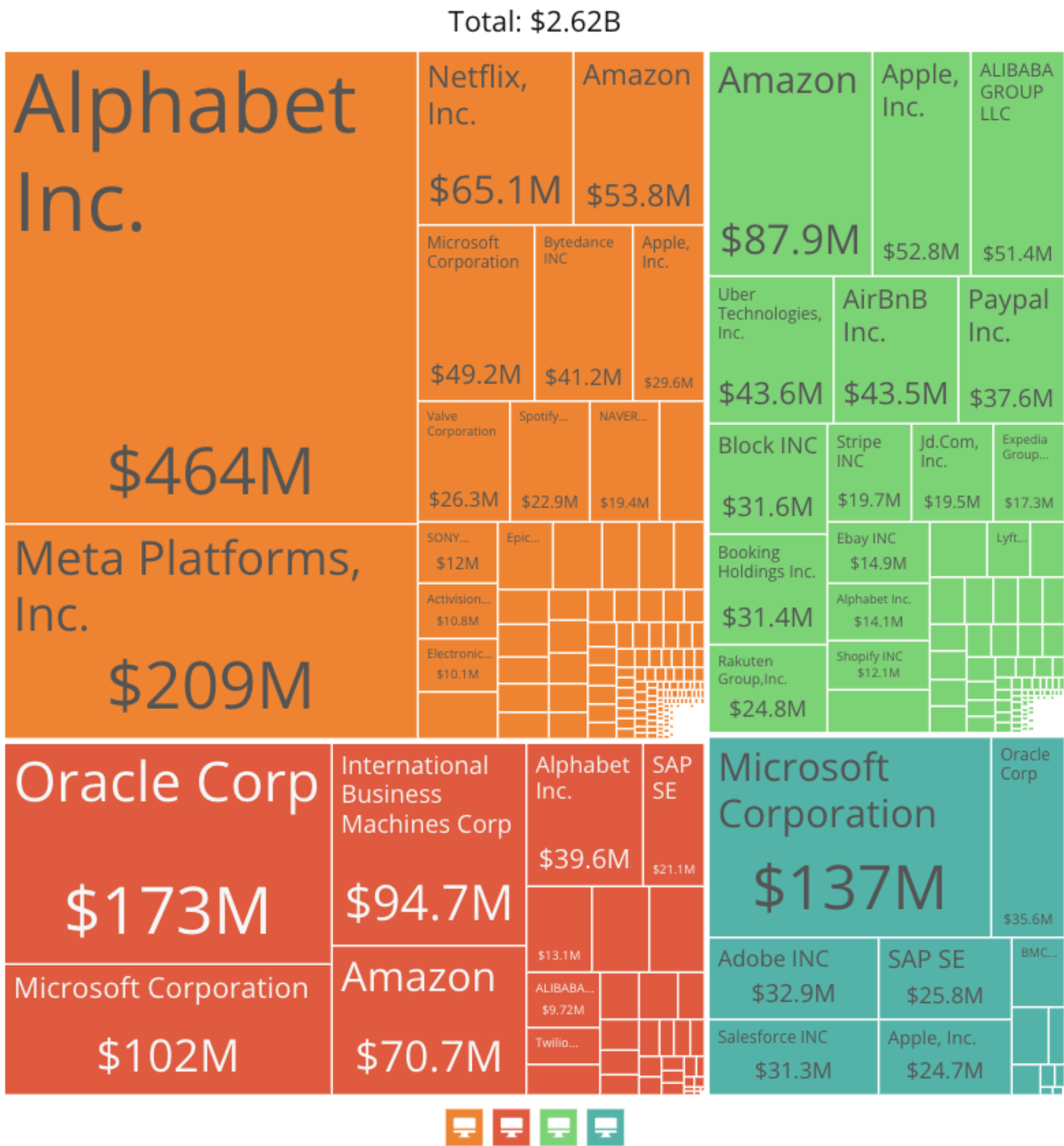


The dataset - Hungary



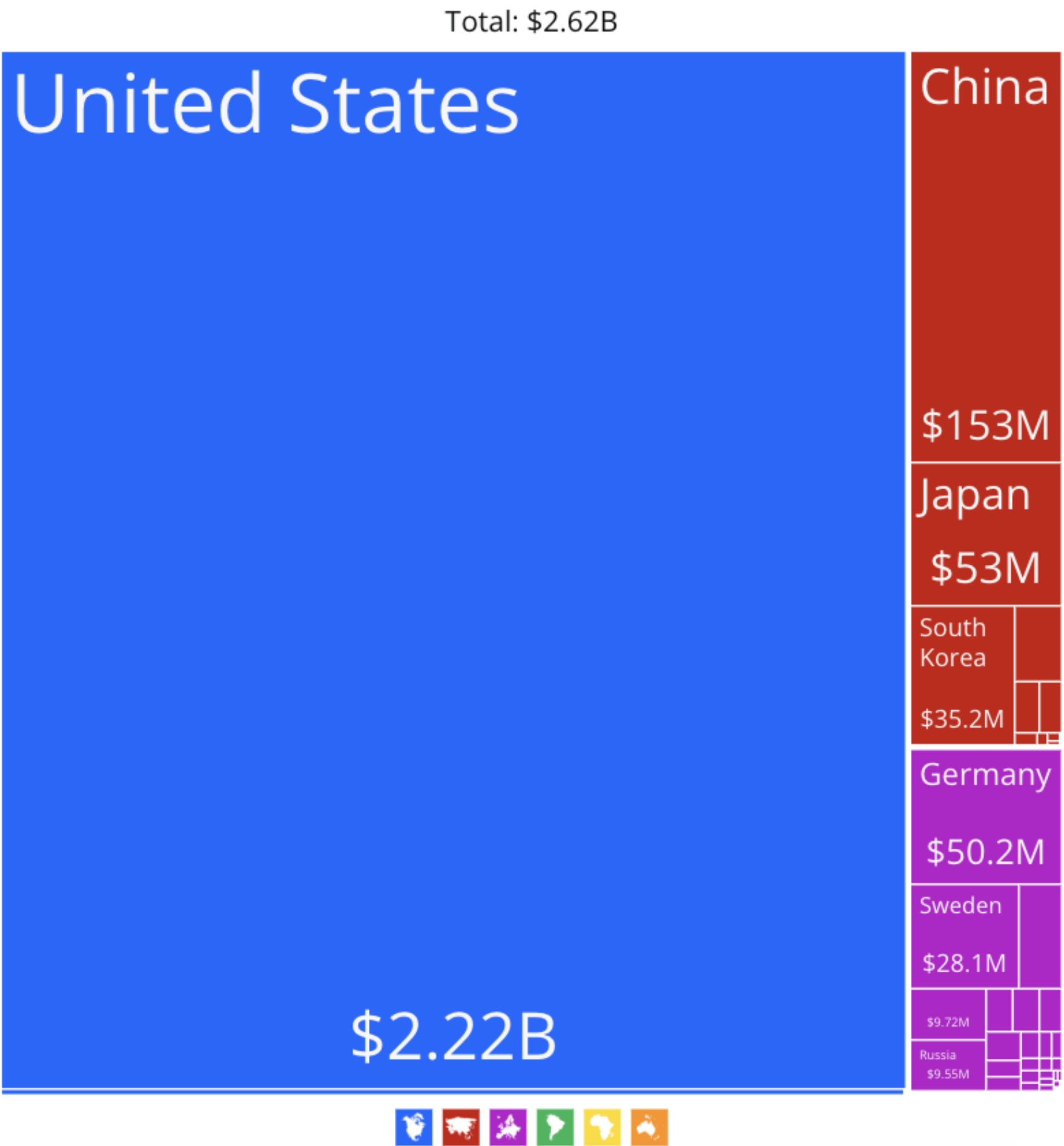
Hungary digital importers

[Click to Select a Firm]

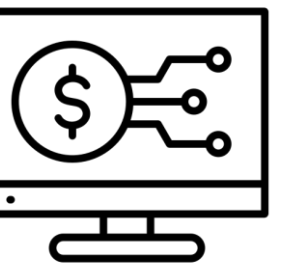


Hungary digital imports origins

[Click to Select a Country]



Limitations



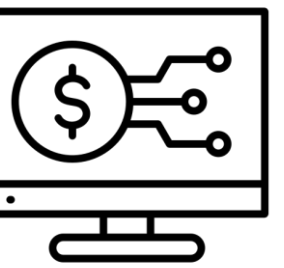
Our estimates are most likely lower bounds:

For instance, we do not see direct transactions between people, only corporate revenues. For instance, if a programmer in India sells a programming service to the UK using Upwork, we only “see” the fee collected by Upwork.

We also provide conservative estimates by maximizing the possible observable domestic consumption (we assume as little trade as possible).

CONFIDENCE INTERVALS - 95% confidence intervals based on a linear regression that predicts the yearly exports of a firm-category pair.

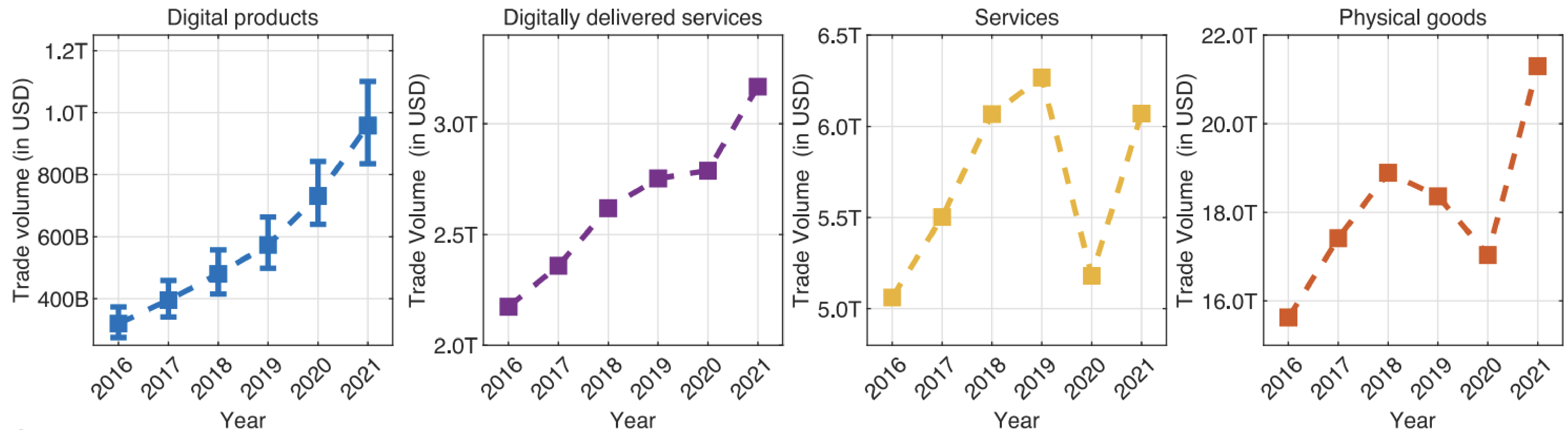
The Geography of Digital Trade



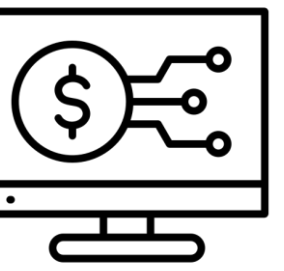
Digital Trade Trends

We estimate that Digital trade grew from USD 320B in 2016 to USD 958B in 2021

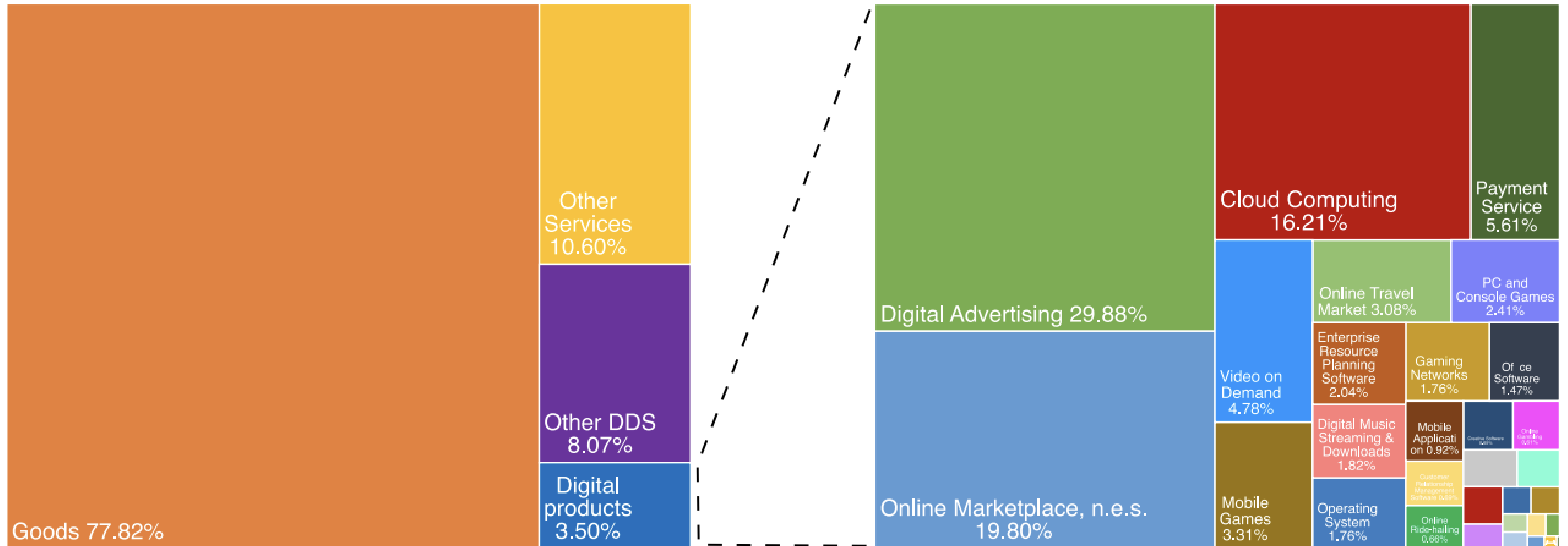
- Digital trade grew during COVID-19, unlike merchandise trade.



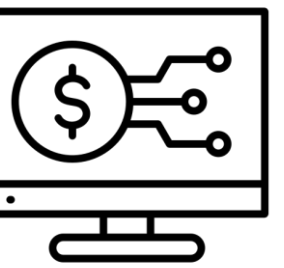
The Geography of Digital Trade



In 2021, digital trade amounted to 3.5% of total world trade

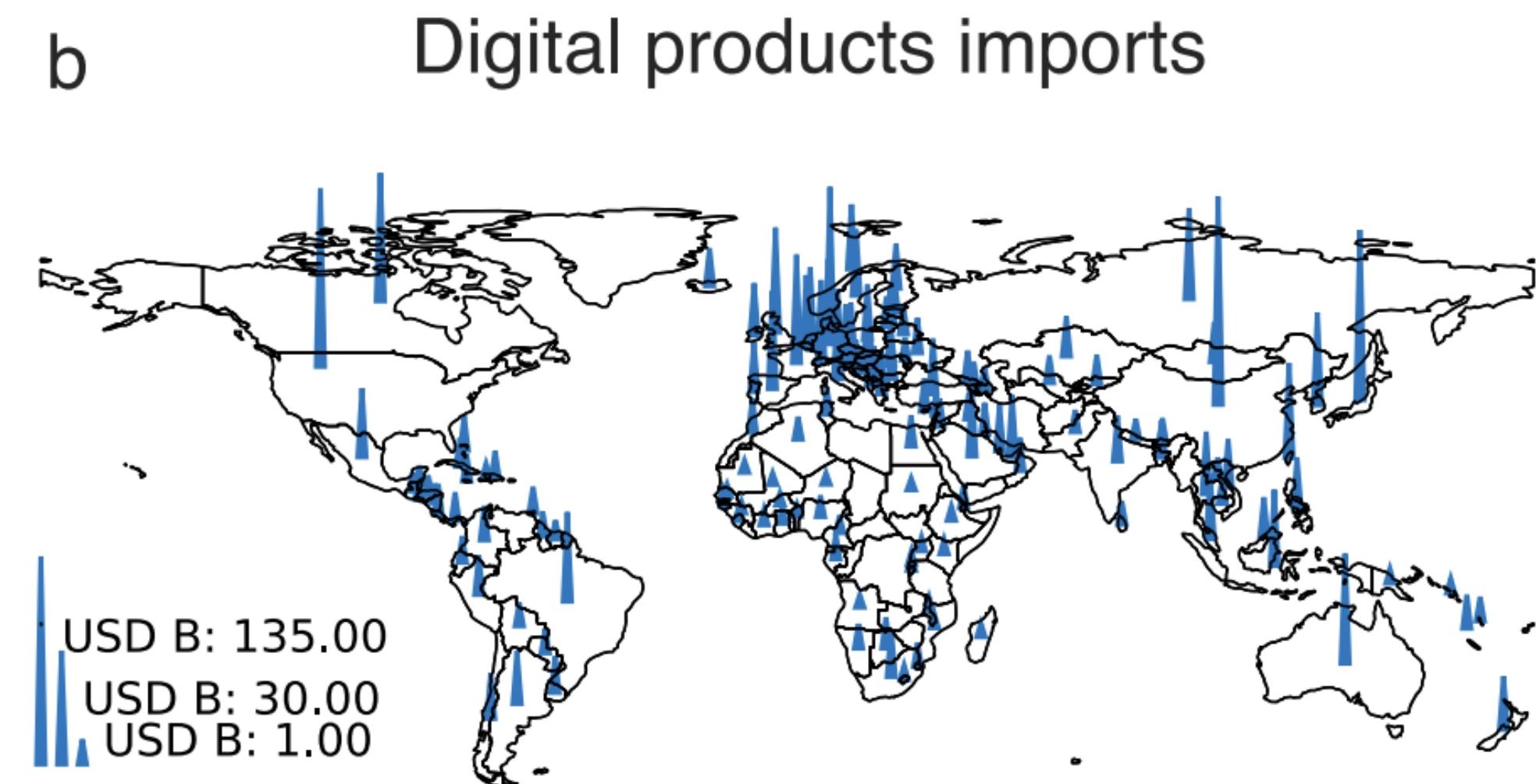


The Geography of Digital Trade

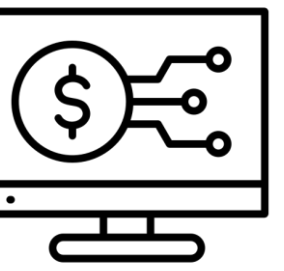


Digital exports are concentrated, whereas digital imports are flatter.

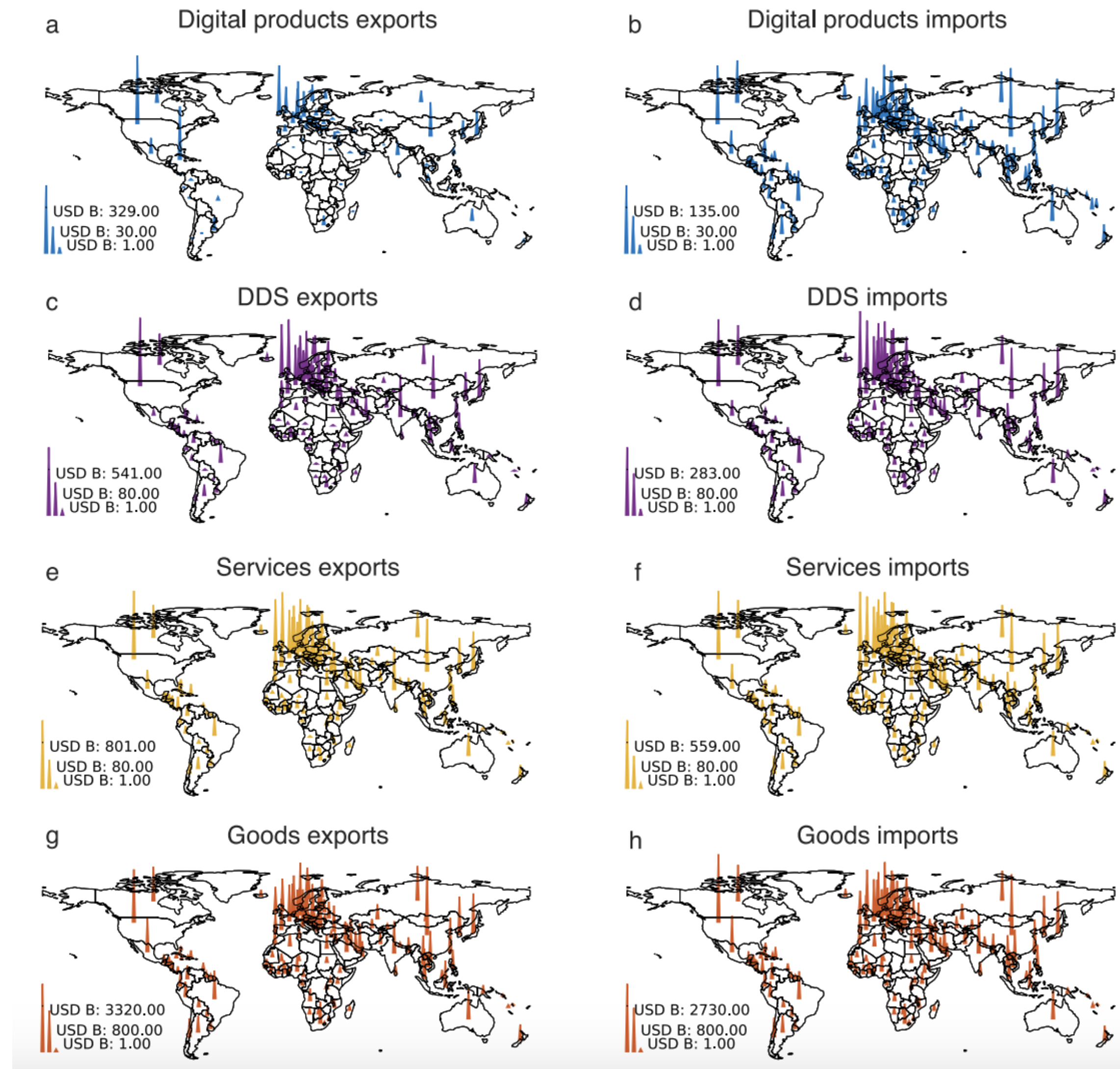
80% of trade in digital products originates in the top 3% of countries, whereas 80% of digital product imports go to less than 20% percent of countries.



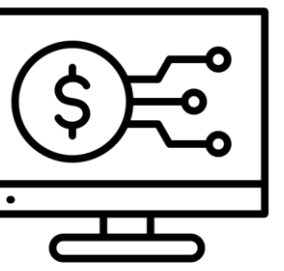
The Geography of Digital Trade



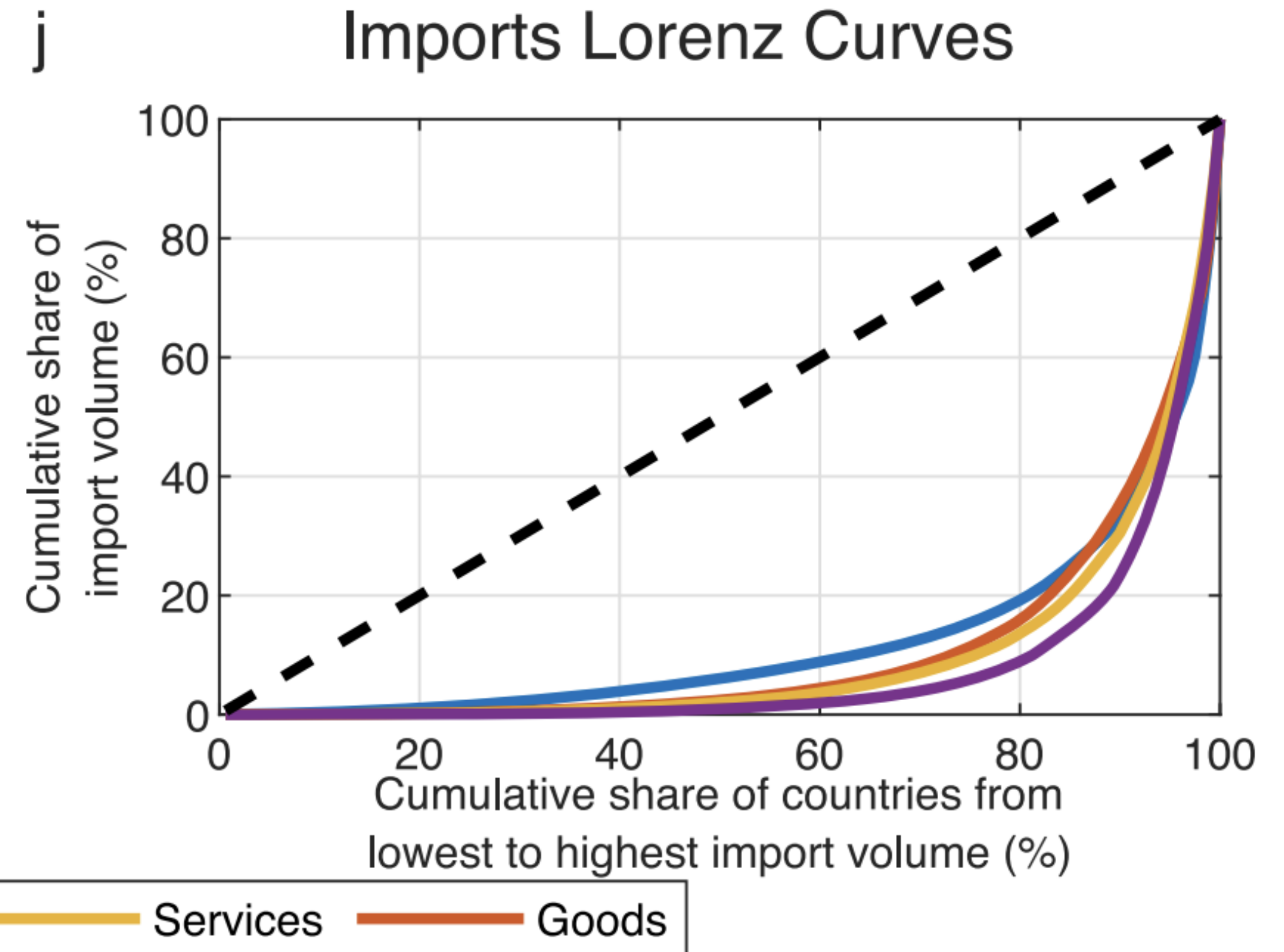
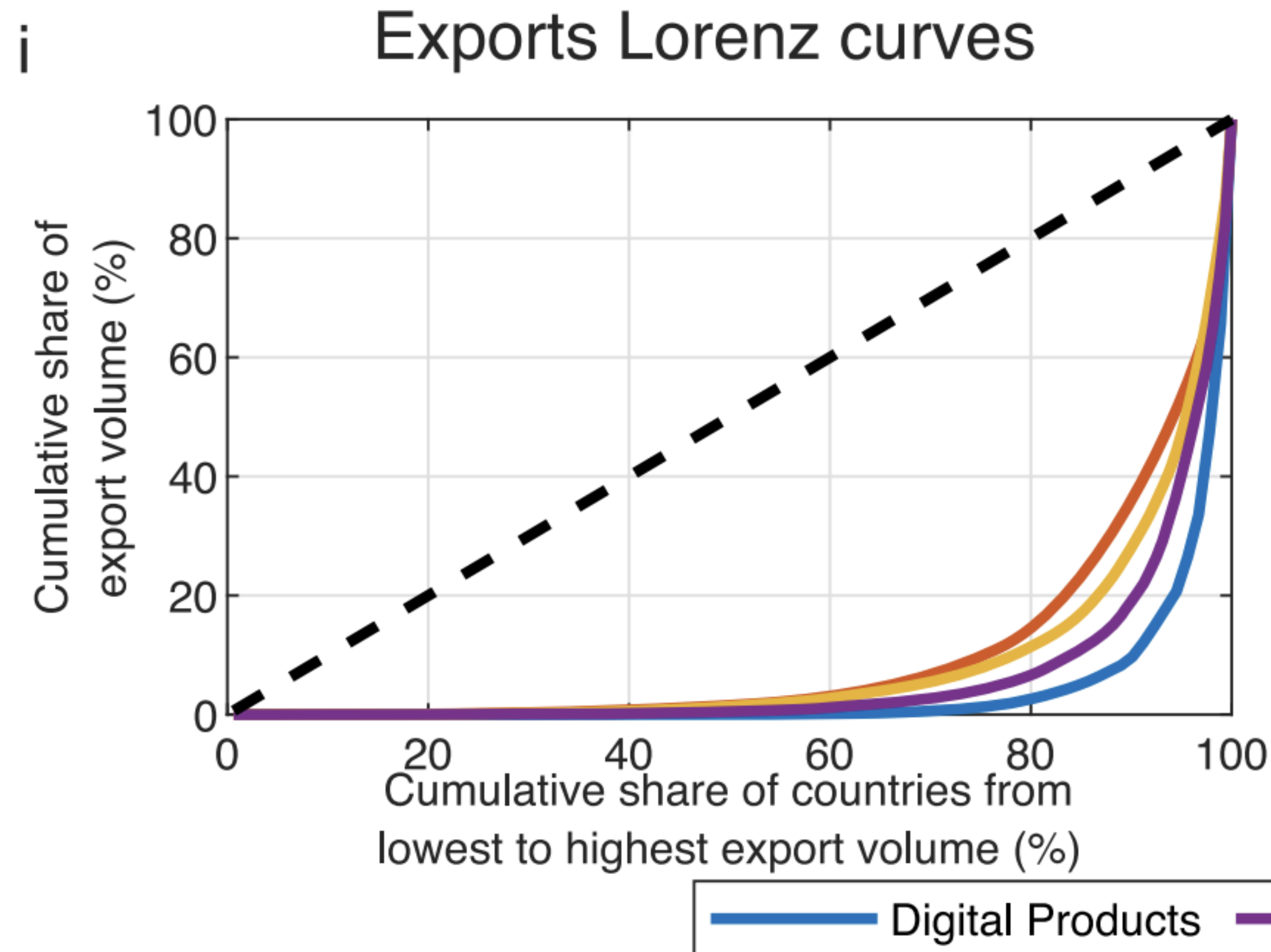
Digital trade is much more concentrated compared to other types of trade



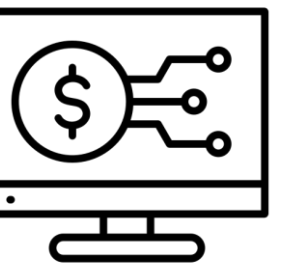
The Geography of Digital Trade



Digital trade is much more concentrated compared to other types of trade



The Geography of Digital Trade

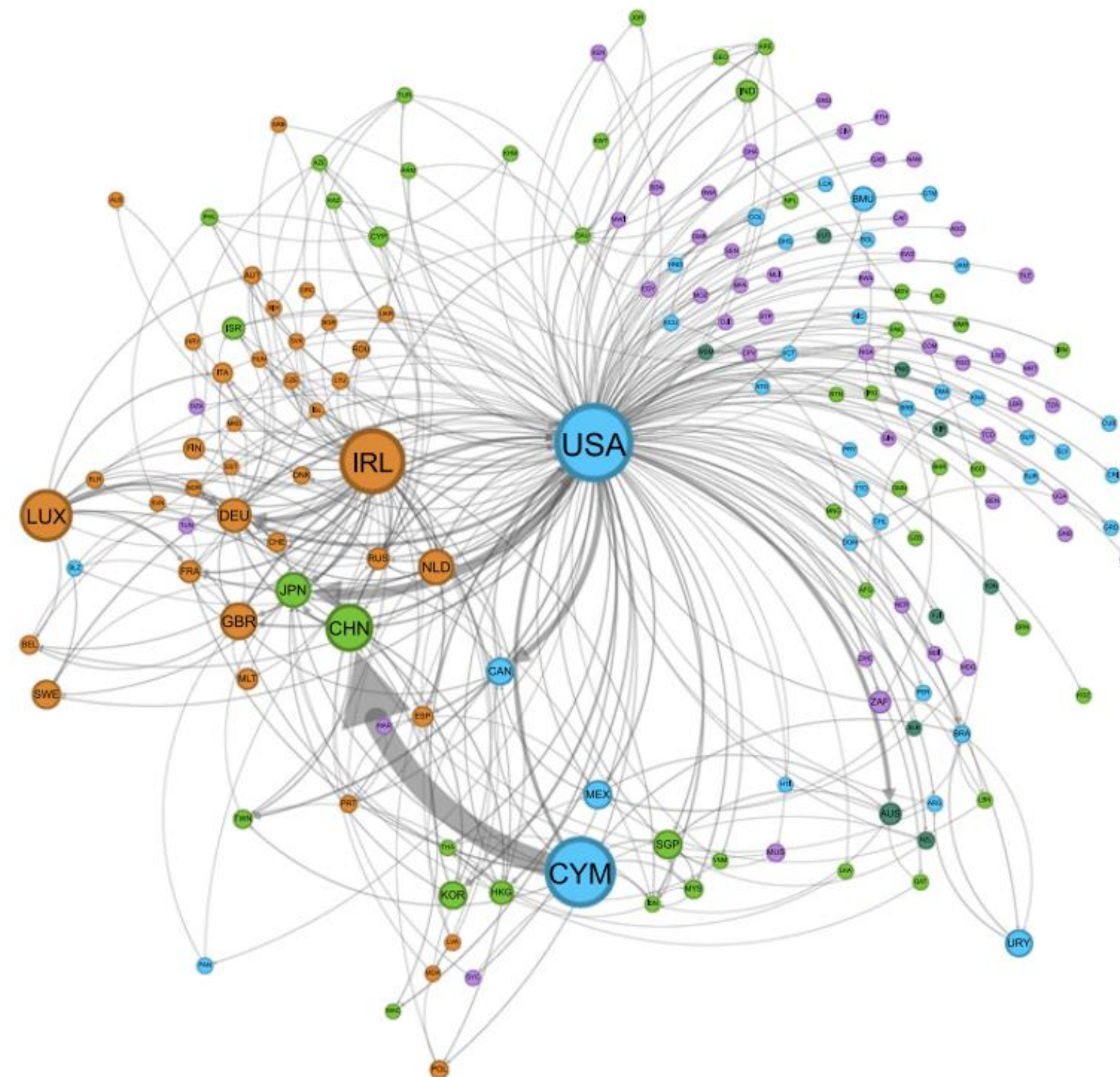


Trade Networks: Differences extend beyond exports and imports...

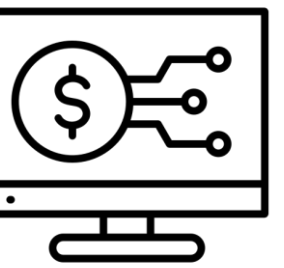
Digital trade is heavily centered on the United States but also several tax havens become central.

a

Digital products

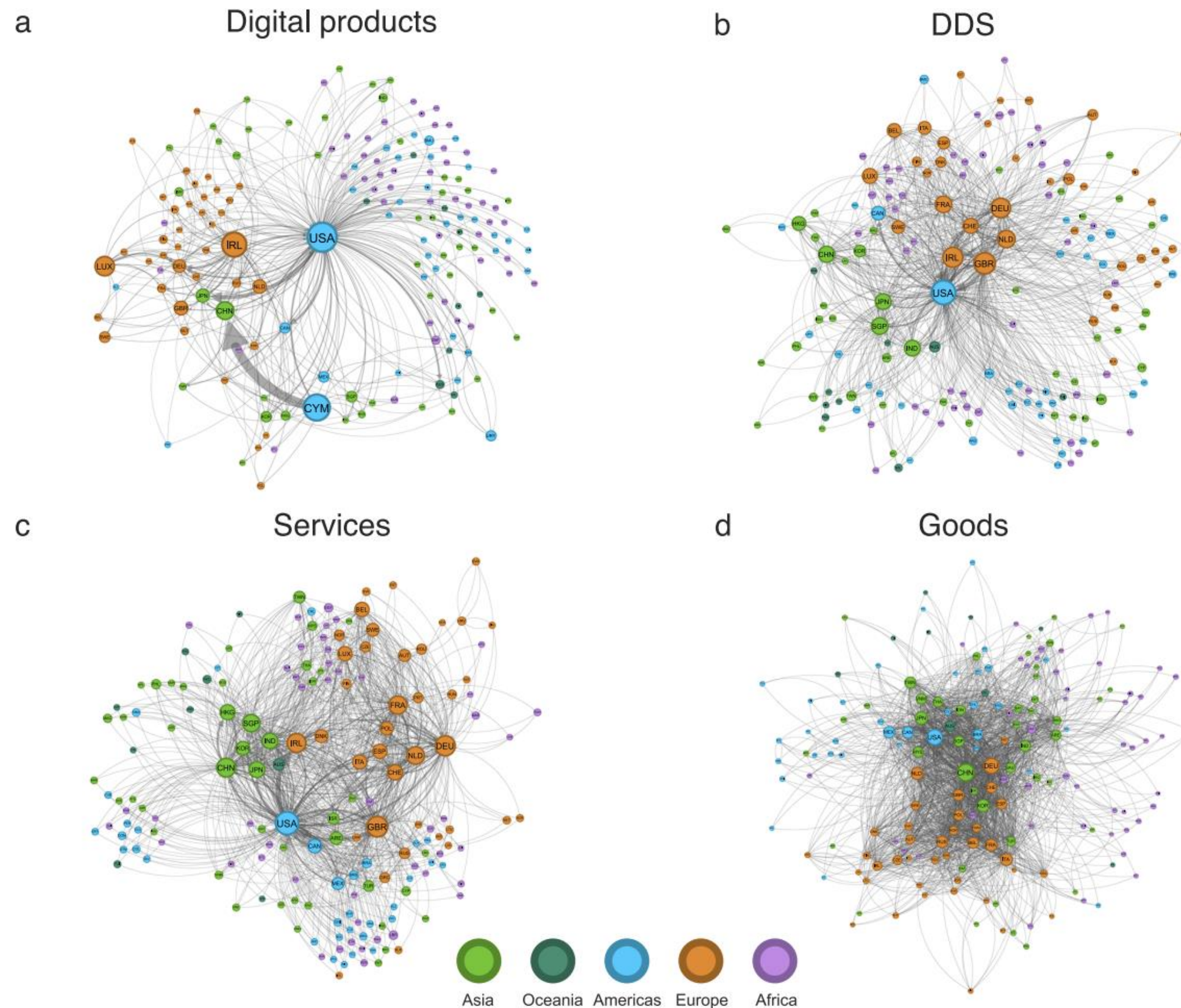


The Geography of Digital Trade

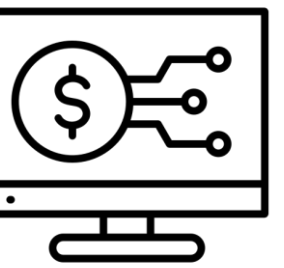


Trade Networks: Differences extend beyond exports and imports...

Digital trade is heavily centered on the United States but also several tax havens become central.

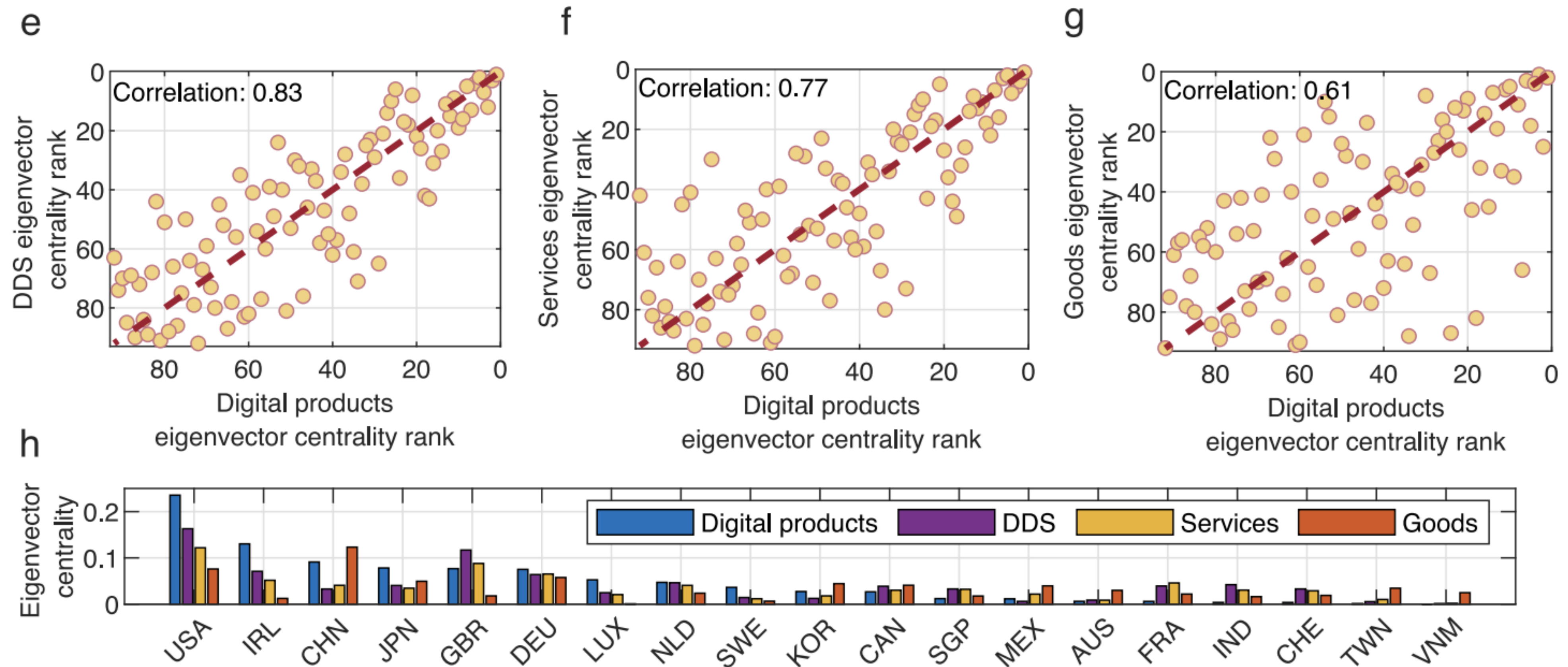


The Geography of Digital Trade

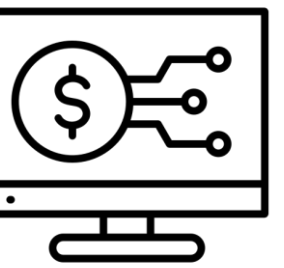


Trade Networks: Differences extend beyond exports and imports...

Digital trade is heavily centered on the United States but also several tax havens become central.

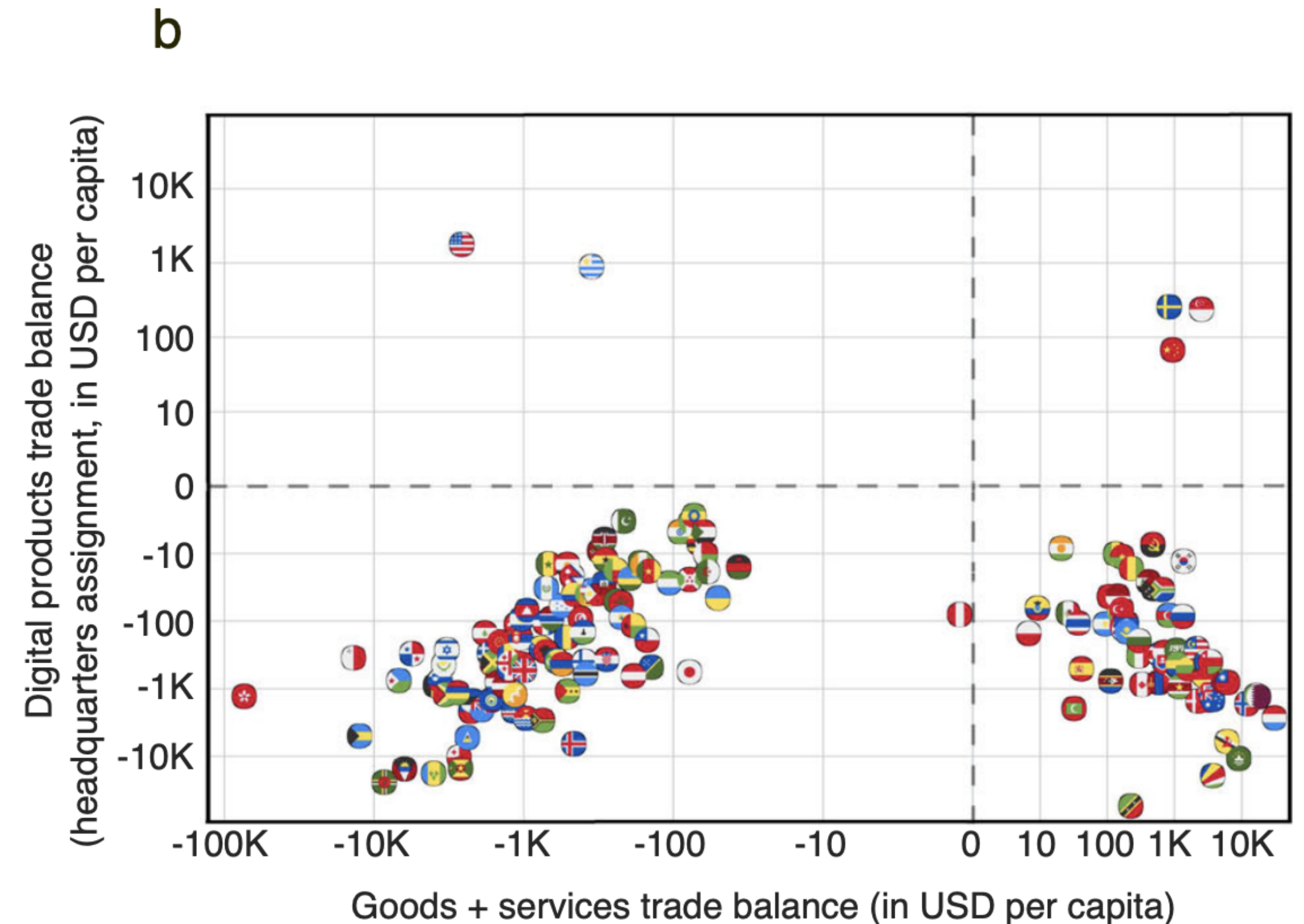
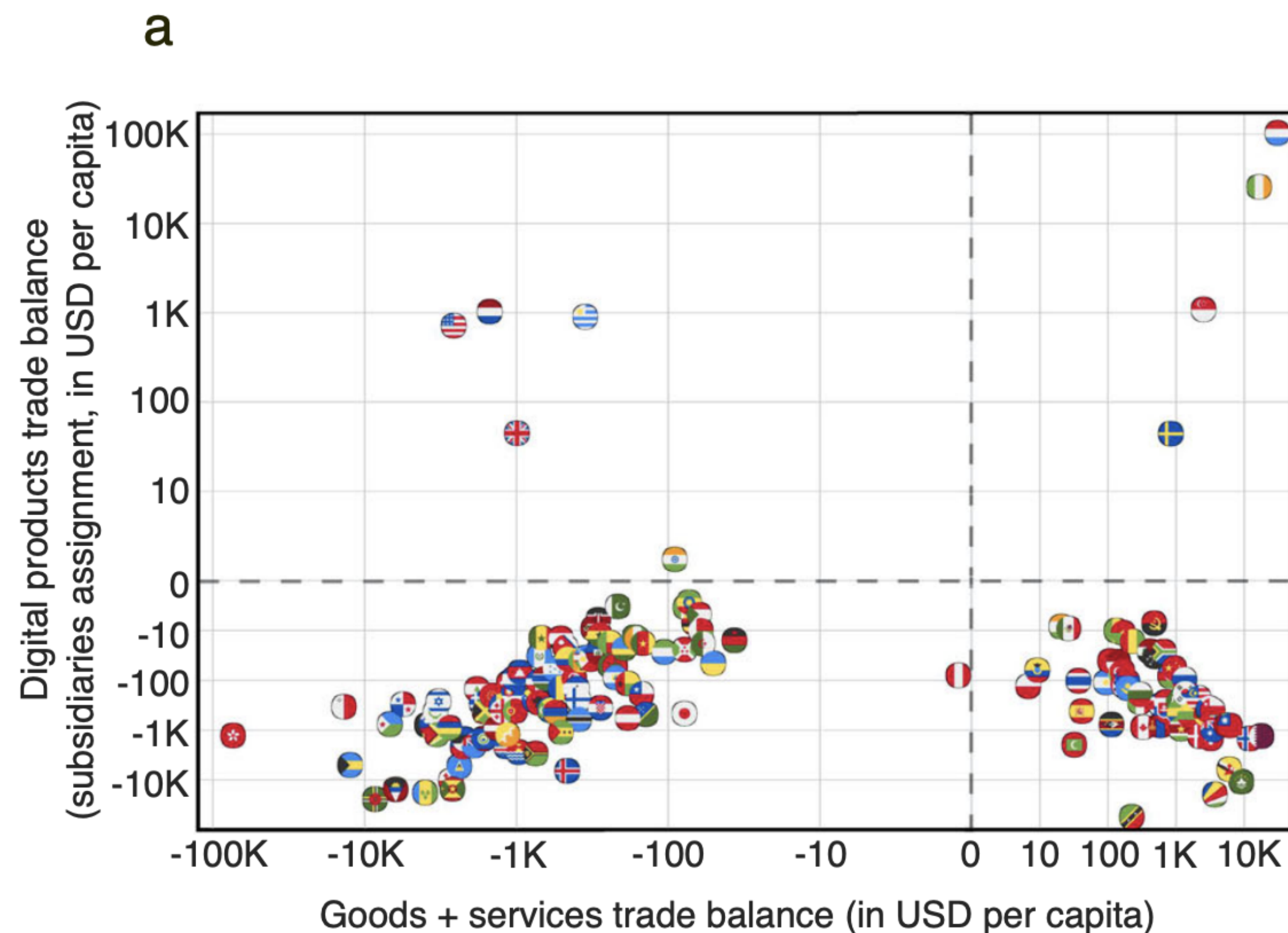


The Implications of Digital Product Trade

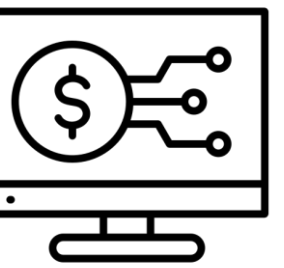


Trade balances

- Many **tax havens** have a positive digital product trade balance.
- Bits exporting countries like the USA have reduced trade deficit.

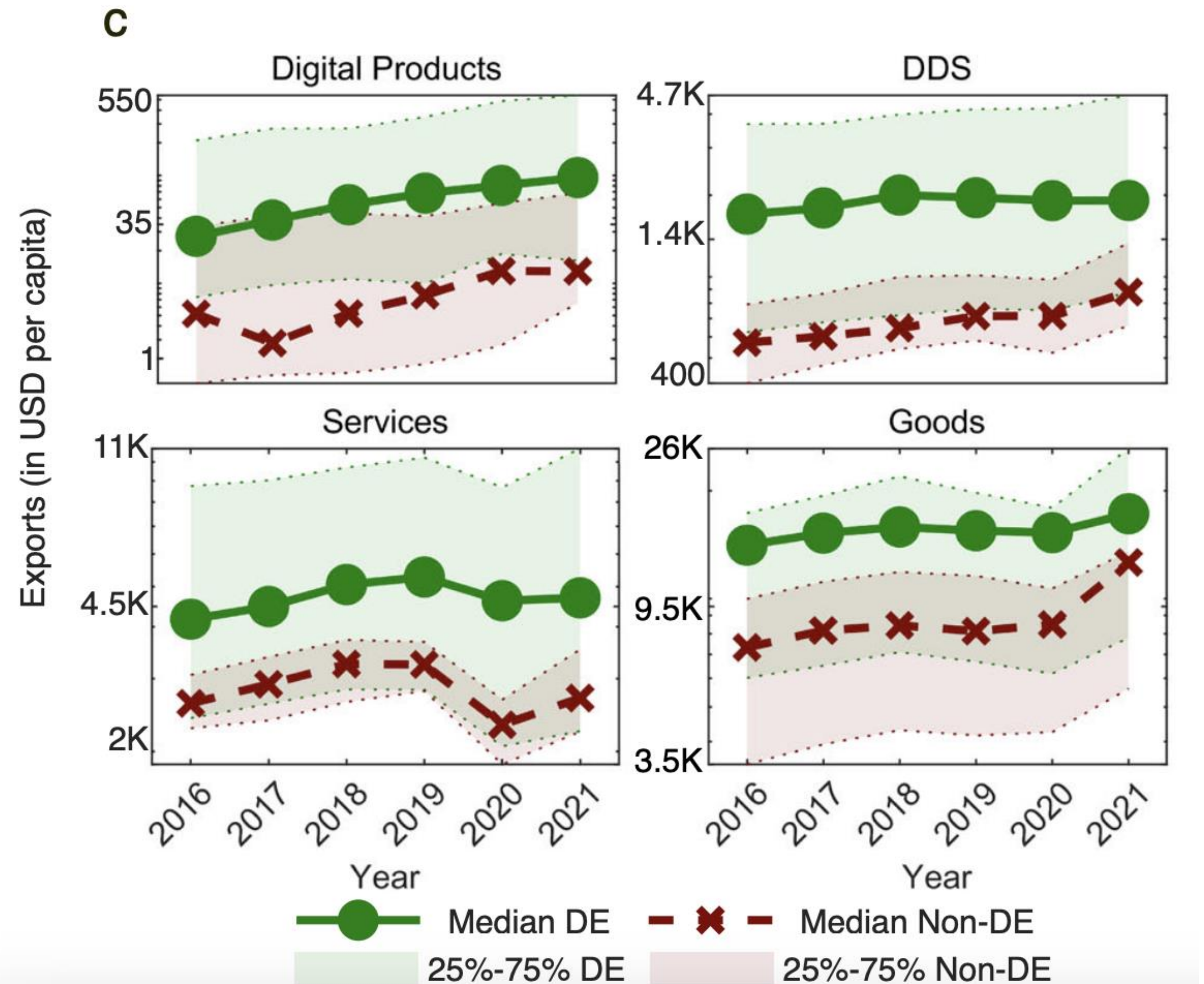


The Geography of Digital Trade

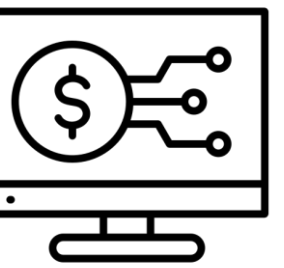


Decoupling and Twin transition

- Focus on high-income countries (WB definition, pop > 1.5M) and ask: **who decoupled growth from emissions between 2016–2019?**
- Compare export structures of decoupled vs non-decoupled: digital products, DDS, services, and goods (25th, median, 75th percentiles).
- Decoupled economies systematically have larger digital and DDS export sectors, pointing to a link between decoupling, digital trade, and the twin transition

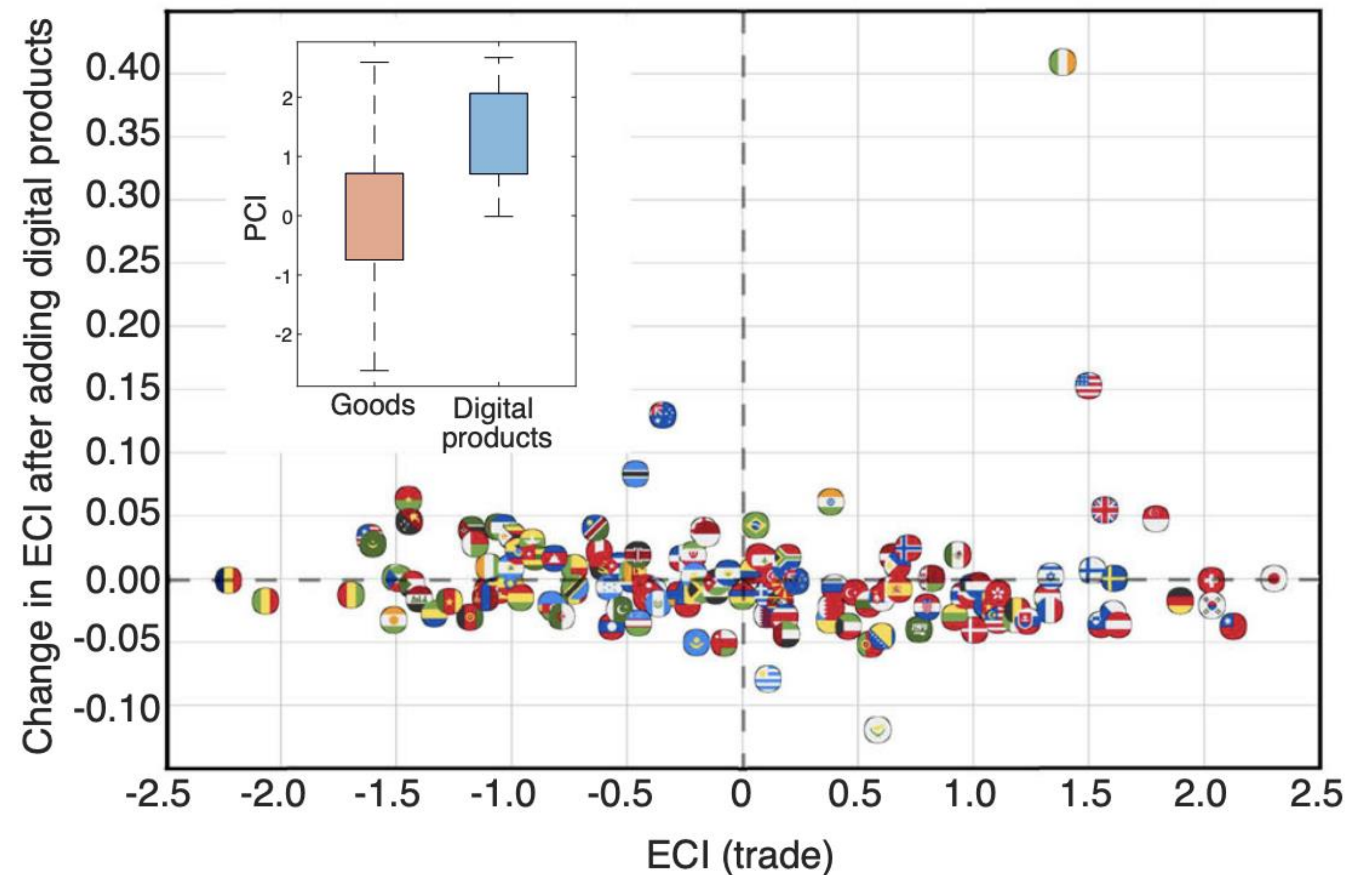


The Geography of Digital Trade

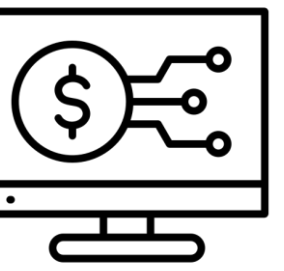


Economic Complexity

- We correct standard economic complexity measures (ECI/PCI) by adding our digital product exports to highly disaggregated HS4 goods trade for 2021.
- Including digital exports lowers complexity for some manufacturing hubs (e.g., Mexico, Slovakia) and raises it for digital leaders (e.g., US, Ireland, Australia), since digital sectors are on average high-complexity.
- The “digital-corrected” ECI performs about as well as the traditional ECI in explaining future growth and emission intensity, even with a short time series.



The Geography of Digital Trade



Discussion

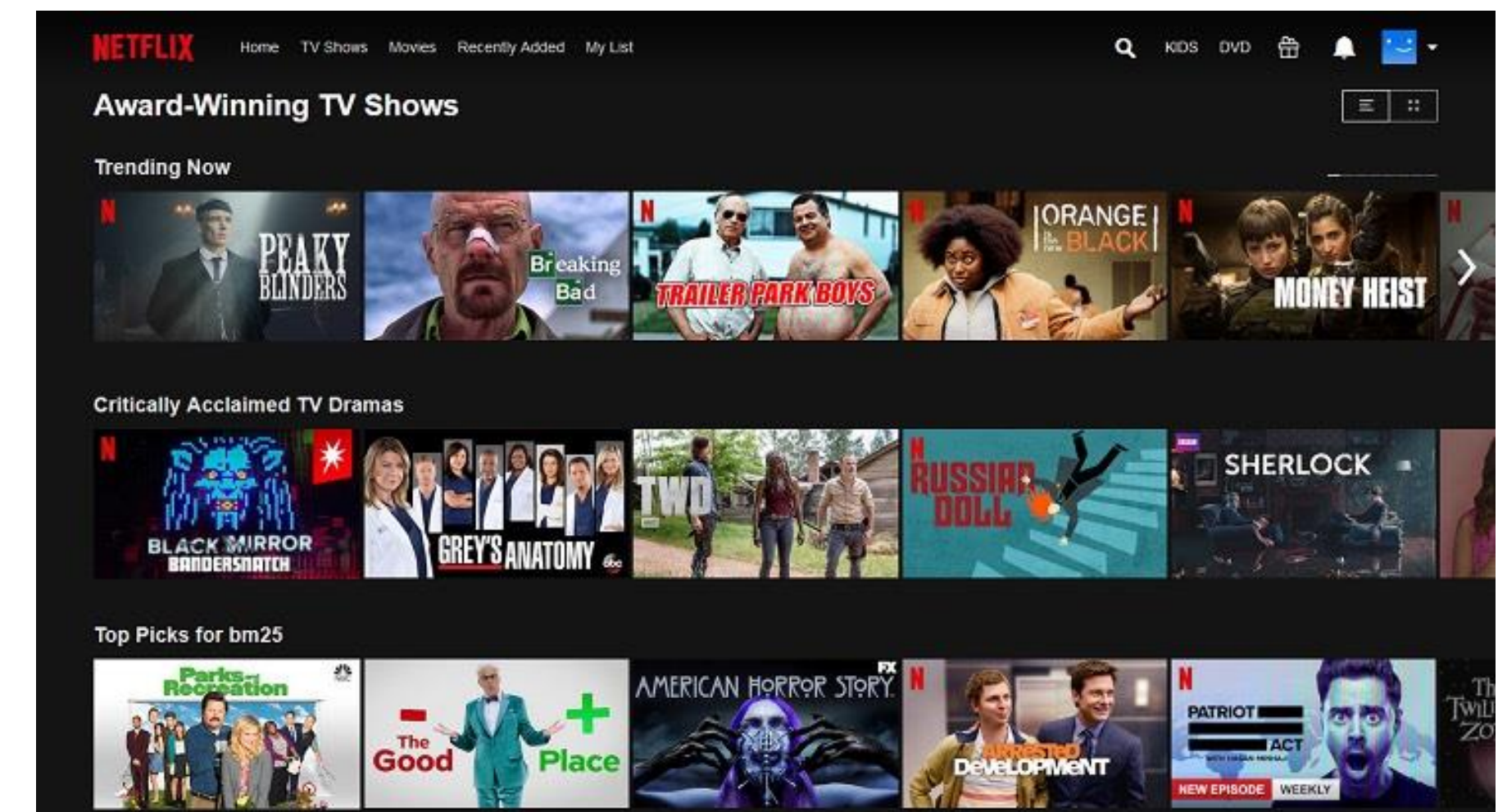
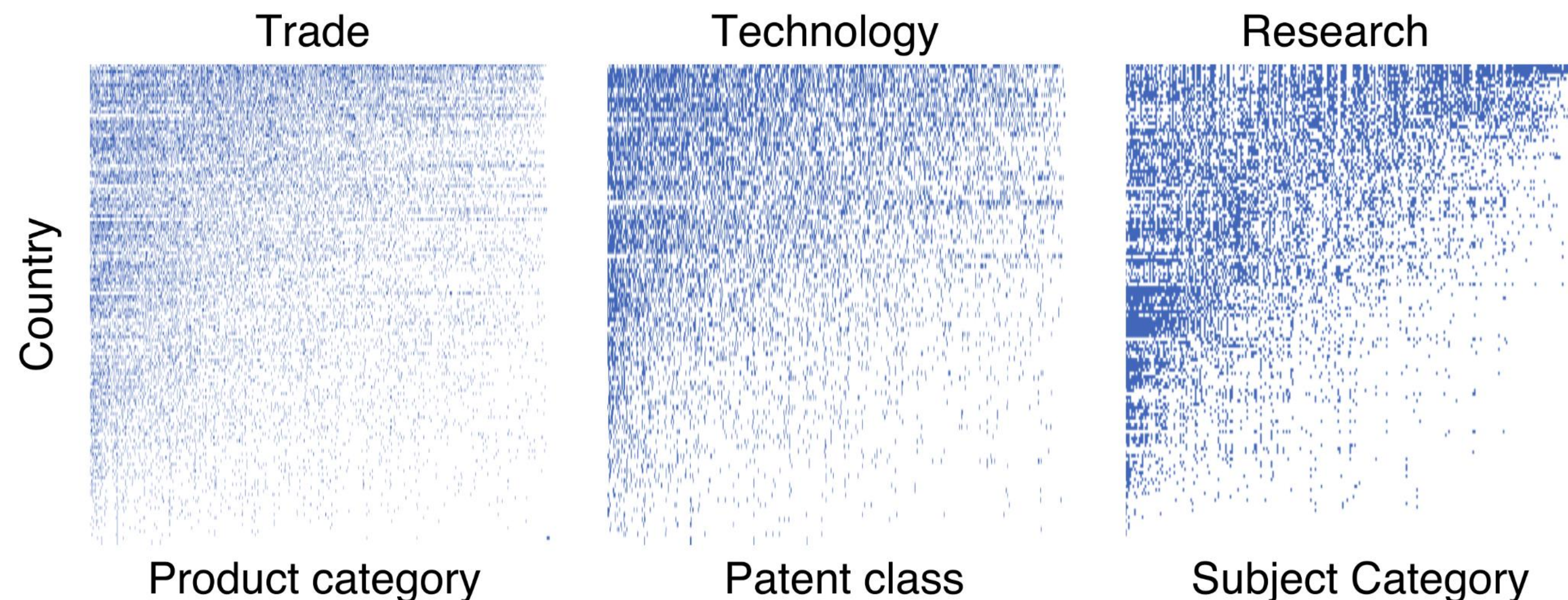
Here we contributed a first dataset to **understand the geography, structure, and implications of digital trade:**

1. Digital trade is growing faster than physical trade.
2. Digital trade is “spiky” in exports (few origins) but “flat” in imports (many destinations)
3. Digital trade reshapes our understanding of trade deficits, reducing the deficit of countries with important physical trade deficits.
4. Countries that have decoupled growth from emissions tend to be growing digital exporters.
5. Digital trade could improve the ability of economic complexity to explain inclusive green growth.



Summary: Measures of Economic Complexity

- **Economic complexity is not just about trade!**
- By balancing multiple targets and opportunities, we can provide a more holistic landscape for inclusive green development, emphasizing that economic complexity transcends trade sophistication.



Future

Many service sectors are still missing...

Second, because the data is collected by customs offices, it includes only goods and not services. This is an important drawback, as services are becoming a rising share of international trade. Unfortunately, the statistical efforts of most countries of the world have not kept up with this reality. Finally, the data does not include information on non-tradable activities. These are an important part of the economic eco-system that allows products and services to be made. Our current research is focused on finding implementable solutions to these limitations, and we hope we will be able to present them in future versions of this Atlas. ●

Future

Many service sectors are still missing...

a)

Alphabet 2023 Annual report

Disaggregated Revenues

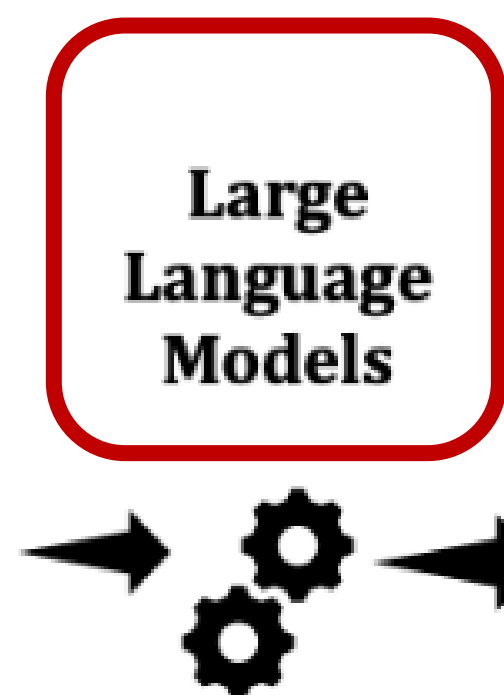
The following table presents revenues disaggregated by type (in millions):

	Year Ended December 31,		
	2021	2022	2023
Google Search & other	\$ 148,851	\$ 162,450	\$ 175,033
YouTube ads	28,845	29,243	31,510
Google Network	31,701	32,780	31,312
Google advertising	209,497	224,473	237,855
Google subscriptions, platforms, and devices	28,032	29,055	34,688
Google Services total	237,529	253,528	272,543
Google Cloud	19,206	26,280	33,088
Other Bets	753	1,068	1,527
Hedging gains (losses)	149	1,960	236
Total revenues	\$ 257,637	\$ 282,836	\$ 307,394

The following table presents revenues disaggregated by geography, based on the addresses of our customers (in millions):

	Year Ended December 31,					
	2021		2022		2023	
United States	\$ 117,854	46 %	\$ 134,814	48 %	\$ 146,286	47 %
EMEA ⁽¹⁾	79,107	31	82,062	29	91,038	30
APAC ⁽¹⁾	46,123	18	47,024	16	51,514	17
Other Americas ⁽¹⁾	14,404	5	16,976	6	18,320	6
Hedging gains (losses)	149	0	1,960	1	236	0
Total revenues	\$ 257,637	100 %	\$ 282,836	100 %	\$ 307,394	100 %

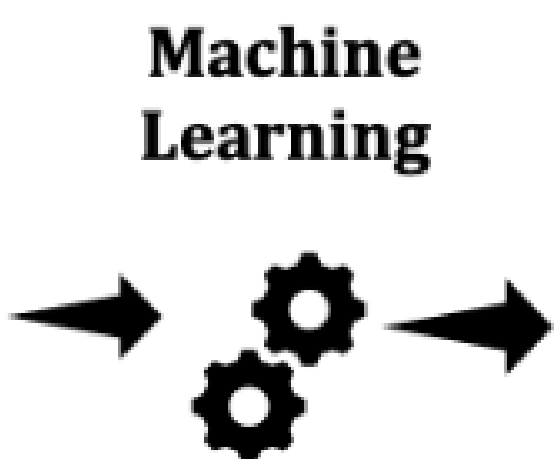
⁽¹⁾ Regions represent Europe, the Middle East, and Africa (EMEA); Asia-Pacific (APAC); and Canada and Latin America ("Other Americas").



b)

Structured Revenue Data

year	firm	origin	destination	category	revenue
2023	Alphabet	USA	USA	Cloud Computing	XB USD
2023	Alphabet	USA	USA	Digital Advertising	XB USD
...	...	...	...	...	...
2023	Alphabet	USA	EU	Digital Advertising	XB USD
2021	Alphabet	USA	EU	Video Streaming	XB USD
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
2023	Alphabet	USA	Other Americas	Cloud Computing	XB USD
2023	Alphabet	USA	Other Americas	Digital Advertising	XB USD
...	...	...	...	...	...



c)

Augmented Revenue Data

year	firm	origin	destination	category	revenue
2023	Alphabet	USA	USA	Cloud Computing	XB USD
2023	Alphabet	USA	USA	Digital Advertising	XB USD
...	...	...	...	...	...
2023	Alphabet	USA	Germany	Digital Advertising	XB USD
2021	Alphabet	USA	North Macedonia	Video Streaming	XB USD
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
2023	Alphabet	USA	Chile	Cloud Computing	XB USD
2023	Alphabet	USA	Brazil	Digital Advertising	XB USD
...	...	...	...	...	...

COMPLEXITYRANKINGS

The Economic Complexity Index (ECI) and the Product Complexity Index (PCI) are, respectively, measures of the relative knowledge intensity of an economy or a product.

Here we provide estimates of economic complexity based on trade data (ECI trade), patent data (ECI technology), and research publication data (ECI research).

These three complexity measures provide a better ability to explain future economic growth than measures of trade complexity alone (more details).

Dataset

ProductCountry

Search

Search

Depth (Trade)

4 Digit6 Digit

Revision (Trade)

Country Rankings



Thank you for your attention!!!

Viktor Stojkoski: vstojkoski@eccf.ukim.edu.mk Economic Complexity: oec.world